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Partial Differential Equation

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COMP 1212

PARTIAL DIFFERENTIAL EQUATION

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CHAPTER 1

CLASSIFICATION AND FORMATION OF PARTIAL DIFFERENTIAL EQUATIONS

1.1. Partial Differential Equations (P.D.E.)

An equation containing one or more partial derivative of an unknown function of two or more independent variables is known as a partial differential equations. we list the following examples of partial differential equations:

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = z + xy \quad \dots(1)$$

$$\left(\frac{\partial z}{\partial x}\right)^2 + \frac{\partial^3 z}{\partial y^3} = 2x \left(\frac{\partial z}{\partial x}\right) \quad \dots(2)$$

$$z \left(\frac{\partial z}{\partial x}\right) + \left(\frac{\partial z}{\partial y}\right) = x \quad \dots(3)$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = xyz \quad \dots(4)$$

$$\frac{\partial^2 z}{\partial x^2} = \left(1 + \frac{\partial z}{\partial y}\right)^{\frac{1}{2}} \quad \dots(5)$$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial z}{\partial y} = 1 \quad \dots(6)$$

1.2. Order of A Partial Differential Equation

The order of a partial differential equation is defined as order of the highest partial derivative occurring in the partial differential equation.

In above examples of partial differential equations (1),(3), and (4) are of the first order, (5) and (6) is of the second order and (2) is of the third order.

1.3. Degree of A Partial Differential Equation

The degree of a partial differential equation is the degree of the highest order derivative which occurs in it after the equation has been rationalized, i.e., made free from radicals and fractions so far as derivative are concerned.

In above examples of partial differential equations (1),(2),(3), and (4), (6) are of first degree while (5) is of second degree.

Notations:

When we consider the case of two independent variable we usually assume them to be x , y and assume z to be the dependent variable. We adopt the following notations throughout the study of partial differential equations

$$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}, \quad r = \frac{\partial^2 z}{\partial x^2}, \quad s = \frac{\partial^2 z}{\partial x \partial y}, \quad t = \frac{\partial^2 z}{\partial y^2}$$

Sometimes the partial differentiations are also denoted by making use of suffixes. Thus we write $u_x = \frac{\partial u}{\partial x}$, $u_y = \frac{\partial u}{\partial y}$, $u_{xx} = \frac{\partial^2 u}{\partial x^2}$, $u_{xy} = \frac{\partial^2 u}{\partial x \partial y}$ and so on.

1.4. Classification of First Order Partial Differential Equations

1.4.1. Linear Equation

A first order partial differential equation is said to be a linear equation if it is linear in p, q and z , i.e., if it is of the form

$$P(x, y)p + Q(x, y)q = R(x, y)z + S(x, y)$$

Examples: $yp - xq = xyz + x$ and $p + q = z + xy$

1.4.2. Semi-Linear Equation

A first order partial differential equation is said to be a semi-linear equation if it is linear in p and q and the coefficient of P and Q the functions of x and y only i.e. if it is of the form

$$P(x, y)p + Q(x, y)q = R(x, y, z)$$

Examples: $e^x p - yxq = xz^2$ and $yp + xq = \frac{x^2 z^2}{y^2}$

1.4.3. Quasi Linear Equation

A first order partial differential equation is said to be a quasi-linear equation if it is linear in p and q , i.e. if it is of the form

$$P(x, y, z)p + Q(x, y, z)q = R(x, y, z).$$

Examples: $(x^2 + z^2)p - xyq = z^3 x + y^2$ and $x^2 zp + y^2 zp = xy$

1.4.4. Non-Linear Equation

Partial differential equations of the form $f(x, y, z, p, q) = 0$ which do not come under the above three types are said to be non-linear equations.

Example: $pq = z$, $p^2 + q^2 = 1$ and $x^2 p^2 + y^2 q^2 = z^2$ are all non-linear partial differential equations.

1.5. Formation of Partial Differential Equation

1.5.1. By Elimination of Arbitrary Constants

Consider an equation $F(x, y, z, a, b) = 0$... (1)

Where a and b denote arbitrary constants. Let z be regarded as function of two independent variables x and y . Differentiating (1) w. r. t x and y partially, we get

$$\frac{\partial F}{\partial x} + p \frac{\partial F}{\partial z} \text{ and } \frac{\partial F}{\partial y} + q \frac{\partial F}{\partial z} = 0 \quad \dots (2)$$

Eliminating two constants a and b from three equations of (1) and (2), we shall obtain an equation of the form $f(x, y, z, p, q) = 0$ which is partial differential equation of the first order.

In a similar manner it can be shown that if there are more arbitrary constants than the number of independent variables, the above procedure of elimination will give rise to partial differential equations of higher order than the first.

Put Your Own Notes

Example: Form the partial differential equation by eliminating arbitrary the constants a & b from $z = ax + by + ab$.

Solution: Given $z = ax + by + ab$... (1)

Differentiating equation (1) partially with respect to x and y we obtain

$$\frac{\partial z}{\partial x} = a = p \quad \dots (2)$$

$$\frac{\partial z}{\partial y} = b = q \quad \dots (3)$$

Substituting p and q for a and b in equation (1), we get the required partial differential equation as $z = px + qy + pq$

Three Situations May Arise

Situation I: When the number of arbitrary constants is less than the number of independent variables, then the elimination of arbitrary constants usually gives rise to more than one partial differential equation of order one.

For Example: Consider $z = ax + y$... (1)

Where a is the only arbitrary constant and x, y are two independent variables.

$$\text{Differentiating (1) partially w. r. t. 'x', we get } \frac{\partial z}{\partial x} = a \quad \dots (2)$$

$$\text{Differentiating (1) partially w. r. t. 'y', we get } \frac{\partial z}{\partial y} = 1 \quad \dots (3)$$

Eliminating a between (1) and (2)

$$z = x \left(\frac{\partial z}{\partial x} \right) + y \quad \dots (4)$$

Since (3) does not contain arbitrary constant, so (3) is also partial differential under consideration. Thus, we get two partial differential equations (3) and (4).

Situation II: When the number of arbitrary constants is equal to the number of independent variables, then the elimination of arbitrary constants shall give rise to a unique partial differential equation of order one.

For Example: Eliminate a and b from $az + b = a^2x + y$... (1)

Differentiating (1) partially w. r. t. x and y , we get

$$a \left(\frac{\partial z}{\partial x} \right) = a^2 \quad \dots (2) \quad a \left(\frac{\partial z}{\partial y} \right) = 1 \quad \dots (3)$$

Eliminating a from (2) and (3), we have $\frac{\partial z}{\partial x} \cdot \frac{\partial z}{\partial y} = 1$, which is the unique partial differential equation of order one.

Situation III: When the number of arbitrary constants is greater than the number of independent variables, then the elimination of arbitrary constants leads to a partial differential equation of order usually greater than one.

Put Your Own Notes

For Example: Eliminate a, b and c from

$$z = ax + by + cxy \quad \dots(1)$$

Differentiating (1) partially w. r. t x and y , we have

$$\frac{\partial z}{\partial x} = a + cy \quad \dots(2)$$

$$\frac{\partial z}{\partial y} = b + cx \quad \dots(3)$$

Differentiating (2) partially w. r. t. x and (3) w. r. t. y , we have

$$\left(\frac{\partial^2 z}{\partial x^2} \right) = 0 \quad \dots(4)$$

$$\left(\frac{\partial^2 z}{\partial y^2} \right) = 0 \quad \dots(5)$$

Differentiating (2) w.r. t y , we have

$$\frac{\partial^2 z}{\partial x \partial y} = c \quad \dots(6)$$

Now, (2) and (3)

$$\Rightarrow x \left(\frac{\partial z}{\partial x} \right) = ax + cxy \text{ and } y \left(\frac{\partial z}{\partial y} \right) = by + cxy$$

$$\therefore x \left(\frac{\partial z}{\partial x} \right) + y \left(\frac{\partial z}{\partial y} \right) = \underline{ax + by + cxy} + cxy$$

$$x \left(\frac{\partial z}{\partial x} \right) + y \left(\frac{\partial z}{\partial y} \right) = z + xy \left(\frac{\partial^2 z}{\partial x \partial y} \right), \text{ (By using (1) and (6))} \quad \dots(7)$$

Thus, we get three partial differential equation given by (4), (5) and (7) which are all of order two.

1.5.2. By Elimination of Arbitrary Function ϕ

Let $\phi(u, v) = 0$, where u and v are functions of x, y and z . $\dots(1)$

We treat z as dependent variable and x and y as independent variables so that

$$\frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q, \quad \frac{\partial y}{\partial x} = 0 \text{ and } \frac{\partial x}{\partial y} = 0. \quad \dots(2)$$

Differentiating (1) partially w.r.t. x , we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial u}{\partial z} \cdot \frac{\partial z}{\partial x} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} \cdot \frac{\partial x}{\partial x} + \frac{\partial v}{\partial y} \cdot \frac{\partial y}{\partial x} + \frac{\partial v}{\partial z} \cdot \frac{\partial z}{\partial x} \right)$$

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0$$

Put Your Own Notes

$$\frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{-\left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z}\right)}{\left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z}\right)} \quad \dots(3)$$

Similarly, differentiating (1) partially w.r.t 'y', we get

$$\frac{\frac{\partial \phi}{\partial u}}{\frac{\partial \phi}{\partial v}} = \frac{-\left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z}\right)}{\left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z}\right)} \quad \dots(4)$$

Eliminating ϕ with the help of (3) & (4), we get

$$\frac{\left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z}\right)}{\left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z}\right)} = \frac{\left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z}\right)}{\left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z}\right)}$$

$$\left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z}\right) \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z}\right) = \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z}\right) \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z}\right)$$

$$Pp + Qq = R \quad \dots(5)$$

Where

$$P = \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial z} - \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial y}, \quad Q = \frac{\partial u}{\partial z} \cdot \frac{\partial v}{\partial x} - \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial z}, \quad R = \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial x}$$

Thus we obtain a linear partial differential equation of first order and of first degree, i.e. in p and q .

Example: Form a partial differential equation by eliminating the arbitrary function ϕ from $\phi(x+y+z, x^2+y^2-z^2)=0$ what is the order of this partial differential equation?

$$\text{Solution: Given } \phi(x+y+z, x^2+y^2-z^2)=0 \quad \dots(1)$$

$$\text{Let } u = x+y+z \text{ and } v = x^2+y^2-z^2 \quad \dots(2)$$

$$\text{Then (1) becomes } \phi(u, v) = 0 \quad \dots(3)$$

Differentiating (3) w.r.t x partially, we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial x} + p \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial x} + p \frac{\partial v}{\partial z} \right) = 0 \quad \dots(4)$$

from (2),

$$\frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = 1, \frac{\partial u}{\partial z} = 1, \frac{\partial v}{\partial x} = 2x, \frac{\partial v}{\partial y} = 2y, \frac{\partial v}{\partial z} = -2z \quad \dots(5)$$

from (4) and (5),

$$\left(\frac{\partial \phi}{\partial u} \right) (1+p) + 2 \left(\frac{\partial \phi}{\partial v} \right) (x-pz) = 0$$

$$\frac{\left(\frac{\partial \phi}{\partial u}\right)}{\left(\frac{\partial \phi}{\partial v}\right)} = \frac{-2(\bar{x} - pz)}{(1+p)} \quad \dots(6)$$

Again, differentiating (3) w.r.t, y partially, we get

$$\frac{\partial \phi}{\partial u} \left(\frac{\partial u}{\partial y} + q \frac{\partial u}{\partial z} \right) + \frac{\partial \phi}{\partial v} \left(\frac{\partial v}{\partial y} + q \frac{\partial v}{\partial z} \right) = 0$$

$$\frac{\partial \phi}{\partial u} (1+q) + 2 \left(\frac{\partial \phi}{\partial v} \right) (y - zq) = 0, \text{ by (5)}$$

$$\frac{\left(\frac{\partial \phi}{\partial u}\right)}{\left(\frac{\partial \phi}{\partial v}\right)} = \frac{-2(y - qz)}{(1+q)} \quad \dots(7)$$

Eliminating ϕ from (6) and (7), we obtain

$$\frac{(x - pz)}{(1+p)} = \frac{(y - qz)}{(1+q)}$$

$$\text{or } (1+q)(x - pz) = (1+p)(y - qz) \text{ or } (y + z)p - (x + z)q = x - y$$

which is the desired partial differential equation of first order.

Example: Form the PDE by eliminating the arbitrary function from $z = f(x+it) + g(x-it)$, where $i = \sqrt{-1}$.

Solution:

$$\text{Given } z = f(x+it) + g(x-it) \quad \dots(1)$$

Differentiating equation (1) twice partially with respect to x and t , we get

$$\frac{\partial z}{\partial x} = f'(x+it) + g'(x-it)$$

$$\frac{\partial^2 z}{\partial x^2} = f''(x+it) + g''(x-it) \quad \dots(2)$$

Here, f' indicates derivative of f with respect to $(x+it)$ and g' indicates derivative of g with respect to $(x-it)$. Also, we have

$$\frac{\partial z}{\partial t} = i[f'(x+it) - g'(x-it)]$$

$$\frac{\partial^2 z}{\partial t^2} = -(f''(x+it) + g''(x-it)) \quad \dots(3)$$

From Equation (2) and (3), we at once, find that

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial t^2} = 0, \quad \dots(4)$$

which is the required PDE.

Note: We observed that by eliminating arbitrary functions, we always produce quasi-linear partial differential equations, only. However, we get both quasi-linear as well as non-linear partial differential equations when we eliminate arbitrary constants.

Put Your Own Notes

CHAPTER 2

LINEAR PARTIAL DIFFERENTIAL EQUATION OF ORDER ONE

2.1 Lagrange's Method

Theorem: The general solution of the quasi-linear partial differential equation

$$Pp + Qq = R \quad \dots(1)$$

$$\text{is } \phi(u, v) = 0, \quad \dots(2)$$

where ϕ is an arbitrary function and $u(x, y, z) = c_1$ and $v(x, y, z) = c_2 \dots(3)$ are two

$$\text{independent solutions of } \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(4)$$

Where c_1 & c_2 are arbitrary constants and at least one of u & v must contain z .

2.1.1. Working Rule For Solving $Pp + Qq = R$ By Lagrange's Method

Step 1: Put the given linear partial differential equation of the first order in the standard form

$$Pp + Qq = R \quad \dots(1)$$

Step 2: Write down Lagrange's auxiliary equations for (1) namely,

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(2)$$

Step 3: Solve (2) by using the well known methods. Let $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ be two linearly independent solutions of equation (2).

Step 4: The general solution (or integral) of (1) can be written in one of the following three equivalent forms:

$$\phi(u, v) = 0 \text{ or } u = \phi(v) \text{ or } v = \phi(u).$$

2.1.2. Four Types of Problems Based on $Pp + Qq = R$

Type-I

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(1)$$

Suppose that one of the variables is either absent or cancels out from any two fractions of given equation (1). Then an integral can be obtained by usual methods. The same method can be repeated with another set of two fractions of given equation (1).

Put Your Own Note

Example: Solve $\left(\frac{y^2 z}{x}\right)p + xzq = y^2$.

Solution: Given $\left(\frac{y^2 z}{x}\right)p + xzq = y^2$... (1)

The Lagrange's auxiliary equations for (1) are

$$\frac{dx}{\left(\frac{y^2 z}{x}\right)} = \frac{dy}{xz} = \frac{dz}{y^2} \quad \dots (2)$$

Taking the first two fractions of (2), we have

$$x^2 dx = y^2 dy \text{ or } x^2 dx - y^2 dy = 0, \quad \dots (3)$$

$$\text{integrating (3), } (1/3)x^3 - (1/3)y^3 = (1/3)c_1 \text{ or } x^3 - y^3 = c_1 \quad \dots (4)$$

Next, taking the first and the last fractions of (2) we get

$$xy^2 dx = y^2 z dz \text{ or } x dx - z dz = 0 \quad \dots (5)$$

$$\text{Integrating (5), } (1/2)x^2 - (1/2)z^2 = (1/2)c_2 \text{ or } x^2 - z^2 = c_2 \quad \dots (6)$$

From (4) and (6), the required general integral is

$$\phi(x^3 - y^3, x^2 - z^2) = 0, \phi \text{ being an arbitrary function.}$$

Type-II

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots (1)$$

Suppose that one of integral of equation (1) is known by using Rule-1 and also suppose that another integral cannot be obtained by using Rule-1. Then one integral known to us is used to find another integral as shown in following solved example.

Note that in the second integral, the constant of integration of first integral should be removed later.

Example: Solve $p + 3q = 5z + \tan(y - 3x)$

Solution: Given $p + 3q = 5z + \tan(y - 3x)$... (1)

The Lagrange's subsidiary equations of (1) are

$$\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z + \tan(y - 3x)} \quad \dots (2)$$

$$\text{Taking the first two fractions, } dy - 3dx = 0 \quad \dots (3)$$

$$\text{Integrating (3), } y - 3x = c_1 \quad \dots (4)$$

$$\text{Using (4), from (2) we get } \frac{dx}{1} = \frac{dz}{5z + \tan c_1} \quad \dots (5)$$

$$\text{Integrating (5), } x - (1/5)\log(5z + \tan c_1) = (1/5)c_2$$

$$\text{or } 5x - \log[5z + \tan(y - 3x)] = c_2, \text{ using (4)} \quad \dots (6)$$

from (4) and (6) the required general integral is

$$5x - \log[5z + \tan(y - 3x)] = \phi(y - 3x), \text{ where } \phi \text{ is an arbitrary function.}$$

Put Your Own Notes

Type-III

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(1)$$

Let P_1, Q_1 and R_1 be functions of x, y and z . Then by a well known principle of algebra, each fraction in (1) will be equal to

$$\frac{P_1 dx + Q_1 dy + R_1 dz}{P_1 P + Q_1 Q + R_1 R} \quad \dots(2)$$

If $P_1 P + Q_1 Q + R_1 R = 0$, then we know that the numerator of (2) is also zero. This gives $P_1 dx + Q_1 dy + R_1 dz = 0$ which can be integrated to give $u_1(x, y, z) = c_1$. This method may be repeated to get another integral $u_2(x, y, z) = c_2$. P_1, Q_1, R_1 are called multipliers. As a special case, these can be constants also. Sometimes only one integral is possible by use of multipliers. In such cases second integral should be obtained by using Rule-1 or Rule-2.

Example: Solve $\{(b-c)/a\} yz p + \{(c-a)/b\} zx q = \{(a-b)/c\} xy$.

Solution: Given $\{(b-c)/a\} yz p + \{(c-a)/b\} zx q = \{(a-b)/c\} xy$ (1)

The Lagrange's auxiliary equations of (1) are

$$\frac{adx}{(b-c)yz} = \frac{bdy}{(c-a)zx} = \frac{cdz}{(a-b)xy} \quad \dots(2)$$

Choosing x, y, z as multipliers, each fraction of (2)

$$= \frac{axdx + bydy + czdz}{xyz[(b-c) + (c-a) + (a-b)]} = \frac{axdx + bydy + czdz}{0}$$

$$\therefore axdx + bydy + czdz = 0 \quad \text{or} \quad 2axdx + 2bydy + 2czdz = 0$$

$$\text{Integrating, } ax^2 + by^2 + cz^2 = c_1 \quad \dots(3)$$

Again, choosing ax, by, cz as multipliers, each fraction of (2)

$$= \frac{a^2 xdx + b^2 ydy + c^2 zdz}{xyz[a(b-c) + b(c-a) + c(a-b)]} = \frac{a^2 xdx + b^2 ydy + c^2 zdz}{0}$$

$$\text{Integrating, } a^2 x^2 + b^2 y^2 + c^2 z^2 = c_2 \quad \dots(4)$$

From (3) and (4) the required general solution is given by

$$\phi(ax^2 + by^2 + cz^2, a^2 x^2 + b^2 y^2 + c^2 z^2) = 0, \text{ where } \phi \text{ is an arbitrary function.}$$

Type-IV

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad \dots(1)$$

Let P_1, Q_1 and R_1 be functions of x, y and z . Then by a well known principle of algebra, each fraction of (1) will be equal to

$$\frac{P_1 dx + Q_1 dy + R_1 dz}{P_1 P + Q_1 Q + R_1 R} \quad \dots(2)$$

Put Your Own Notes

Suppose the numerator of (2) is exact differential of the denominator of (2). Then (2) can be combined with a suitable fraction in (1) to give an integral. However, in some problems, another set of multipliers P_2, Q_2 and R_2 are so chosen that the fraction

$$\frac{P_2 dx + Q_2 dy + R_2 dz}{P_2 P + Q_2 Q + R_2 R} \quad \dots(3)$$

is such that its numerator is exact differential of denominator. Fractions (2) and (3) are then combined to give an integral. This method may be repeated in some problems to get another integral. Sometimes only integral is possible by using this Rule-4. In such cases second integral should be obtained by using Rule-1 or Rule-2 or Rule-3.

Example: Solve $(y+z)p + (z+x)q = x+y$

Solution: Here the Lagrange's auxiliary equations are

$$\frac{dx}{y+z} = \frac{dy}{z+x} = \frac{dz}{x+y} \quad \dots(1)$$

Choosing 1, -1, 0 as multipliers, each fraction of (1)

$$= \frac{dx - dy}{(y+z) - (z+x)} = \frac{d(x-y)}{-(x-y)} \quad \dots(2)$$

Again, choosing 0, 1, -1 as multipliers, each fraction of (1)

$$= \frac{dy - dz}{(z+x) - (x+y)} = \frac{d(y-z)}{-(y-z)} \quad \dots(3)$$

Finally, choosing 1, 1, 1 as multipliers, each fraction of (1)

$$= \frac{dx + dy + dz}{(y+z) + (z+x) + (x+y)} = \frac{d(x+y+z)}{2(x+y+z)} \quad \dots(4)$$

$$(2), (3) \text{ and } (4) \Rightarrow \frac{d(x-y)}{-(x-y)} = \frac{d(y-z)}{-(y-z)} = \frac{d(x+y+z)}{2(x+y+z)} \quad \dots(5)$$

Taking the first two fractions of (5), we have

$$\frac{d(x-y)}{x-y} = \frac{d(y-z)}{y-z}$$

Integrating, $\log(x-y) = \log(y-z) + \log c_1$

$$\text{or } \log\left\{\frac{(x-y)}{(y-z)}\right\} = \log c_1 \text{ or } (x-y)/(y-z) = c_1 \quad \dots(6)$$

Taking the first and the third fractions of (5), we have

$$2 \frac{d(x-y)}{(x-y)} + \frac{d(x+y+z)}{x+y+z} = 0$$

Integrating, $2\log(x-y) + \log(x+y+z) = \log c_2$

$$\text{Or } (x-y)^2 (x+y+z) = c_2 \quad \dots(7)$$

From (6) and (7), the required general solution is

$$\phi\left\{(x-y)^2 (x+y+z), (x-y)/(y-z)\right\} = 0, \phi \text{ being an arbitrary function.}$$

Put Your Own Notes

2.2 Cauchy Problem For First Order Quasi-Linear PDE or Integral Surfaces Passing Through A Given Curve

In the last section we obtained general integral $Pp + Qq = R$. We shall now present two methods of using such a general solution for getting in the integral surface which passes through a given curve.

$$\text{Let } Pp + Qq = R \quad \dots(1)$$

be the given equation. Let its auxiliary equations give the following two independent solutions.

$$u(x, y, z) = c_1 \text{ and } v(x, y, z) = c_2 \quad \dots(2)$$

Suppose we wish to obtain the integral surface which passes through the curve whose equation in parametric form is given by

$$x = x(t), y = y(t), z = z(t) \quad \dots(3)$$

Where t is a parameter. Then (2) may be expressed as

$$u[x(t), y(t), z(t)] = c_1, v[x(t), y(t), z(t)] = c_2 \quad \dots(4)$$

We eliminate single parameter t from the equation of (4) and get a relation involving c_1 and c_2 .

Finally, we replace c_1 and c_2 with help of (2) and obtain the required integral surface.

Example: Find the integral surface of the linear partial differential equations

$$x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z \text{ which contains the straight line } x + y = 0, z = 1.$$

$$\text{Solution: Given } x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z \quad \dots(1)$$

Lagrange's auxiliary equations of (1) are

$$\frac{dx}{x(y^2 + z)} = \frac{dy}{-y(x^2 + z)} = \frac{dz}{(x^2 - y^2)z} \quad \dots(2)$$

Solving we get

$$xyz = c_1, x^2 + y^2 - 2z = c_2 \quad \dots(3)$$

Taking t as parameter, the given equation of the straight line $x + y = 0, z = 1$ can be put in parametric form.

$$x = t, y = -t, z = 1 \quad \dots(4)$$

Using (4), (3) may be re-written as

$$-t^2 = c_1, 2t^2 - 2 = c_2$$

$$\text{These give } 2(-c_1) - 2 = c_2 \text{ or } 2c_1 + c_2 + 2 = 0. \quad \dots(5)$$

Putting values of c_1 and c_2 from (3) in (5), the desired integral surface is

$$2xyz + x^2 + y^2 - 2z + 2 = 0$$

Put Your Own Notes

2.3 Existence and Uniqueness of Integral Surface Passing Through a Given Curve

Given the Partial differential equation $Pp + Qq = R$ with the initial curve $\Gamma: x_0(t), y_0(t), z_0(t)$... (1)

Unique Solution: If $\frac{P(x_0, y_0, z_0)}{\frac{dx_0}{dt}} \neq \frac{Q(x_0, y_0, z_0)}{\frac{dy_0}{dt}}$

$$\text{or } \Delta = \begin{vmatrix} P(x_0(t), y_0(t), z_0(t)) & Q(x_0(t), y_0(t), z_0(t)) \\ \frac{dx_0}{dt} & \frac{dy_0}{dt} \end{vmatrix} \neq 0$$

then (1) has unique solution

No-Solution: If $\frac{P(x_0, y_0, z_0)}{\frac{dx_0}{dt}} = \frac{Q(x_0, y_0, z_0)}{\frac{dy_0}{dt}} \neq \frac{R(x_0, y_0, z_0)}{\frac{dz_0}{dt}}$ then (1) has no solution

Infinite Solution: If $\frac{P(x_0, y_0, z_0)}{\frac{dx_0}{dt}} = \frac{Q(x_0, y_0, z_0)}{\frac{dy_0}{dt}} = \frac{R(x_0, y_0, z_0)}{\frac{dz_0}{dt}} = \mu$, then (1) has infinite solutions.

Example: Consider the Cauchy's problem $u_x - u_y = 2$ check whether equation has unique solution passing through the curve $(2s, s, 2s)$

Solution: Given $P(x_0(s), y_0(s), z_0(s)) = 1$, $Q(x_0(s), y_0(s), z_0(s)) = -1$ and $R(x_0(s), y_0(s), z_0(s)) = 2$

$$x_0(s) = 2s$$

$$y_0(s) = s$$

$$z_0(s) = 2s$$

$$\text{Now we define } \Delta = \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix} = 1 + 2 = 3 \neq 0 \text{ (unique solution)}$$

Example: $u_x - u_y = u$ passes through $(s, s, \sin s)$ then number of solution of this partial differential equation $0 < s < 1$

Solution: The given partial differential $u_x - u_y = u$... (1)

Compare (1) with $Pp + Qq = R$

$$P = 1, Q = -1, R = u, x_0 = s, y_0 = s, u_0 = \sin s$$

$$\frac{dx_0}{ds} = 1, \frac{dy_0}{ds} = 1, \frac{du_0}{ds} = \cos s$$

$$\frac{P(x_0, y_0, u_0)}{\frac{dx_0}{ds}} = \frac{Q(x_0, y_0, u_0)}{\frac{dy_0}{ds}} \neq \frac{R(x_0, y_0, u_0)}{\frac{du_0}{ds}}$$

$$\frac{1}{1} = \frac{-1}{1} \neq \frac{\sin s}{\cos s}$$

Hence (1) has no solution, equation (1) has zero number of solutions.

Put Your Own Notes

2.4 Surfaces Orthogonal to a Given System of Surfaces

Let $f(x, y, z) = C$... (1)

represent a system of surfaces, where C is parameter. Suppose we wish to obtain a system of surfaces which cut each of (1) at right angles. Then the direction ratios of the normal at the point (x, y, z) to (1) which passes through that point are $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$

Let the surface $z = \phi(x, y)$... (2)

cut each surface of (1) at right angles. Then the normal at (x, y, z) to (2) has direction ratios $\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, -1$ i.e. $p, q, -1$. Since normals at (x, y, z) to (1) and (2) are at right angles, we have

$$p \frac{\partial f}{\partial x} + q \frac{\partial f}{\partial y} - \frac{\partial f}{\partial z} = 0 \quad \text{or} \quad p \frac{\partial f}{\partial x} + q \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} \quad \dots (3)$$

which is of the form $Pp + Qq = R$.

Conversely, we easily verify that any solution of (3) is orthogonal surface of (1).

Example: Find the surface which intersects the surfaces of the system $z(x+y) = c(3z+1)$ orthogonally and which passes through the circle $x^2 + y^2 = 1, z = 1$.

Solution: The given system of surfaces is given by

$$f(x, y, z) = \frac{z(x+y)}{3z+1} = C \quad \dots (1)$$

$$\therefore \frac{\partial f}{\partial x} = \frac{z}{3z+1}, \frac{\partial f}{\partial y} = \frac{z}{3z+1}, \frac{\partial f}{\partial z} = (x+y) \frac{1(3z+1) - z \cdot 3}{(3z+1)^2} = \frac{x+y}{(3z+1)^2}$$

The required orthogonal surface is solution of

$$p \frac{\partial f}{\partial x} + q \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} \quad \text{Or} \quad \frac{z}{3z+1} p + \frac{z}{3z+1} q = \frac{x+y}{(3z+1)^2}$$

$$\text{Or} \quad z(3z+1)p + z(3z+1)q = x+y \quad \dots (2)$$

Lagrange's auxiliary equations for (2) are

$$\frac{dx}{z(3z+1)} = \frac{dy}{z(3z+1)} = \frac{dz}{x+y} \quad \dots (3)$$

Taking the first two fractions of (3), we get

$$dx - dy = 0 \quad \text{So that} \quad x - y = C_1 \quad \dots (4)$$

Choosing $x, y, -z(3z+1)$ as multipliers, each fraction of (3)

$$= \frac{xdx + ydy - z(3z+1)dz}{0}$$

$$\therefore xdx + ydy - 3z^2 dz - zdz = 0$$

Put Your Own Notes

Integrating $(1/2)x^2 + (1/2)y^2 - 3 \times (z^3/3) - (1/2)z^2 = (1/2)C_2$.

$$\text{Or } x^2 + y^2 - 2z^3 - z^2 = C_2 \quad \dots(5)$$

Hence any surface which is orthogonal to (1) has equation of the form

$$x^2 + y^2 - 2z^3 - z^2 = \phi(x-y), \phi \text{ being an arbitrary function} \quad \dots(6)$$

In order to get the desired surface passing through the circle $x^2 + y^2 = 1, z = 1$ we must choose $\phi(x-y) = -2$. Thus, the required particular surface is $x^2 + y^2 - 2z^3 - z^2 = -2$

2.5 The Linear Partial Differential Equation With n Independent Variables

Let $x_1, x_2, \dots, x_n$ be the n independent variables and let $p_1 = \frac{\partial z}{\partial x_1}, p_2 = \frac{\partial z}{\partial x_2}, \dots, p_n = \frac{\partial z}{\partial x_n}$, where z is the dependent variable. Consider the general linear partial differential equation with n independent variables.

$$P_1 p_1 + P_2 p_2 + \dots + P_n p_n = R, \quad \dots(1)$$

Where $P_1, P_2, \dots, P_n$ are function of $x_1, x_2, \dots, x_n$. Let $u_1 = c_1, u_2 = c_2, \dots, u_n = c_n$ be any n independent integrals of the auxiliary equations.

$$\frac{dx_1}{P_1} = \frac{dx_2}{P_2} = \dots = \frac{dx_n}{P_n} \quad \dots(2)$$

$$\text{Then the general solution of (1) is given by } \phi(u_1, u_2, \dots, u_n) = 0 \quad \dots(3)$$

Note that the above procedure is generalization of Lagrange's method.

Example: Solve $x_2 x_3 p_1 + x_3 x_1 p_2 + x_1 x_2 p_3 = 0$

Solution: Re-writing the given equation in standard form

$$x_2 x_3 p_1 + x_3 x_1 p_2 + x_1 x_2 p_3 = -x_1 x_2 x_3. \quad \dots(2)$$

The auxiliary equations of (2) are

$$\frac{dx_1}{x_2 x_3} = \frac{dx_2}{x_3 x_1} = \frac{dx_3}{x_1 x_2} = \frac{dz}{-x_1 x_2 x_3} \quad \dots(3)$$

Taking the first and the fourth fractions of (3), we get

$$x_1 dx_1 + dz = 0 \text{ so that } x_1^2 + 2z = c_1 \quad \dots(4)$$

Taking 1st and 2nd fractions of (3), we get

$$x_1 dx_1 = x_2 dx_2 \text{ so that } x_1^2 - x_2^2 = c_2 \quad \dots(5)$$

Finally, 2nd and 3rd fraction of (3) give

$$x_2 dx_2 = x_3 dx_3 \text{ so that } x_2^2 - x_3^2 = c_3 \quad \dots(6)$$

Hence the required general integral is

$$\phi(x_1^2 + 2z, x_1^2 - x_2^2, x_2^2 - x_3^2) = 0, \phi \text{ being an arbitrary function.}$$

CHAPTER-3

NON-LINEAR PARTIAL DIFFERENTIAL EQUATIONS

3.1. Types of Solutions

A solution or integral of a differential equation is a relation between the variables, by means of which and the derivatives obtained there from the equation is satisfied. Let us now discuss various classes of integrals of a partial differential equation of order one.

3.1.1. Complete Integral (C.I.) or Complete Solution (C. S.)

Let us consider a relation $\phi(x, y, z, a, b) = 0$... (1)

In which x, y, z are variables such that z is dependent on x and y .

Differentiating (1) partially w. r. t x and y

respectively, we obtain

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z} p = 0 \text{ and } \frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z} q = 0, \quad \dots (2)$$

Since there are two arbitrary constants (namely a and b) connected by the above three equations, these can be eliminated and there will appear a relation of the form $f(x, y, z, p, q) = 0$... (3)

Which is a partial differential equation of order one.

Suppose, that (1) has been derived from (3), by using some method; then the integral (1), which has as many arbitrary constants as there are independent variables, is called the complete integral of (3).

Or

A two-parameter family of solutions

$$z = F(x, y, a, b) \quad \dots (2)$$

is called a 'complete integral' of (1) if in the region considered, the rank of

the matrix $M = \begin{pmatrix} F_a & F_{xa} & F_{ya} \\ F_b & F_{xb} & F_{yb} \end{pmatrix}$ is two.

3.1.2. General Integral (G.I.) or General Solution (G.S.)

Assume that in $k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t} ip$, one of the constants is a function of the other, say $b = F(a)$, then (1) becomes $\phi(x, y, z, a, F(a)) = 0$ (5)

Now (5) represents one of the families of surfaces given by the system (1). As before, the equation of the envelope of the family of surfaces given by (5) must also satisfy (3). Again the equation so obtained will be distinct from that of the envelope of the surfaces, and it is not a particular integral. It is known as the general integral and is obtained by eliminating a between

$$\phi(x, y, z, a, F(a)) = 0 \text{ and } \frac{\partial \phi}{\partial a} = 0. \quad \dots (6)$$

Since other relations may appear in the process of getting the singular integral, it is necessary to test that the equation of general integral satisfies the given differential equation.

Particular Integral: A particular integral of (3) is obtained by giving particular values to a and b in (1) which is the complete integral of (3)

3.1.3. Singular Integral (S. I.) or Singular Solution (S.S.)

We know that the locus of all the points whose co-ordinates along with the values of p and q satisfy (3), represent the doubly infinite system of surfaces given by (1). The system is doubly infinite, since there are two constants a and b and each of these can take an infinite number of values. Since the envelope of all the surfaces given by (1) is touched at each of its points by some one of these surfaces, the coordinates of any point on the envelope along with the values of p and q belonging to the envelope at that point must also satisfy (3). Hence we conclude that the equation of the envelope is a solution of (3). The envelope of the surface given by (2) is obtained by eliminating a and b between the equations

$$\phi(x, y, z, a, b) = 0, \quad \frac{\partial \phi}{\partial a} = 0 \text{ and } \frac{\partial \phi}{\partial b} = 0. \quad \dots (4)$$

The relation between x, y and z so obtained is called the singular integral. In general, it is distinct from the complete integral. However, in exceptional cases it may be contained in the complete integral, that is, singular integral may be obtained by giving particular values to the constants in the complete integral. Since other relations may appear in the process of getting the singular integral, it is necessary to test that the equation of singular integral satisfies the given differential equation.

3.2. Method of Getting Singular Integral Directly From The Partial Differential Equation of First Order

Let the given partial differential equation be $f(x, y, z, p, q) = 0$... (1)

whose complete integral is of the form $\phi(x, y, z, a, b) = 0$... (2)

where ' a ' and ' b ' are arbitrary constants.

The singular integral of (1) is obtained by eliminating ' a ' and ' b ' between equation (2)

$$\frac{\partial \phi}{\partial a} = 0 \quad \dots (3)$$

$$\text{and } \frac{\partial \phi}{\partial b} = 0 \quad \dots (4)$$

Put Your Own Notes

The values of z, p, q derived from (2) when substituted in (1) will reduce it into an identity and the substitution of the values of p and q (but not of z) will in general render (1) equivalent to the integral equation. By using this substitution p and q are replaced by functions of x, y, z, a and b in (1). It follows that the singular integral is given by (1) and the equations obtained on differentiating (1) partially w. r. t. ' a ' and ' b ', namely the equations

$$\frac{\partial f}{\partial p} \frac{\partial p}{\partial a} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial a} = 0 \quad \dots(5)$$

$$\frac{\partial f}{\partial p} \frac{\partial p}{\partial b} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial b} = 0 \quad \dots(6)$$

If $\frac{\partial f}{\partial p} \neq 0$ and $\frac{\partial f}{\partial q} \neq 0$, (5) & (6) hold if $\frac{\partial p}{\partial a} \frac{\partial q}{\partial b} - \frac{\partial p}{\partial b} \frac{\partial q}{\partial a} = 0$

showing that there exists a functional relation between p and q which does not contain ' a ' and ' b ' explicitly. Let this functional relation be

$$\psi(p, q) = 0. \quad \dots(7)$$

If both the constants ' a ' and ' b ' occur in p and q (which does not always happen), then (7) shows that one of them is a function of the other and the equations using them give general integral which is not now required.

Equation (5) and (6) are also true if

$$\frac{\partial f}{\partial p} = 0 \quad \dots(8) \quad \text{and} \quad \frac{\partial f}{\partial q} = 0. \quad \dots(9)$$

Elimination of p and q from (1), (7) and (8) will yield a relation between x, y, z free from ' a ' and ' b '. If this relation satisfies the given differential equation (1), it must be the singular integral.

Example: Consider

$$f(x, y, z, p, q) = z - px - qy - p^2 - q^2 = 0 \quad \dots(15)$$

The two-parameter family of planes

$$z = F(x, y, a, b) = ax + by + a^2 + b^2, \quad \dots(16)$$

is a complete integral of (15), since the matrix

$$\begin{pmatrix} x+2a & 1 & 0 \\ y+2b & 0 & 1 \end{pmatrix} \text{ is of rank two and these planes satisfy the p.d.e (15).}$$

Let us now take $b = \sqrt{1-a^2}$. Then

$$z = F(x, y, a, \sqrt{1-a^2}) = ax + \sqrt{1-a^2} y + 1,$$

$$\frac{\partial F}{\partial a} = x - \frac{ay}{\sqrt{1-a^2}} = 0$$

On eliminating a , we get

$$(z-1)^2 = (x^2 + y^2).$$

This is a particular solution of the given partial differential equation.

If we take $b = a$, then $z = ax + ay + 2a^2$

$$\frac{\partial F}{\partial a} = 0 \Rightarrow x + y = -4a.$$

On eliminating a , the envelope is

$$8z = -(x + y)^2.$$

This is another particular solution of the given p.d.e.

Now from equation (16), the conditions $F_a = 0$ and $F_b = 0$ become

$$F_a = (x + 2a) = 0 \quad \dots(17)$$

$$F_b = (y + 2b) = 0, \quad \dots(18)$$

Respectively. On eliminating a and b between equation (16), (17) and (18) we obtain the singular integral

$$4z = -(x^2 + y^2), \quad \dots(19)$$

Which is a paraboloid of revolution.

Note: The singular integral can also be obtained directly by eliminating p and q between (15) and

$$f_p = -x - 2p = 0, \quad \dots(20)$$

$$f_q = -y - 2q = 0. \quad \dots(21)$$

3.3. Compatible System of First-Order Equations

Consider the first order partial differential equations

$$f(x, y, z, p, q) = 0 \quad \dots(1)$$

$$\text{and } g(x, y, z, p, q) = 0 \quad \dots(2)$$

Equation (1) and (2) are known as compatible if they have atleast one common solution.

To find condition for (1) and (2) to be compatible.

$$\text{Let } J = \text{Jacobian of } f \text{ and } g \equiv \frac{\partial(f, g)}{\partial(p, q)} \neq 0. \quad \dots(3)$$

Then (1) and (2) can be solved to obtain the explicit expressions for p and q given by

$$p = \phi(x, y, z) \text{ and } q = \psi(x, y, z). \quad \dots(4)$$

The condition that the pair of equations (1) and (2) should be compatible reduces then to the condition that the system of equations (4) should be completely integrable, i.e., that the equation

$$dz = p dx + q dy \text{ or } \phi dx + \psi dy - dz = 0, \text{ using (4)} \quad \dots(5)$$

should be integrable (5) is integrable if

$$\phi \left(\frac{\partial \psi}{\partial z} - 0 \right) + \psi \left(0 - \frac{\partial \phi}{\partial z} \right) + (-1) \left(\frac{\partial \phi}{\partial y} - \frac{\partial \psi}{\partial x} \right) = 0$$

$$\text{Which is equivalent to } \frac{\partial \psi}{\partial x} + \phi \frac{\partial \psi}{\partial z} = \frac{\partial \phi}{\partial y} + \psi \frac{\partial \phi}{\partial z} \quad \dots(6)$$

Substituting from equations (4) in (1) and differentiating w. r. t. 'x' and 'z' respectively, we get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial p} \frac{\partial \phi}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial \psi}{\partial x} = 0 \quad \dots(7)$$

$$\text{and } \frac{\partial f}{\partial z} + \frac{\partial f}{\partial p} \frac{\partial \phi}{\partial z} + \frac{\partial f}{\partial q} \frac{\partial \psi}{\partial z} = 0 \quad \dots(8)$$

$$\text{From (7) and (8), } \frac{\partial f}{\partial x} + \phi \frac{\partial f}{\partial z} + \frac{\partial f}{\partial p} \left(\frac{\partial \phi}{\partial x} + \phi \frac{\partial \phi}{\partial z} \right) + \frac{\partial f}{\partial q} \left(\frac{\partial \psi}{\partial x} + \phi \frac{\partial \psi}{\partial z} \right) = 0. \quad \dots(9)$$

Similarly (2) yields

$$\frac{\partial g}{\partial x} + \phi \frac{\partial g}{\partial z} + \frac{\partial g}{\partial p} \left(\frac{\partial \phi}{\partial x} + \phi \frac{\partial \phi}{\partial z} \right) + \frac{\partial g}{\partial q} \left(\frac{\partial \psi}{\partial x} + \phi \frac{\partial \psi}{\partial z} \right) = 0. \quad \dots(10)$$

$$\text{Solving (9) and (10), } \frac{\partial \psi}{\partial x} + \phi \frac{\partial \psi}{\partial z} = \frac{I}{J} \left\{ \frac{\partial(f,g)}{\partial(x,p)} + \phi \frac{\partial(f,g)}{\partial(z,p)} \right\} \quad \dots(11)$$

Again, substituting from equations (4) in (1) and differentiating w. r. t. 'y' and 'z' and proceeding as before, we obtain

$$\frac{\partial \phi}{\partial y} + \psi \frac{\partial \phi}{\partial z} = -\frac{I}{J} \left\{ \frac{\partial(f,g)}{\partial(y,q)} + \psi \frac{\partial(f,g)}{\partial(z,q)} \right\} \quad \dots(12)$$

Substituting from equations (11) and (12) in (1) and replacing ϕ, ψ by p, q respectively, we obtain

$$\frac{I}{J} \left\{ \frac{\partial(f,g)}{\partial(x,p)} + p \frac{\partial(f,g)}{\partial(z,p)} \right\} = -\frac{I}{J} \left\{ \frac{\partial(f,g)}{\partial(y,q)} + q \frac{\partial(f,g)}{\partial(z,q)} \right\} \text{ or } [f, g] = 0, \quad \dots(13)$$

$$\text{Where } [f, g] \equiv \frac{\partial(f,g)}{\partial(x,p)} + p \frac{\partial(f,g)}{\partial(z,p)} + \frac{\partial(f,g)}{\partial(y,q)} + q \frac{\partial(f,g)}{\partial(z,q)} \quad \dots(14)$$

A Particular Case

Theorem: To show that first order partial differential equations $p = P(x, y)$

and $q = Q(x, y)$ are compatible if and only if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

$$\text{Proof: Given } \frac{\partial z}{\partial x} = p = P(x, y) \text{ and } \frac{\partial z}{\partial y} = q = Q(x, y) \quad \dots(1)$$

$$\text{Since } dz = \left(\frac{\partial z}{\partial x} \right) dx + \left(\frac{\partial z}{\partial y} \right) dy = p dx + q dy, \quad \dots(2)$$

It follows that the given partial differential equations (1) are compatible if and only if the single differential equation is integrable.

$$dz = Pdx + Qdy \quad \dots(3)$$

Since P and Q are functions of two variables x and y , hence $Pdx + Qdy$ is an exact differential if and only if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$. Therefore (3) is integrable if

$$\text{and only if } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Remark 1. If $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, then the system of two given partial differential equations (1) is compatible and hence these will possess a common solution.

Remark 2. If $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$, then the system of two given partial differential equations (1) is not compatible and hence these equations possess no solution.

Example: Show that the equations $xp = yq$ and $z(xp + yq) = 2xy$ are compatible and solve them

Solution. Let $f(x, y, z, p, q) = xp - yq = 0 \quad \dots(1)$

and $g(x, y, z, p, q) = z(xp + yq) - 2xy = 0 \quad \dots(2)$

$$\therefore \frac{\partial(f, g)}{\partial(x, p)} = \begin{vmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial p} \end{vmatrix} = \begin{vmatrix} p & x \\ zp - 2y & xz \end{vmatrix} = 2xy,$$

$$\frac{\partial(f, g)}{\partial(z, p)} = \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial p} \end{vmatrix} = \begin{vmatrix} 0 & x \\ xp + yq & xz \end{vmatrix} = -x^2p - xyq,$$

$$\frac{\partial(f, g)}{\partial(y, q)} = \begin{vmatrix} \frac{\partial f}{\partial y} & \frac{\partial f}{\partial q} \\ \frac{\partial g}{\partial y} & \frac{\partial g}{\partial q} \end{vmatrix} = \begin{vmatrix} -q & -y \\ zq - 2x & zy \end{vmatrix} = -2xy$$

$$\text{and } \frac{\partial(f, g)}{\partial(z, p)} = \begin{vmatrix} \frac{\partial f}{\partial z} & \frac{\partial f}{\partial p} \\ \frac{\partial g}{\partial z} & \frac{\partial g}{\partial p} \end{vmatrix} = \begin{vmatrix} 0 & -y \\ xp + yq & zy \end{vmatrix} = xyp + y^2q$$

$$\begin{aligned} \therefore [f, g] &= \frac{\partial(f, g)}{\partial(x, p)} + p \frac{\partial(f, g)}{\partial(z, p)} + \frac{\partial(f, g)}{\partial(y, q)} + q \frac{\partial(f, g)}{\partial(z, q)} = 2xy - x^2p^2 - xyqp - 2xy + xypq + y^2q^2 \\ &= -xp(xp + yq) + yq(xp + yq) = -(xp - yq)(xp + yq) = 0, \text{ using (1)} \end{aligned}$$

Hence (1) and (2) are compatible.

Put Your Own Notes

Solving (1) and (2) for p and q , $p = \frac{y}{z}$ and

$$q = \frac{x}{z} \quad \dots(3)$$

Using (3) in $dz = pdx + qdy$, we have $dz = \left(\frac{y}{z}\right)dx + \left(\frac{x}{z}\right)dy$ or $z dz = d(xy)$.

Integrating, $\frac{z^2}{2} = xy + \frac{c}{2}$ or $z^2 = 2xy + c$, where c is an arbitrary constant.

3.4. Charpit's Method (General Method of Solving Partial Differential Equations of Order One But of Any Degree)

Let the given partial equation differential of first order and non-linear in p and q be $f(x, y, z, p, q) = 0$...(1)

We know that $dz = pdx + qdy$(2)

The next step consists in finding another relation $F(x, y, z, p, q) = 0$...(3)

such that when the values of p and q obtained by solving (1) and (3), are substituted in (2), it becomes integrable. The integration of (2) will give the complete integral of (1).

In order to obtain (3), differentiate partially (1) and (3) with respect to x and y and get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = 0, \quad \dots(4)$$

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x} = 0, \quad \dots(5)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q + \frac{\partial f}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = 0, \quad \dots(6)$$

$$\text{and } \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} q + \frac{\partial F}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial y} = 0. \quad \dots(7)$$

Eliminating $\frac{\partial p}{\partial x}$ from (4) and (5), we get

$$\left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x}\right) \frac{\partial F}{\partial p} - \left(\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x}\right) \frac{\partial f}{\partial p} = 0$$

$$\text{or } \left(\frac{\partial f}{\partial x} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial x} \frac{\partial f}{\partial p}\right) + \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial z} \frac{\partial f}{\partial p}\right) p + \left(\frac{\partial f}{\partial q} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial q} \frac{\partial f}{\partial p}\right) \frac{\partial q}{\partial x} = 0. \dots(8)$$

Similarly, eliminating $\frac{\partial q}{\partial y}$ from (6) and (7), we get

$$\left(\frac{\partial f}{\partial y} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial y} \frac{\partial f}{\partial q}\right) + \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial z} \frac{\partial f}{\partial q}\right) q + \left(\frac{\partial f}{\partial p} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial p} \frac{\partial f}{\partial q}\right) \frac{\partial p}{\partial y} = 0 \quad \dots(9)$$

Since $\frac{\partial q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial p}{\partial y}$, the last term in (8) is the same as that in (9), except for a minus sign and hence they cancel on adding (8) and (9).

Therefore, adding (8) and (9) and rearranging the terms, we obtain

$$\left(\frac{\partial}{\partial x} + p\frac{\partial}{\partial z}\right)\frac{\partial F}{\partial p} + \left(\frac{\partial}{\partial y} + q\frac{\partial}{\partial z}\right)\frac{\partial F}{\partial q} + \left(-p\frac{\partial}{\partial p} - q\frac{\partial}{\partial q}\right)\frac{\partial F}{\partial z} + \left(\frac{\partial}{\partial p}\right)\frac{\partial F}{\partial x} + \left(\frac{\partial}{\partial q}\right)\frac{\partial F}{\partial y} = 0 \quad \dots(10)$$

This is a linear equation of the first order to obtain the desired function F . Integral of (10) is obtained by solving the auxiliary equations.

$$\frac{dp}{\left(\frac{\partial}{\partial x} + p\frac{\partial}{\partial z}\right)} = \frac{dq}{\left(\frac{\partial}{\partial y} + q\frac{\partial}{\partial z}\right)} + \frac{dz}{-p\left(\frac{\partial}{\partial p}\right) - q\left(\frac{\partial}{\partial q}\right)} = \frac{dx}{\frac{\partial}{\partial p}} = \frac{dy}{\frac{\partial}{\partial q}} = \frac{dF}{0} \quad \dots(11)$$

Since any of the integrals of (11) will satisfy (10), an integral of (11) which involves p or q (or both) will serve along with the given equation to find p and q . In practice, however, we shall select the simplest integral.

Note: In what follows we shall use the following standard notations:

$$\frac{\partial f}{\partial x} = f_x, \quad \frac{\partial f}{\partial y} = f_y, \quad \frac{\partial f}{\partial z} = f_z, \quad \frac{\partial f}{\partial p} = f_p, \quad \frac{\partial f}{\partial q} = f_q$$

Therefore, Charpit's auxiliary equations (11) may be re-written as

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0} \quad \dots(12)$$

3.4.1. Working Rule While Using Charpit's Method

Step 1. Transfer all terms of the given equation to L.H.S. and denote the entire expression by f .

Step 2. Write down the Charpit's auxiliary equations (11) or (12).

Step 3. Using the value of f in step 1 write down the values of $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \dots, i.e., f_x, f_y, \dots$ etc. occurring in step 2 and put these in Charpit's equations (11) or (12).

Step 4. After simplifying the step 3, select two proper fractions so that the resulting integral may come out to be the simplest relation involving at least one of p and q .

Step 5. The simplest relation of step 4 is solved along with the given equation to determine p and q . Put these values of p and q in $dz = p dx + q dy$ which on integration gives the complete integral of the given equation.

The Singular and General integrals may be obtained in the usual manner.

Remark: Sometimes Charpit's equations give rise to $p = a$ and $q = b$, where a and b are constants. In such cases, putting $p = a$ and $q = b$ in the given equation will give the required complete integral.

Example: Find a complete integral of $px + qy = pq$.

Solution: Here given equation is $f(x, y, z, p, q) = px + qy - pq = 0 \quad \dots(1)$

Charpit's auxiliary equations are

$$\frac{dp}{(\partial f / \partial x) + p(\partial f / \partial z)} = \frac{dq}{(\partial f / \partial y) + q(\partial f / \partial z)} = \frac{dz}{-p(\partial f / \partial p) - q(\partial f / \partial q)}$$

$$= \frac{dx}{-\partial f / \partial p} = \frac{dy}{-\partial f / \partial q}$$

$$\text{or } \frac{dp}{-(x-q)} = \frac{dy}{-(y-q)} = \frac{dz}{-p(x-q)-q(y-p)} = \frac{dp}{p+p.0} = \frac{dq}{q+q.0} \quad \dots(2)$$

Taking the last two fractions of (2), $(1/p)dp = (1/q)dq$.

$$\text{Integrating } \log p = \log q + \log a \quad \text{or } p = aq. \quad \dots(3)$$

Substituting this value of p in (1), we have

$$aqx + qy - aq^2 = 0 \quad \text{or } aq = ax + y, \text{ as } q \neq 0 \quad \dots(4)$$

$$\therefore \text{From (3) and (4), } q = (ax + y)/a \text{ and } p = ax + y \quad \dots(5)$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = (ax + y)dx + [(ax + y)/a]dy \quad \text{or } adz = (ax + y)(adx + dy)$$

$$\text{or } adz = (ax + y)d(ax + y) = u du, \text{ where } u = ax + y$$

$$\text{integrating, } az = (1/2)u^2 + b = (1/2)(ax + y)^2 + b,$$

Which is a complete integral, a and b being arbitrary constants.

Example: Find the complete integral of the partial differential equation $(p^2 + q^2)x = pz$ and reduce the solution which passes through the curve $x = 0, z^2 = 4y$.

$$\text{Solution: Let } f(x, y, q, p, q) = (p^2 + q^2)x - pz = 0. \quad \dots(1)$$

$$\text{Chapit's auxiliary equations are } \frac{dp}{f_x + pf_x} = \frac{dq}{f_y + pf_y} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\text{giving } \frac{dp}{q^2} = \frac{dq}{(-pq)}, \text{ by (1) or } 2pdp + 2qdp = 0.$$

$$\text{Integrating, } p^2 + q^2 = a^2, \text{ where } a \text{ is an arbitrary constant.} \quad \dots(2)$$

$$\text{Solving (1) and (2), } p = \frac{a^2 x}{z} \text{ and } q = \left(\frac{a}{z}\right) \times \sqrt{(z^2 - a^2 x^2)}. \quad \dots(3)$$

$$\therefore dz = p dx + q dy = \frac{a^2 x dx}{z} + \frac{a \sqrt{(z^2 - a^2 x^2)} dy}{z} \quad \text{or } \frac{z dz - a^2 x dx}{\sqrt{(z^2 - a^2 x^2)}} = a dy.$$

Putting $z^2 - a^2 x^2 = t$ so that $2(z dz - a^2 x dx) = dt$, we get

$$\left(\frac{1}{2\sqrt{t}}\right) dt = a dy \quad \text{or } \left(\frac{1}{2}\right) \times t^{-1/2} dt = a dy$$

$$\text{Integrating, } t^{1/2} = ay + b \quad \text{or } \sqrt{(z^2 - a^2 x^2)} = ay + b, \text{ as } t = z^2 - a^2 x^2$$

$$\text{Or } z^2 - a^2 x^2 = (ay + b)^2 \quad \text{or } z^2 = a^2 x^2 + (ay + b)^2 \quad \dots(4)$$

Which is complete integral.

The parametric equations of the given curve $x=0, z^2=4y$ are given by

$$x=0, y=t^2, z=2t \quad \dots(5)$$

Therefore the intersections of (1) and (2) are determined by

$$4t^2 = (at^2 + b)^2 \text{ or } a^2t^4 + 2(ab-2)t^2 + b^2 = 0 \quad \dots(6)$$

Equation (6) has equal roots if its discriminant = 0, i.e., if

$$4(ab-2)^2 - 4a^2b^2 = 0 \text{ or } a^2b^2 = 1 \text{ so that } b = \frac{1}{a}$$

Hence from (4), the appropriate one parameter sub-system is given by

$$z^2 = a^2x^2 + \left(ay + \frac{1}{a}\right)^2 \text{ or } a^4(x^2 + y^2) + a^2(2y - z^2) + 1 = 0,$$

which is a quadratic equation in parameter 'a'. Therefore, this has form its envelope surface

$$(2y - z^2)^2 - 4(x^2 + y^2) = 0 \text{ or } (2y - z^2)^2 = 4(x^2 + y^2) \quad \dots(7)$$

The desired solution is given by the function z defined by equation (7).

Example: Find a complete, singular and general integrals of $(p^2 + q^2)y = qz$.

Solution: Here the given equation is

$$f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0. \quad \dots(1)$$

Charpit's auxiliary equations are

$$\frac{dp}{f_x + pf_x} = \frac{dq}{f_y + pf_y} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

$$\text{or } \frac{dp}{-pq} = \frac{dq}{p^2} = \frac{dz}{-2p^2y + qz - 2q^2y} = \frac{dx}{-2py} = \frac{dy}{-2qy + z} \quad \dots(2)$$

Taking the first two fractions, we get $2pdp + 2qdp = 0$ so that

$$p^2 + q^2 = a^2 \quad \dots(3)$$

Using (3), (1) gives $a^2y = qz$ or $q = \frac{a^2y}{z}$.

Putting this value of q is (3), we get

$$p = \sqrt{(a^2 - q^2)} = \sqrt{a^2 - \left(\frac{a^4y^2}{z^2}\right)} = \frac{a}{z} \sqrt{(z^2 - a^2y^2)}.$$

Now putting these values of p and q in $dz = pdx + qdy$, we have

$$dz = \frac{a}{z} \sqrt{(z^2 - a^2y^2)} dx + \frac{a^2y dy}{z} \text{ or } \frac{zdz - a^2y dy}{\sqrt{(z^2 - a^2y^2)}} = a dx.$$

$$\text{Integrating, } (z^2 - a^2y^2)^{\frac{1}{2}} = ax + b \text{ or } z^2 - a^2y^2 = (ax + b)^2, \quad \dots(4)$$

Which is a required complete integral, a, b being arbitrary constants.

Singular Integral . Differentiating (4) partially w. r. t. a and b , we have

$$0 = 2ay^2 + 2(ax+b)x \dots (5)$$

$$\text{and } 0 = 2(ax+b) \dots (6)$$

Eliminating a and b between (4), (5) and (6), we get $z=0$ which clearly satisfies (1) and hence it is the singular integral.

General Integral . Replacing b by $\phi(a)$ in (4), we get

$$z^2 - a^2y^2 = [ax + \phi(a)]^2 \dots (7)$$

Differentiating (7) partially w.r.t.

$$a, -2ay^2 = 2[ax + \phi(a)][x + \phi'(a)] \dots (8)$$

General integral is obtained by eliminating a from (7) and (8).

Example: Find a complete and singular integrals of $2xz - px^2 - 2qxy + pq = 0$.

Solution: Here given equation is

$$f(x, y, z, p, q) = 2xz - px^2 - 2qxy + pq = 0 \dots (1)$$

Charpit's auxiliary equations are

$$\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{\frac{\partial f}{\partial p}} = \frac{dy}{\frac{\partial f}{\partial q}}$$

$$\text{or } \frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 + 2xyq - 2pq}, \text{ by (1)}$$

The second fraction gives $dq=0$ so that $q=a$

$$\text{Putting } q=a \text{ in (1), we get } p = \frac{2x(z-ay)}{(x^2-a)}$$

Putting values p and q in $dz = p dx + q dy$, we get

$$dz = \frac{2x(z-ay)}{x^2-a} dx + a dy \quad \text{or} \quad \frac{dz - a dy}{z - ay} = \frac{2x dx}{x^2 - a}$$

$$\text{Integrating, } \log(z - ay) = \log(x^2 - a) + \log b$$

$$\text{or } z - ay = b(x^2 - a) \quad \text{or} \quad z = ay + b(x^2 - a), \dots (2)$$

which is the complete integral, a and b being arbitrary constants.

Differentiating (2) partially with respect to a and b , we get

$$0 = y - b \quad \text{and} \quad 0 = x^2 - a \dots (3)$$

$$\text{Solving (3) for } a \text{ and } b, a = x^2 \text{ and } b = y. \dots (4)$$

Substituting the values of a and b , given by (4) in (2), we get $z = x^2y$, which is the required singular integral.

Example: Find the complete integral, general integral and the singular integral of $2(z + xp + yq) = yp^2$

Solution: Given equation is $f(x, y, z, p, q) = 2(z + px + qy) - yp^2 = 0 \dots (1)$

Charpit's auxiliary equation are

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + qf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dp}{-f_p} = \frac{dp}{-f_q}$$

$$\text{or } \frac{dp}{2p + 2p} = \frac{dq}{2q - p^2 + 2q} = \frac{dz}{-p(2x - 2yp) - q \times 2y} = \frac{dx}{-(2x - 2yp)} = \frac{dy}{-2y},$$

by (1)

$$\text{Taking the first and the last fractions, } \frac{dp}{4p} = \frac{dy}{-2y} \text{ or } \frac{dp}{p} = \frac{-2dy}{y}$$

$$\text{Integrating, } \log p + 2 \log y = \log a \text{ or } py^2 = a. \dots (2)$$

$$\text{Solving (1) and (2) for } p \text{ and } q, p = \frac{a}{y^2} \text{ and } q = -\frac{z}{y} - \frac{ax}{y^3} + \frac{a^2}{2y^4}.$$

$$\therefore dz = p dx + q dy = \frac{a}{y^2} dx + \left[-\frac{z}{y} - \frac{ax}{y^3} + \frac{a^2}{2y^4} \right] dy$$

Multiplying both sides by y and re-arranging, we get

$$(y dz + z dy) - a \left(\frac{y dx - x dy}{y^2} \right) - \frac{a^2}{2y^3} dy = 0$$

$$\text{or } d(yz) - ad \left(\frac{x}{y} \right) - \frac{a^2}{2} y^{-3} dy = 0.$$

$$\text{Integrating, } yz - a \left(\frac{x}{y} \right) + \left(\frac{a^2}{4y^2} \right) = b, \dots (3)$$

where a, b being arbitrary constants.

General integral: Replacing b by $\phi(a)$ in (3), we get

$$yz - a \left(\frac{x}{y} \right) + \left(\frac{a^2}{4y^2} \right) = \phi(a) \dots (4)$$

$$\text{Differentiating (4) partially w. r. t. 'a', } -\left(\frac{x}{y} \right) + \left(\frac{a}{2y^2} \right) = \phi'(a) \dots (5)$$

Then the general integral is obtained by eliminating a from (4) and (5)

Singular integral. Differentiating (3) partially w. r. t. 'a' and 'b', we get

$$-\left(\frac{x}{y} \right) + \left(\frac{a}{2y^2} \right) = 0 \dots (6)$$

$$0 = 1 \dots (7)$$

Relation (7) is absurd and hence there is no singular solution of the given equation.

3.4.2. Special Methods of Solutions Applicable To Certain Standard Forms:

We now consider equations in which p and q occur other than in the first degree, that is non-linear equations. We have already discussed the Charpit's method. We now discuss four standard forms to which many equations can be reduced, and for which a complete integral can be obtained by inspection or by other shorter methods.

Standard Form I: Only p and q present

Under this standard form, we consider equations of the form

$$f(p, q) = 0 \quad \dots(1)$$

Charpit's auxiliary equations are

$$\frac{dp}{f_x + pf_z} = \frac{dq}{f_y + pf_z} = \frac{dz}{-pf_p - qf_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0}$$

$$\text{giving } \frac{dp}{0} = \frac{dq}{0}, \text{ by (1)}$$

$$\text{Taking the first ratio, } dp = 0 \text{ so that } p = \text{constant} = a, \text{ say} \quad \dots(2)$$

$$\text{Substituting in (1), we get } f(a, q) = 0, \text{ giving } q = \text{constant} = b, \text{ say,} \quad \dots(3)$$

$$\text{Where } b \text{ is such that } f(a, b) = 0, \quad \dots(4)$$

$$\text{Then, } dz = p dx + q dy = a dx + b dy, \text{ using (2) and (3)}$$

$$\text{Integrating, } z = ax + by + c \quad \dots(5)$$

Where c is an arbitrary constant, (5) together with (4) give the required solution.

Now solving (4) for b , suppose we obtain $b = F(a)$, say.

Putting this value of b in (5), the complete integral of (1) is

$$z = ax + yF(a) + c, \quad \dots(6)$$

Which contains two arbitrary constants a and c which are equal to the number of independent variables, namely x and y .

The singular integral of (1) is obtained by eliminating a and c between the complete integral (6) and the equations obtained by differentiating (6) partially w. r. t. a and c ; i.e., between

$$z = ax + yF(a) + c, \quad 0 = x + yF'(a) \text{ and } 0 = 1. \quad \dots(7)$$

Since the last equation in (7) is meaningless, we conclude that the equations of standard form I have no singular solution.

In order to find the general integral of (1), we first take $c = \phi(a)$ in (6), ϕ being an arbitrary function and obtain $z = ax + yF(a) + \phi(a)$. $\dots(8)$

Now, we differentiate (8) partially with respect to a and get

$$0 = x + yF'(a) + \phi'(a). \quad \dots(9)$$

Eliminating a between (8) and (9), we get the general solution of (1).

Remark: Sometimes change of variables can be employed to transform a given equation to standard form I.

SOLVED EXAMPLE

Example: Find a complete integral of the equation $\sqrt{p} + \sqrt{q} = 1$.

Solution: The given PDE is of the form $f(p, q) = 0$. Therefore, let us assume the solution in the form $z = ax + by + c$

Where $\sqrt{a} + \sqrt{b} = 1$ or $b = (1 - \sqrt{a})^2$

Hence, the complete integral is found to be $z = ax + (1 - \sqrt{a})^2 y + c$.

Example: Solve $pq = k$, where k is a constant.

Solution: Given that $pq = k$ (1)

Since (1) is of the form $f(p, q) = 0$, its solution is $z = ax + by + c$, ... (2)

Where $ab = k$ or $b = \frac{k}{a}$, on putting a for p and b for q in (1)

$\therefore$ From (2), the complete integral is $z = ax + \left(\frac{k}{a}\right)y + c$, ... (3)

Which contains two arbitrary constants a and c .

For singular solution, differentiating (3) partially with respect to a and c , we get $0 = x - \left(\frac{k}{a^2}\right)y$ and $0 = 1$. But $0 = 1$ is absurd. Hence there is no singular solution of (1).

To find the general solution, put $c = \phi(a)$ in (3). Then, we get

$z = ax + \left(\frac{k}{a}\right)y + \phi(a)$ (4)

Differentiating (4) partially with respect to ' a ', we get

$0 = x - \left(\frac{k}{a^2}\right)y + \phi'(a)$ (5)

Eliminating a from (4) and (5), we get the required general solution.

Example: Solve $p^2 + q^2 = m^2$, where m is a constant.

Solution: Given that $p^2 + q^2 = m^2$, ... (1)

Since (1) is of the form $f(p, q) = 0$, its solution is

$z = ax + by + c$, ... (2)

Where $p^2 + q^2 = m^2$, or $b = (m^2 - a^2)^{\frac{1}{2}}$, on putting a for p and b for q in (1).

$\therefore$ Form (2), the complete integral is $z = ax + y(m^2 - a^2)^{\frac{1}{2}} + c$, ... (3)

Which contains two arbitrary constants a and c .

For singular solution, differentiating (3) partially with respect to a and c , we get $0 = x - \frac{ay}{(m^2 - a^2)^{\frac{1}{2}}}$ and $0 = 1$. But $0 = 1$ is absurd. Hence there is no singular solution of (1).

To find the general solution, put $c = \phi(a)$ in (3). Then, we get

$$z = ax + y(m^2 - a^2)^{\frac{1}{2}} + \phi(a). \quad \dots(4)$$

Differentiating (4) partially with respect to 'a', we get

$$0 = a - \frac{ay}{(m^2 - a^2)^{\frac{1}{2}}} + \phi'(a). \quad \dots(5)$$

Eliminating a from (4) and (5), we get the required general solution.

Standard Form II : Clairaut Equation.

A first order partial differential equation is said to be of Clairaut form if it can be written in the form

$$z = px + qy + f(p, q). \quad \dots(1)$$

$$\text{Let } F(x, y, z, p, q) \equiv px + qy + f(p, q) - z \quad \dots(2)$$

Charpit's auxiliary equations are

$$\frac{dp}{\left(\frac{\partial F}{\partial x}\right) + p\left(\frac{\partial F}{\partial p}\right)} = \frac{dq}{\left(\frac{\partial F}{\partial y}\right) + q\left(\frac{\partial F}{\partial q}\right)} = \frac{dz}{-p\left(\frac{\partial F}{\partial p}\right) - q\left(\frac{\partial F}{\partial q}\right)} = \frac{dx}{-\frac{\partial F}{\partial p}} = \frac{dy}{-\frac{\partial F}{\partial q}}$$

or $\frac{dp}{0} = \frac{dq}{0} = \frac{dz}{-px - qy - p\left(\frac{\partial f}{\partial p}\right) - q\left(\frac{\partial f}{\partial q}\right)} = \frac{dx}{-x - \left(\frac{\partial f}{\partial p}\right)} = \frac{dy}{-y - \left(\frac{\partial f}{\partial q}\right)}, \text{ by (1)}$

Then, first and second fractions $\Rightarrow dp = 0$ and $dq = 0 \Rightarrow p = a$ and $q = b$.

Substituting these values in (1), the complete integral is

$$z = ax + by + f(a, b)$$

Remark 1: Observe that the complete integral of (1) is obtained by merely replacing p and q by a and b respectively. Singular and general integrals can be obtained by usual methods.

Remark 2: Sometimes change of variables can be employed to transform a given equation to standard form II.

Example: Find the complete integral of the equation

$$z = px + qy + \sqrt{1 + p^2 + q^2}$$

Solution: The given PDE is in the Clairaut's form Hence, its complete integral is

$$z = ax + by + \sqrt{1 + a^2 + b^2}.$$

Example: Find the complete and singular integrals of the $z = px + qy + \log(pq)$

Solution: The complete integral is $z = ax + by + \log(ab)$

or $z = ax + by + \log a + \log b$, a, b being arbitrary constants ... (1)

Differentiating (1) partially with respect to a and b , we get

$$0 = x + \left(\frac{1}{a}\right) \text{ and } 0 = y + \left(\frac{1}{b}\right) \text{ so that } a = -\frac{1}{x} \text{ and } b = -\frac{1}{y} \quad \dots (2)$$

Eliminating a and b from (1) and (2), the required singular integral is

$$z = -1 - 1 + \log\left(\frac{1}{xy}\right) \text{ or } z = -2 - \log(xy).$$

Example: Find the complete and singular integrals of the $z = px + qy - 2\sqrt{pq}$.

Solution: The complete integral is $z = ax + by - 2\sqrt{ab}$ (1)

Differentiating (1) partially with respect to a and b , we get

$$0 = x - \frac{2b}{2\sqrt{ab}} \text{ and } 0 = y - \left(\frac{2a}{2\sqrt{ab}}\right) \text{ so that } x = \sqrt{\frac{b}{a}} \text{ and } y = \sqrt{\frac{a}{b}}. \dots (2)$$

Now, using (1) $x - z = x - (ax + by - 2\sqrt{ab}) = \sqrt{\frac{b}{a}} - a\sqrt{\frac{b}{a}} - b\sqrt{\frac{a}{b}} + 2\sqrt{ab}$,
using (2)

$$\therefore x - z = \sqrt{\left(\frac{b}{a}\right)}. \quad \dots (3)$$

Similarly, using (1)

$$y - z = y - (ax + by - 2\sqrt{ab}) = \sqrt{\frac{a}{b}} - a\sqrt{\frac{b}{a}} - b\sqrt{\frac{a}{b}} + 2\sqrt{ab}$$

$$\therefore y - z = \sqrt{\left(\frac{a}{b}\right)}. \quad \dots (4)$$

From (3) and (4),

Which is singular integral as it satisfies the given equation.

Standard Form III: Only p, q and z are Present.

Under this standard form we consider differential equation of the form

$$f(p, q, z) = 0 \quad \dots (1)$$

Charpit's auxiliary equations are

$$\frac{dp}{\left(\frac{\partial f}{\partial x}\right) + p\left(\frac{\partial f}{\partial z}\right)} = \frac{dq}{\left(\frac{\partial f}{\partial y}\right) + q\left(\frac{\partial f}{\partial z}\right)} = \frac{dz}{-p\left(\frac{\partial f}{\partial p}\right) - q\left(\frac{\partial f}{\partial q}\right)} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$$

$$\text{or } \frac{dp}{p\left(\frac{\partial f}{\partial z}\right)} = \frac{dq}{q\left(\frac{\partial f}{\partial z}\right)} = \frac{dz}{-p\left(\frac{\partial f}{\partial p}\right) - q\left(\frac{\partial f}{\partial q}\right)} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}, \text{ using (1)}$$

Taking the first two ratios, $\left(\frac{1}{p}\right)dp = \left(\frac{1}{q}\right)dq$

Integrating, $q = ap$, a being an arbitrary constant. ... (2)

Now, $dz = p dx + q dy = p dx + ap dy$, using (2)

or $dz = p(dx + a dy) = p d(x + ay) = p du$, ... (3)

Where $u = x + ay$ (4)

Now, (3) $\Rightarrow p = \frac{dz}{du}$ and so by (2) $q = ap = a\left(\frac{dz}{du}\right)$ (5)

Which is an ordinary differential equation of first order. Solving (5), we get z as a function of u . Complete integral is then obtained by replacing u by $(x + ay)$.

Working Rule For Solving Equations of The Form

$f(p, q, z) = 0$ (1)

Step 1. Let $u = x + ay$, where a is an arbitrary constant. ... (2)

Step II. Replace p and q by $\frac{dz}{du}$ and $a\left(\frac{dz}{du}\right)$ respectively in (1) and solve the resulting ordinary differential equation of first order by usual methods.

Step III. Replace u by $x + ay$ in the solution obtained in step II.

Remark 1: Sometimes change of variables can be employed to reduce a given equation in the standard form III.

Remark 2: Singular and general integrals are obtained by well known methods.

SOLVED EXAMPLE

Example: Find the complete integral of $p(1 + q) = qz$.

Solution: Let us assume the solution in the form

$u = x + ay$, where a is an arbitrary constant.

Then, $p = \frac{dz}{du}$, $q = a \frac{dz}{du}$

Substituting these values in the given PDE, we get

$$\frac{dz}{du} \left(1 + a \frac{dz}{du}\right) = az \frac{dz}{du}$$

That is, $a \frac{dz}{du} = az - 1$ or $a \frac{dz}{az - 1} = du$

On integration, we find $\ln(az - 1) = u + c = x + ay + c$

Which is the required complete integral.

Example: Find a complete integral of $9(p^2z + q^2) = 4$.

Solution: Given equation is $9(p^2z + q^2) = 4$, ... (1)

which is of the form $f(p, q, z) = 0$. Let $u = x + ay$, where a is an arbitrary constant. Now, replacing p and q by $\frac{dz}{du}$ and $a\left(\frac{dz}{du}\right)$ respectively in (1), we get

$$9\left[z\left(\frac{dz}{du}\right)^2 + a^2\left(\frac{dz}{du}\right)^2\right] = 4 \text{ or } \left(\frac{dz}{du}\right)^2 = \frac{4}{9(z + a^2)}$$

or $du = \pm\left(\frac{3}{2}\right) \times (z + a^2)^{\frac{1}{2}} dz$, separating variables u and z .

$$\text{Integrating, } u + b = \pm\left(\frac{3}{2}\right) \times \frac{(z + a^2)^{\frac{3}{2}}}{\frac{3}{2}} \text{ or } u + b = \pm(z + a^2)^{\frac{3}{2}}$$

or $(u + b)^2 = (z + a^2)^3$ or $(x + ay + b)^2 = (z + a^2)^3$, as $u = x + ay$

which is a complete integral containing two arbitrary constants a and b .

Example: Find complete and singular integrals of $4(1 + z^3) = 9z^4 pq$.

Solution: The given equation is of the form $f(p, q, z) = 0$. Let $u = x + ay$, a being an arbitrary constant. Replacing p by $\frac{dz}{du}$ and q by $a\left(\frac{dz}{du}\right)$ in the given equation, we have

$$4(1 + z^3) = 9z^4 a\left(\frac{dz}{du}\right)^2 \text{ or } \pm \frac{3\sqrt{a}z^2}{(1 + z^3)^{\frac{1}{2}}} dz = 2 du$$

$$\text{or } \pm\left(\frac{\sqrt{a}}{t}\right) \times 2t dt = 2 du, \text{ putting } 1 + z^3 = t^2 \text{ so that } 3z^2 dz = 2t dt$$

$$\text{Integrating, } \pm\sqrt{at} = u + b \text{ or } \pm\sqrt{a}(1 + z^3)^{\frac{1}{2}} = x + ay + b$$

$$\text{or } a(1 + z^3) = (x + ay + b)^2, \quad \dots(1)$$

which is a complete integral containing two arbitrary constants a and b .

Singular Integral: Differentiating (1) partially w. r. t. a and b by turn, we get

$$1 + z^3 = 2y(x + ay + b) \quad \dots(2)$$

$$\text{and } 0 = 2(x + ay + b). \quad \dots(3)$$

Eliminating a and b from (1), (2) and (3), the singular integral is

$$1 + z^3 = 0. \quad \dots(4)$$

From (4), $p = \frac{\partial z}{\partial x} = 0$ and $q = \frac{\partial z}{\partial y} = 0$. Thus these values of p and q

together with $1 + z^3 = 0$ satisfy the given equation. Hence $1 + z^3 = 0$ is the required singular integral.

Standard form IV: Equation of The Form $f_1(x, p) = f_2(y, q)$. i.e. a form in which z does not appear and the terms containing x and p are on one side and those containing y and q on the other side.

Let $F(x, y, z, p, q) = f_1(x, p) - f_2(y, q) = 0$ (1)

Then Charpit's auxiliary equations are

$$\frac{dp}{\left(\frac{\partial F}{\partial x}\right) + p\left(\frac{\partial F}{\partial z}\right)} = \frac{dq}{\left(\frac{\partial F}{\partial y}\right) + q\left(\frac{\partial F}{\partial z}\right)} = \frac{dz}{-p\left(\frac{\partial F}{\partial p}\right) - q\left(\frac{\partial F}{\partial q}\right)} = \frac{dx}{-\frac{\partial F}{\partial p}} = \frac{dy}{-\frac{\partial F}{\partial q}}$$

or $\frac{dp}{\frac{\partial f_1}{\partial x}} = \frac{dq}{-\frac{\partial f_2}{\partial y}} = \frac{dz}{-p\left(\frac{\partial f_1}{\partial p}\right) - q\left(\frac{\partial f_2}{\partial q}\right)} = \frac{dx}{-\frac{\partial f_1}{\partial p}} = \frac{dy}{-\frac{\partial f_2}{\partial q}}$, by (1)

Taking the first and the fourth ratios, we have

$$\left(\frac{\partial f_1}{\partial p}\right) dp + \left(\frac{\partial f_1}{\partial x}\right) dx = 0 \text{ or } df_1 = 0.$$

Integrating, $f_1 = a$, a being an arbitrary constant.

$\therefore (1) \Rightarrow f_1(x, p) = f_2(y, q) = a$ (2)

Now, (2) $\Rightarrow f_1(x, p) = a$ and $f_2(y, q) = a$... (3)

Form (3), on solving for p and q respectively, we get

$p = F_1(x, a)$, say and $q = F_2(y, a)$, say ... (4)

Substituting these values in $dz = p dx + q dy$, we get

$$dz = F_1(x, a) dx + F_2(y, a) dy.$$

Integrating, $z = \int F_1(x, a) dx + \int F_2(y, a) dy + b$,

Which is a complete integral containing two arbitrary constants a and b

Remark.1: Sometimes change of variables can be employed to reduce a given equation in the standard form IV.

Remark.2: Singular and general integral are obtained by well known methods.

SOLVED EXAMPLE

Example: Find the complete integral of the PDE: $p^2 y(1+x^2) = qx^2$

Solution: The given PDE is of separable type and can be rewritten as

$$\frac{p^2(1+x^2)}{x^2} = \frac{q}{y} = a \text{ (say), where } a \text{ is an arbitrary constant.}$$

Then,

$$p = \frac{\sqrt{ax}}{\sqrt{1+x^2}}, \quad q = ay$$

Substituting these values of p and q in

$$dz = p dx + q dy,$$

We get $dz = \frac{\sqrt{ax}}{\sqrt{1+x^2}} dx + ay dy$

On integration, we obtain $z = \sqrt{a}\sqrt{1+x^2} + \frac{a}{2}y^2 + b$

which is the complete integral of the given PDE.

Example: Find a complete integral of $x(1+y)p = y(1+x)q$.

Solution: Separating p and x from q and y , the given equation reduces to

$$\frac{(xp)}{(1+x)} = \frac{(yq)}{(1+y)}$$

Equating each side to an arbitrary constant a , we have

$$\frac{xp}{1+x} = a \text{ and } \frac{yq}{1+y} = a \text{ so that } p = a\left(\frac{1+x}{x}\right) \text{ and } q = a\left(\frac{1+y}{y}\right).$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{a(1+x)}{x} dx + \frac{a(1+y)}{y} dy \text{ or } dz = a\left(\frac{1}{x} + 1\right) dx + a\left(\frac{1}{y} + 1\right) dy.$$

Integrating, $z = a(\log x + x) + a(\log y + y) + b = a(\log xy + x + y) + b$,

which is a complete integral containing two arbitrary constants a and b .

3.5. Curves & Surfaces in Three Dimensions

A point in 3-dimensions connected by relation $f(x, y, z) = 0$... (1)

Lies on surface $\Rightarrow$ equation (1) is an equation of surface S

* If a point with coordinates (x, y, z) lies on two surfaces s_1 & s_2 given by $f(x, y, z) = 0$ & $g(x, y, z) = 0$... (2)

Then the surfaces s_1 & s_2 intersect in curve C .

Therefore locus of point, whose coordinates satisfy a pair of relations of type (2) is a curve in space.

* Any three equations of the form $x = f_1(t)$, $y = f_2(t)$ & $z = f_3(t)$, ... (3)

in which t is a continuous variable may be regarded as parameter equation of a curve.

* If we take length of the curve as measured from some fixed point as parameter and P any point on

the curve $x = x(s)$, $y = y(s)$, $z = z(s)$... (4)

We can find that direction cosines of the tangent to the curves (4) at point P are given as

Put Your Own Notes

$$\left(\frac{dx}{ds}, \frac{dy}{ds}, \frac{dz}{ds} \right) \quad \dots(5)$$

* Assume that curve (4) lies on the surface s with equation $F(x, y, z) = 0$

Therefore point $P(x(s), y(s), z(s))$ lies on surface s

$$\text{i.e. } F(x(s), y(s), z(s)) = 0 \quad \dots(6)$$

$$\Rightarrow \frac{\partial F}{\partial x} \frac{dx}{ds} + \frac{\partial F}{\partial y} \frac{dy}{ds} + \frac{\partial F}{\partial z} \frac{dz}{ds} = 0 \quad \dots(7)$$

From equation (5) & (7) it is clear that tangent to curve C at P is perpendicular to the line whose direction ratios are

$$\left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) \quad \dots(8)$$

Obviously these are d.r's of normal to the surface at P

Example: Surface $S: z = f(x, y)$

$$\Rightarrow F(x, y, z) = f(x, y) - z = 0$$

$$\Rightarrow F_x = p, F_y = q, F_z = -1$$

$\Rightarrow$ Hence d.c's of normal to surface at (x, y, z) are

$$\left(\frac{p}{\sqrt{p^2 + q^2 + 1}}, \frac{q}{\sqrt{p^2 + q^2 + 1}}, \frac{-1}{\sqrt{p^2 + q^2 + 1}} \right)$$

* Using (5) & (7) we can write the equation of tangent plane at point $P(x, y, z)$ to the surface S_1 with equation $F(x, y, z) = 0$ is

$$(X-x) \frac{\partial F}{\partial x} + (Y-y) \frac{\partial F}{\partial y} + (Z-z) \frac{\partial F}{\partial z} = 0 \quad \dots(9)$$

Where x, y, z are any other point on the tangent plane. Similarly equation of tangent plane to the surface S_2 with equation $G(x, y, z) = 0$ at the point $P(x, y, z)$ is

$$(X-x) \frac{\partial G}{\partial x} + (Y-y) \frac{\partial G}{\partial y} + (Z-z) \frac{\partial G}{\partial z} = 0 \quad \dots(10)$$

* The intersection of tangent planes given by (9) & (10) is the tangent at P to the curve C which is intersection of surface S_1 & S_2 . From (9) & (10) equation of tangent line is

$$\frac{X-x}{\frac{\partial F}{\partial y} \frac{\partial G}{\partial z} - \frac{\partial F}{\partial z} \frac{\partial G}{\partial y}} = \frac{Y-y}{\frac{\partial F}{\partial z} \frac{\partial G}{\partial x} - \frac{\partial F}{\partial x} \frac{\partial G}{\partial z}} = \frac{Z-z}{\frac{\partial F}{\partial x} \frac{\partial G}{\partial y} - \frac{\partial F}{\partial y} \frac{\partial G}{\partial x}}$$

$\Rightarrow$ direction ratios of tangent line are

$$\left\{ \frac{\partial(F, G)}{\partial(y, z)}, \frac{\partial(F, G)}{\partial(z, x)}, \frac{\partial(F, G)}{\partial(x, y)} \right\}$$

Put Your Own Notes

3.6. Cauchy's Method of Characteristics for Solving Non-Linear

P.D.E.

Let non- linear P.D.E is: $F\left(x, y, z, \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}\right) = 0$... (1)

The plane passing through point $P(x_0, y_0, z_0)$ with its normal parallel to the direction n defined by direction ratios $(p_0, q_0, -1)$ is uniquely specified by set of numbers $D(x_0, y_0, z_0, p_0, q_0) \rightarrow$ called plane element of space). Conversely any such set of 5 real number defines a plane in 3-D. In particular a plane element $(x_0, y_0, z_0, p_0, q_0)$ whose components satisfy on equation

$$F(x, y, z, p, q) = 0 \quad \dots (2)$$

is called an integral element of equation (2) at (x_0, y_0, z_0) .

It is possible to solve (2) to obtain an expression

$$q = G(x, y, z, p) \quad \dots (3)$$

Keeping x_0, y_0 & z_0 fixed and varying p , we obtain a set of plane elements $\{x_0, y_0, z_0, p, G(x_0, y_0, z_0, p)\}$ which depend on single parameter p . As p varies, we obtain a set of plane elements all of which pass through point P and which therefore envelope a cone with vertex P , the cone so generated is called elementary cone of equation (2) at point P .

Consider a surface s with equation $z = g(x, y)$... (4)

If $g(x, y)$, $g_x(x, y)$ & $g_y(x, y)$ are continuous in certain region R of xy plane, then tangent plane at each point of s determines a plane element of type

$$\{x_0, y_0, g(x_0, y_0), g_x(x_0, y_0), g_y(x_0, y_0)\} \quad \dots (5)$$

called the tangent element of surface s at point $\{x_0, y_0, g(x_0, y_0)\}$

Theorem: A necessary & sufficient condition that a surface be an integral surface of a P.D.E. is that at each point its tangent element should touch the elementary cone of the equation.

Proof: Let surface s be $z = g(x, y)$... (1)

A curve c with parameter equation ... (2)

lies on surface (1) if $z(t) = g(x(t), y(t)) \quad \forall t \in I$ (say)

If P_0 is a point on the curve corresponding to value t_0 , then the direction ratios of the tangent line $P_0 P_1$ are $\{x'(t_0), y'(t_0), z'(t_0)\}$. This direction will be $\perp$ to the direction $(p_0, q_0, -1)$ if $z'(t_0) = p_0 x'_0(t_0) + q_0 y'_0(t_0)$

So, any set $\{x(t), y(t), z(t), p(t), q(t)\}$... (3)

of 5 real number satisfying the condition

$$z'(t) = p(t)x'(t) + q(t)y'(t) \quad \dots (4)$$

Defines a strip at the point (x, y, z) of curve c .

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If such a strip is also an integral element of equation

$$F(x, y, z, p, q) = 0 \quad \dots(5)$$

We say that it is an integral strip of equation (5)

Therefore set of from (3) in an integral strip of equation (2) provided they satisfy conditions (4) and conditions.

$$F\{x(t), y(t), z(t), p(t), q(t)\} \equiv 0 \quad \dots(6)$$

$$\forall t \in I.$$

If at each point the curve (2) touches a generator of elementary cone, we say that the corresponding strip in a characteristic strip the point $(x+dx, y+dy, z+dz)$ lies in the tangent plane to the elementary cone at P if

$$dz = p dx + q dy \quad \dots(7)$$

where p, q satisfy (5)

Differentiating (7) w. r. t. p we get

$$0 = dx + \frac{dq}{dp} dy \quad \dots(8)$$

$$\text{from (5)} \Rightarrow \frac{\partial F}{\partial p} + \frac{\partial F}{\partial q} \frac{dq}{dp} = 0 \quad \dots(9)$$

Solving (7), (8) & (9) for ratios of dy, dz to dx we have

$$\frac{dx}{F_p} = \frac{dy}{F_q} = \frac{dz}{pF_p + qF_q} \quad \dots(10)$$

$\Rightarrow$ Along a characteristic strip, $x'(t), y'(t), z'(t)$ must be proportional to F_p, F_q & $pF_p + qF_q$ respectively.

If we choose parameter t in such a way that

$$\left. \begin{aligned} x'(t) &= F_p \\ y'(t) &= F_q \end{aligned} \right\} \quad \dots(11)$$

$$z'(t) = pF_p + qF_q \quad \dots(12)$$

Along a characteristic strip p in a function of t so that $p'(t) = \frac{\partial p}{\partial x} \frac{dx}{dt} + \frac{\partial p}{\partial y} \frac{dy}{dt}$

$$= \frac{\partial p}{\partial x} \frac{\partial F}{\partial p} + \frac{\partial p}{\partial y} \frac{\partial F}{\partial q}$$

$$= \frac{\partial p}{\partial x} \frac{\partial F}{\partial p} + \frac{\partial q}{\partial x} \frac{\partial F}{\partial q}$$

$$\text{Since } \frac{\partial p}{\partial y} \equiv \frac{\partial q}{\partial x}$$

Differentiating equation (5) w.r.t. x we get

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x} = 0$$

so that on a characteristic strip

$$p'(t) = -(F_x + pF_z) \quad \dots(13)$$

Similarly we may obtain

$$q'(t) = -(F_y + qF_z) \quad \dots(14)$$

Now we have a system of following O.D.E. to determine characteristic strip.

$$\left. \begin{aligned} x'(t) &= F_p, y'(t) = F_q, z'(t) = pF_p + qF_q, p'(t) = -(F_x + pF_z) \\ &\& q'(t) = -(F_y + qF_z) \end{aligned} \right\} \quad \dots(15)$$

these are called characteristic equations of D.E. (5)

Theorem: Along every characteristic strip of equation $F(x, y, z, p, q) = 0$ the function $F(x, y, z, p, q)$ is a constant.

Proof: Along a characteristic strip we have

$$\begin{aligned} \frac{d}{dt} F(x(t), y(t), z(t), p(t), q(t)) &= F_x x' + F_y y' + F_z z' + F_p p' + F_q q' \\ &= F_x F_p + F_y F_q + F_z (pF_p + qF_q) - F_p (F_x + pF_z) - F_q (F_y + qF_z) \\ &= 0 \end{aligned}$$

$\Rightarrow F(x, y, z, p, q) = k$ a constant, along the strip.

Theorem: If a characteristic strip contains at least one integral element of $F(x, y, z, p, q) = 0$, it is an integral strip of equation $F(x, y, z, z_x, z_y) = 0$

Proof: Let solution of PDE, $F(x, y, z, z_x, z_y) = 0 \quad \dots(1)$

Passes through curve Γ whose equations are

$$x = \theta(v), y = \phi(v), z = \psi(v) \quad \dots(2)$$

$$\text{then in the solution } x = x(p_0, q_0, x_0, y_0, z_0, t_0, t) \text{ etc} \quad \dots(3)$$

of the characteristic equation (15) we may take

$$x_0 = \theta(v), y_0 = \phi(v), z_0 = \psi(v)$$

as initial values of x, y, z . The corresponding initial values of p_0, q_0 are determined by

$$\psi'(v) = p_0 \theta'(v) + q_0 \phi'(v)$$

$$F(\theta(v), \phi(v), \psi(v), p_0, q_0) = 0$$

If we substitute the values of x_0, y_0, z_0, p_0, q_0 and the appropriate value of t_0 in equation (3), we find that x, y, z , can be expressed in terms of two parameters t & v to give

$$x = X(v, t), y = Y(v, t) \& z = Z(v, t) \quad \dots(4)$$

Eliminating v, t from these three equation, we get a relation

$$\psi(x, y, z) = 0$$

which is equation of integral surface of equation (1) through curve Γ .

Example: Find solution of equation $z = \frac{1}{2}(p^2 + q^2) + (p-x)(q-y)$

Which passes through the x -axis

Solution: Here $F(x, y, z, p, q) = z - \frac{1}{2}p^2 - \frac{1}{2}q^2 - (p-x)(q-y) = 0$... (1)

Initial values are: $x_0 = v, y_0 = 0, z_0 = 0, p_0 = 0, q_0 = 2v, t_0 = 0$ characteristic equation are

$$\frac{dx}{dt} = F_p = -p - q + y \quad \dots (2)$$

$$\frac{dy}{dt} = F_q = -q - p + x \quad \dots (3)$$

$$\frac{dz}{dt} = pF_p + qF_q = p(-p - q + y) + q(-q - p + x) = -p^2 - q^2 - 2pq + py + qx \quad \dots (4)$$

$$\frac{dp}{dt} = -F_x - pF_z = -q + y - p \quad \dots (5)$$

$$\frac{dq}{dt} = -F_y - qF_z = -p + x - q \quad \dots (6)$$

From (2) & (5) $\Rightarrow \frac{dx}{dt} = \frac{dp}{dt} \Rightarrow x = p + c_1$

at $t = 0, x_0 = v, p_0 = 0 \Rightarrow c_1 = v$

Hence $x = p + v$... (7)

Similarly from (3) & (6) we have $y = q - 2v$... (8)

From (4) we have $-\frac{dp}{dt} - \frac{dq}{dt} + \frac{dx}{dt} = +q + p - y + p + q - x - p - q + y$

$$\Rightarrow \frac{d}{dt}(-p - q + x) = p + q - x$$

$$\Rightarrow p + q - x = ve^{-t} \quad \dots (9)$$

Similarly $\frac{d}{dt}(p + q - y) = -p - q + y \Rightarrow p + q - y = 2ve^{-t}$... (10)

from (7) & (8) $p + q = x + y + v$

So equation (9) & (10) becomes

$$\left. \begin{aligned} y + v &= ve^{-t} &\Rightarrow y &= v(e^{-t} - 1) \\ \& \quad x + v &= 2ve^{-t} &\Rightarrow x &= v(2e^{-t} - 1) \\ \text{Therefore } p &= 2v(e^{-t} - 1) \& \quad q &= v(e^{-t} + 1) \end{aligned} \right\} \quad \dots (11)$$

Therefore (4) becomes

$$\frac{dz}{dt} = 2v(e^{-t} - 1)(-2ve^{-t}) + v(e^{-t} + 1)(-ve^{-t})$$

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$$= -4v^2(e^{-2t} - e^{-t}) - v^2(e^{-2t} + e^{-t})$$

$$= -5v^2e^{-2t} - 3v^2e^{-t}$$

$$z = \frac{5}{2}v^2e^{-2t} + 3v^2e^{-t} + c_2$$

$$\text{at } t_0 = 0, z_0 = 0, \Rightarrow c_2 = -\frac{5}{2}v^2 + 3v^2 = +\frac{v^2}{2}$$

$$(4) \Rightarrow z = \frac{5}{2}v^2e^{-2t} + 3v^2e^{-t} + \frac{v^2}{2} \quad \dots(12)$$

From (11) we have

$$x = v(2e^{-t} - 1)$$

$$y = v(e^{-t} - 1)$$

$$\Rightarrow x - y = ve^{-t}$$

$$\therefore y = ve^{-t} - v = x - y - v \Rightarrow v = 2y - x$$

$$\therefore x - y = ve^{-t} \Rightarrow e^{-t} = \frac{x - y}{2y - x}$$

Therefore (12) becomes

$$\begin{aligned} z &= (2y - x)^2 \left[\frac{5}{2} \times \frac{(x - y)^2}{(2y - x)^2} + 3 \frac{(x - y)}{(2y - x)} + \frac{1}{2} \right] \\ &= (2y - x)^2 \times \left[\frac{5(x - y)^2 + 6(x - y)(2y - x) + (2y - x)^2}{2(2y - x)^2} \right] \\ &= \frac{1}{2} [5x^2 + 5y^2 - 10xy + 18y - 6x^2 - 12y^2 + 4y^2 + x^2 - 4xy] \\ &\Rightarrow z = \frac{1}{2} [-3y^2 + 4yx] = \frac{1}{2} y [4x - 3y] \end{aligned}$$

Example: Find the characteristics of equation $pq = z$ & determine the integral surface which passes through parabola $x = 0, y^2 = z$.

$$\text{Solution: Given P.D.E. } F(x, y, z, p, q) = z - pq = 0 \quad \dots(1)$$

Initial value is $x_0 = 0, t_0 = 0, y_0 = v, z_0 = v^2, p_0 = \frac{v}{2}, q_0 = 2v$ characteristic equations are

$$\frac{dx}{dt} = F_p = -q \quad \dots(2)$$

$$\frac{dy}{dt} = F_q = -p \quad \dots(3)$$

$$\frac{dz}{dt} = pF_q + qF_p = -2pq \quad \dots(4)$$

$$\frac{dp}{dt} = -F_x - pF_z = -p \quad \dots(5)$$

$$\frac{dq}{dt} = -F_y - qF_z = -q \quad \dots(6)$$

$$\text{From (2) \& (6)} \quad \frac{dx}{dt} = \frac{dq}{dt} \Rightarrow x = q + c_1 \Rightarrow x = q - 2v \quad \dots(7)$$

$$(3) \& (5) \Rightarrow \frac{dy}{dt} = \frac{dp}{dt} \Rightarrow y = p + c_2 \Rightarrow y = p + \frac{v}{2} \quad \dots(8)$$

$$\text{Also } \frac{dp}{dt} + \frac{dq}{dt} = -q - p \Rightarrow \frac{d}{dt}(p+q) = -(p+q)$$

$$\Rightarrow p+q = c_3 e^{-t} \Rightarrow c_3 = \frac{5}{2}v$$

$$\text{i.e. } p+q = \frac{5}{2}ve^{-t} \quad \dots(9)$$

$$\Rightarrow p-q = c_4 e^{-t}$$

$$\Rightarrow c_4 = -\frac{3}{2}v$$

$$\text{Hence } p-q = -\frac{3}{2}ve^{-t} \quad \dots(10)$$

$$\text{Solving (9) \& (10) we get } p = \frac{v}{2}e^{-t}, q = 2ve^{-t} \quad \dots(11)$$

$$\text{From equation (4)} \Rightarrow \frac{dz}{dt} = -2pq = -2v^2 e^{-2t}$$

$$\Rightarrow z = v^2 e^{-2t} + c_5 \Rightarrow c_5 = 0$$

$$\text{Hence } z = v^2 e^{-2t} \quad \dots(12)$$

$$\text{From equation (7), (8) \& (11) Q we have; } x = 2v(e^{-t} - 1) \& y = \frac{v}{2}(e^{-t} + 1)$$

$$v = \frac{4y-x}{4} \text{ And } e^{-t} = \frac{4y+x}{4y-x}$$

Therefore equation (12) becomes

$$z = \frac{(4y-x)^2}{16} \times \frac{(4y+x)^2}{(4y-x)^2}$$

$$\Rightarrow 16z = (4y+x)^2$$

Example: Determine the characteristic strip of equation $z = p^2 - q^2$, and find the integral surface passing through parabola $x^2 + 4z = 0, y = 0$

$$\text{Solution: Given P.D.E.; } F(x, y, z, p, q) = z - p^2 + q^2 = 0 \quad \dots(1)$$

$$\text{Initial values are, } t_0 = 0, x_0 = v, z_0 = \frac{-v^2}{4}, p_0 = -\frac{v}{2}, q_0 = \frac{v}{\sqrt{2}}, y_0 = 0$$

Characteristic are

$$\frac{dx}{dt} = F_p = -2p$$

$$\frac{dy}{dt} = F_q = 2q$$

$$\frac{dz}{dt} = -2p^2 + 2q^2$$

$$= pF_p + qF_q$$

$$\frac{dp}{dt} = -F_x - pF_z = -p$$

$$\frac{dq}{dt} = -F_y - qF_z = -q$$

$$\text{Here } \frac{dx}{dt} = 2 \frac{dp}{dt} \Rightarrow x = 2p + c_1 \Rightarrow x = 2(p + v) \quad \dots(2)$$

$$\frac{dy}{dt} = -2 \frac{dq}{dt} \Rightarrow y = -2q + c_2 \Rightarrow y = -2q + \sqrt{2}v \quad \dots(3)$$

$$\text{Also } \frac{dp}{dt} + \frac{dq}{dt} = -(p+q)$$

$$\frac{d}{dt}(p+q) = -(p+q) \Rightarrow p+q = c_3 e^{-t}$$

$$\text{Initial values: } \Rightarrow \frac{v}{2} + \frac{v}{\sqrt{2}} = c_3 \Rightarrow c_3 = \frac{v}{2}(\sqrt{2}-1)$$

$$\therefore p+q = \frac{v}{2}(\sqrt{2}-1)e^{-t} \quad \dots(4)$$

$$\text{Also } \frac{d}{dt}(p-q) = -p+q = -(p-q)$$

$$\Rightarrow p-q = c_4 e^{-t} \Rightarrow -\frac{v}{2} - \frac{v}{\sqrt{2}} = c_4 \Rightarrow c_4 = -\frac{v}{2}(\sqrt{2}+1)$$

$$\therefore p-q = -\frac{v}{2}(\sqrt{2}+1)e^{-t} \quad \dots(5)$$

Solving (4) & (5) we get

$$2p = \frac{v}{2}e^{-t}[\sqrt{2}-1-\sqrt{2}-1] \Rightarrow p = -\frac{v}{2}e^{-t}$$

$$\text{and } 2q = \frac{v}{2}e^{-t}[\sqrt{2}-1+\sqrt{2}+1] \Rightarrow q = \frac{v}{\sqrt{2}}e^{-t}$$

Substituting in equation (2) & (3) we get

$$x = 2p + 2v = -ve^{-t} + 2v = v(2 - e^{-t}) \quad \dots(6)$$

$$y = -2q + \sqrt{2}v = -\sqrt{2}ve^{-t} + \sqrt{2}v = \sqrt{2}v(1 - e^{-1}) \quad \dots(7)$$

How to choose initial values:

Given that $y_0 = 0$

Let $x_0 = v$ then $x^2 + 4z = 0$ gives, $z_0 = -\frac{v^2}{4}$, by

differentiating $x^2 + 4z = 0$, we get, $p_0 = -\frac{v}{2}$ and

substituting these values in given P.D.E we get

initial value $q_0 = \frac{v}{\sqrt{2}}$.

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Using these values we have

$$\begin{aligned}\frac{dz}{dt} &= -2p^2 + 2q^2 = -\frac{v^2}{2}e^{-2t} + v^2e^{-2t} \\ &= v^2e^{-2t}\left(1 - \frac{1}{2}\right) = \frac{v^2}{2}e^{-2t} \quad z = -\frac{v^2}{4}e^{-2t} + c_5\end{aligned}$$

$$\text{at } t_0 = 0, z_0 = -\frac{v^2}{4} \Rightarrow c_5 = 0$$

$$\therefore z = -\frac{v^2}{4}e^{-2t} + c_5 \quad \dots(8)$$

Solving (6) & (7) for v & e^{-t} we get

$$x - 2v = -ve^{-t}$$

$$\& \ y - \sqrt{2}v = -\sqrt{2}ve^{-t} \Rightarrow y - \sqrt{2}v = \sqrt{2}(x - 2v) \Rightarrow \sqrt{2}v = -y + \sqrt{2}x$$

$$\Rightarrow v = \frac{-y + \sqrt{2}x}{\sqrt{2}}$$

$$\Rightarrow e^{-t} = -\frac{x}{v} + 2 = \frac{-\sqrt{2}x}{-y + \sqrt{2}x} + 2 = \frac{-\sqrt{2}x - 2y + 2\sqrt{2}x}{-y + \sqrt{2}x}$$

$$\Rightarrow e^{-t} = \frac{-2y + \sqrt{2}x}{-y + \sqrt{2}x}$$

Using these value in (8) solution becomes

$$z = -\frac{1}{4} \frac{(-y + \sqrt{2}x)^2}{2} \times \frac{(-2y + \sqrt{2}x)^2}{(-y + \sqrt{2}x)^2} = -\frac{1}{8} \times (\sqrt{2})^2 (x - \sqrt{2}y)^2$$

$$\Rightarrow 4z + (x - \sqrt{2}y)^2 = 0$$

it satisfies the given P.D.E.

$$\text{Example: Given P.D.E. } F(x, y, z, p) = p^2 + q^2 - 4z = 0 \quad \dots(1)$$

$$\text{Initial surface } z = x^2 + 1 \quad y = 0$$

$$\text{Solution: Initial values are } t_0 = 0, y_0 = 0, x_0 = v, z_0 = 1 + v^2, p_0 = 2v, q_0 = 2.$$

Characteristics equation are:

$$\frac{dx}{dt} = F_p = 2p$$

$$\frac{dp}{dt} = -F_x - pF_z = 4p$$

$$\frac{dy}{dt} = F_q = 2q$$

$$\frac{dq}{dt} = -F_y - qF_z = 4q$$

$$\frac{dz}{dt} = pF_p + qF_q = 2p^2 + 2q^2$$

$$\text{Here } \frac{dx}{dt} = \frac{1}{2} \frac{dp}{dt} \Rightarrow x = \frac{1}{2} p + c_1 \Rightarrow x = \frac{1}{2} p \quad \dots(2)$$

$$\frac{dy}{dt} = \frac{1}{2} \frac{dq}{dt} \Rightarrow y = \frac{q}{2} + c_2 \Rightarrow y = \frac{q}{2} - 1 \quad \dots(3)$$

$$\frac{dp}{dt} + \frac{dq}{dt} = 4(p+q)$$

$$\Rightarrow p+q = c_3 e^{4t} \Rightarrow (p+q) = 2(v+1)e^{4t} \quad \dots(4)$$

$$\text{also } \frac{d}{dt}(p-q) = 4(p-q) \Rightarrow p-q = c_4 e^{4t} \Rightarrow c_4 = 2(v-1)$$

$$\therefore p-q = 2(v-1)e^{4t} \quad \dots(5)$$

$$\text{Solving (4) \& (5) we get } 2p = 4ve^{4t} \Rightarrow p = 2ve^{4t}$$

$$\& 2q = 4e^{4t} \Rightarrow q = 2e^{4t}$$

$$\text{Hence } \frac{dz}{dt} = 2(p^2 + q^2) = 2[4v^2 e^{8t} + 4e^{8t}] = 8[v^2 + 1]e^{8t}$$

$$\Rightarrow z = (v^2 + 1)e^{8t} + c_5 \Rightarrow 1 + v^2 = (v^2 + 1) + c_5 \Rightarrow c_5 = 0$$

$$\therefore z = (v^2 + 1)e^{8t} \quad \dots(6)$$

Also using values of p & q , equation (2) & (3) become

$$x = ve^{4t} \& y = e^{4t} - 1$$

$$\Rightarrow e^{4t} = y + 1$$

$$\text{Hence } v = \frac{x}{y+1}$$

Therefore equation (6) become

$$z = \left[\frac{x^2}{(y+1)^2} + 1 \right] (y+1)^2$$

$$\Rightarrow z = x^2 + (y+1)^2$$

Put Your Own Notes

CHAPTER-4

LINEAR PARTIAL DIFFERENTIAL EQUATIONS WITH CONSTANT COEFFICIENTS AND EQUATIONS WITH VARIABLE COEFFICIENTS

4.1 Homogeneous and Non-Homogeneous Linear Equations With Constant Coefficients

A partial differential equation in which the dependent variable and its derivatives appear only in the first degree and are not multiplied together, their coefficients being constants or functions of x and y , is known as a linear partial differential equation. The general form of such an equation is

$$\begin{aligned} & \left(A_0 \frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1} \partial y} + \dots + A_n \frac{\partial^n z}{\partial y^n} \right) \\ & \left(B_n \frac{\partial^{n-1} z}{\partial x^{n-1}} + B_1 \frac{\partial^{n-1} z}{\partial x^{n-1}} + \dots + B_{n-1} \frac{\partial^{n-1} z}{\partial y^{n-1}} \right) \\ & + \dots + \left(M_0 \frac{\partial z}{\partial x} + M_1 \frac{\partial z}{\partial y} \right) + N_0 z = f(x, y) \end{aligned} \quad \dots(1)$$

Where the coefficients $A_0, A_1, \dots, A_n, B_0, B_1, \dots, B_{n-1}, M_0, M_1$ and N_0 are constants or functions of x and y . If $A_0, A_1, \dots, A_n, B_0, B_1, \dots, B_{n-1}, M_0, M_1$ and N_0 are all constants, then (1) is called a linear partial differential equation with constant coefficients.

For convenience $\frac{\partial}{\partial x}$ and $\frac{\partial}{\partial y}$ will be denoted by D (or D_x) and D' (or D_y) respectively. Then (1) can be rewritten as

$$\begin{aligned} & [(A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D^n) (B_0 D^{n-1} + B_1 D^{n-2} D' + \dots + B_{n-1} D^{n-1}) \\ & + \dots + (M_0 D + M_1 D') + N_0] z = f(x, y), \end{aligned} \quad \dots(2)$$

$$\text{Or briefly, } F(D, D') z = f(x, y) \quad \dots(3)$$

When all the derivatives appearing in (1) are of the same order, then the resulting equation is called a linear homogeneous partial differential equation with constant coefficients and it is then of the

$$\text{Form } (A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D^n) z = f(x, y) \quad \dots(4)$$

On the other hand, when all the derivatives in (1) are not of the same order, then it is called a non-homogeneous linear partial differential equation with constant coefficients.

4.2 Working Rule For Finding Complementary Function of Linear Homogeneous Partial Differential Equation With Constant Coefficients

Step 1: Put the given equation in standard form

$$(A_0 D^n + A_1 D^{n-1} D' + \dots + A_n D^n) z = f(x, y) \quad \dots(1)$$

Step 2: Replacing D by m and D' by 1 in the coefficient of z , we obtain auxiliary equation (A.E.) for (1) as

$$A_0 m^n + A_1 m^{n-1} + \dots + A_n = 0 \quad (2)$$

Step 3: Solve (2) for m , following cases will arise.

Case (i): Let $m = m_1, m_2, \dots, m_n$ (distinct roots)

$$\text{Then, } C.F. = \phi_1(y + m_1 x) + \phi_2(y + m_2 x) + \dots + \phi_n(y + m_n x),$$

Where $\phi_1, \phi_2, \dots, \phi_n$ are arbitrary functions

Case (ii): Let $D = m$ (repeated n times). Then corresponding to these roots

$$C.F. \text{ then } = \phi_1(y + mx) + x\phi_2(y + mx) + \dots + x^{n-1}\phi_n(y + mx)$$

Case (iii): Corresponding to a non-repeated factor D on L.H.S. of (2) the part of C.F. is taken as $\phi(y)$.

Case (iv): Corresponding to a repeated factor D^m on L.H.S. of (2) the part of C.F. is taken as

$$\phi_1(y) + x\phi_2(y) + x^2\phi_3(y) + \dots + x^{m-1}\phi_m(y).$$

Case (v): Corresponding to a non-repeated factor D' on L.H.S. of (2), the part of C.F. is taken as $\phi(x)$.

Case (vi): Corresponding to a repeated factor D'^m on L.H.S. of (2) the part of C.F. is taken as

$$\phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \dots + y^{m-1}\phi_m(x).$$

Alternative working Rule For Finding C.F.

Let the given differential equation be $F(D, D')z = f(x, y)$. Factorize $F(D, D')$ into linear factors of the form $(bD - aD')$. Then we use the following results:

(i) Corresponding to each non-repeated factor $(bD - aD')$, the part of C.F. is taken as $\phi(by + ax)$.

(ii) Corresponding to a repeated factor $(bD - aD')^m$, the part of C.F. is taken as $\phi_1(by + ax) + x\phi_2(by + ax) + x^2\phi_3(by + ax) + \dots + x^{m-1}\phi_m(by + ax)$.

(iii) Corresponding to a non-repeated factor D , part of C.F. is taken as $\phi(y)$.

(iv) Corresponding to a repeated factor D^m , the part of C.F. is taken as

$$\phi_1(y) + x\phi_2(y) + x^2\phi_3(y) + \dots x^{m-1}\phi_m(y)$$

(v) Corresponding to a non-repeated factor D' , part of C.F. is taken as $\phi(x)$.

(vi) Corresponding to a repeated factor D'^m , the part of C.F. is taken as

$$\phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \dots + y^{m-1}\phi_m(x)$$

Example: Solve $2r + 5s + 2t = 0$

Solution: Now, $r = \left(\frac{\partial^2 z}{\partial x^2}\right) = D^2 z$, $s = \left(\frac{\partial^2 z}{\partial x \partial y}\right) = DD'z$ and $t = \frac{\partial^2 z}{\partial y^2} = D'^2 z$.

Hence the given

$$\text{Equation can be re-written as } (2D^2 + 5DD' + 2D'^2)z = 0 \quad \dots(1)$$

Its auxiliary equation is $2m^2 + 5m + 2 = 0$ or $(2m+1)(m+2) = 0$.

So $m = -1/2, -2$ and hence the general solution of (1) is

$$z = \phi_1(y - x/2) + \phi_2(y - 2x), \phi_1 \text{ and } \phi_2 \text{ being arbitrary functions.} \quad \dots(2)$$

Let $\phi_1(y - x/2) = \phi_1\{(1/2)(2y - x)\} = \psi_1(2y - x)$. Then (2) becomes

$$z = \psi_1(2y - x) + \phi_2(y - 2x), \psi_1 \text{ and } \phi_2 \text{ being arbitrary functions.}$$

Example: Solve $\left(\frac{\partial^4 z}{\partial x^4}\right) - \left(\frac{\partial^4 z}{\partial y^4}\right) = 0$

Solution: Re-writing the given equation is $(D^4 - D'^4)z = 0 \quad \dots(1)$

Its auxiliary equation is $m^4 - 1 = 0$ or $(m^2 - 1)(m^2 + 1) = 0$

So that $m = 1, -1, i, -i$ where $i = \sqrt{-1}$

Hence the general solution of (1) is

$$z = \phi_1(y - x) + \phi_2(y + x) + \phi_3(y + ix) + \phi_4(y - ix),$$

Where ϕ_1, ϕ_2, ϕ_3 and ϕ_4 are arbitrary functions.

4.3 Method of Finding Particular Integrating (P.I.) of a Linear Homogeneous Partial Differential Equation

$$F(D, D')z = f(x, y).$$

Method-I. When $f(x, y)$ is of the form $\phi(ax + by)$.

If $F(D, D')$ be homogeneous function of D and D' of degree n , the

$$\frac{1}{F(D, D')} \phi^{(n)}(ax + by)$$

$$= \frac{1}{F(a, b)} \phi(ax + by),$$

Provided $F(a, b) \neq 0$, $\phi^{(n)}$ being the n th derivative of ϕ w.r.t. $ax + by$ as a whole.

Exceptional case: When $F(a, b) = 0$, then the above theorem does not hold good. In such a case the new method is based on the following theorem. Note that $F(a, b) = 0$ if and only if $(bD - aD')$ is a factor $F(D, D')$.

$$\frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

Working Rule For Finding Particular Integral Where $f(x, y) = \phi(ax + by)$

Formula (i): When $F(a, b) \neq 0$ and $F(D, D')$ is a homogeneous function of degree n , then we have

$$P.I. = \frac{1}{F(D, D')} \phi(ax + by) = \frac{1}{F(a, b)} \iint \dots \int f(v) dv dv \dots dv, \text{ where } v = ax + by$$

note that R.H.S. contains a multiple integral of n th order.

$$\frac{1}{(bD - aD')^n} \phi(ax + by) = \frac{x^n}{b^n n!} \phi(ax + by).$$

Example: Solve $(D^2 + 3DD' + 2D'^2)z = x + y$

Solution: The auxiliary equation of the given equation is

$$m^2 + 3m + 2 = 0 \text{ giving } m = -1, -2.$$

$\therefore$ C.F. = $\phi_1(y - x) + \phi_2(y - 2x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{Now, } P.I. = \frac{1}{D^2 + 3DD' + 2D'^2} (x + y)$$

$$\Rightarrow \frac{1}{1^2 + 3.1.1 + 2.1^2} \iint v dv dv, \text{ Where } v = x + y$$

$$= \frac{1}{6} \int \left(\frac{v^2}{2} \right) dv = (1/6) \times (v^3 / 6) = (1/36)(x + y)^3.$$

$$\text{Or } z = \phi_1(y - x) + \phi_2(y - 2x) + (1/36)(x + y)^3.$$

Example: $r + s - 2t = (2x + y)^{1/2}$

Solution: The given equation can be re-written as

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 2 \frac{\partial^2 z}{\partial y^2} = (2x + y)^{1/2} \text{ or } (D^2 + DD' - 2D'^2)z = (2x + y)^{1/2}$$

Its auxiliary equation is $m^2 + m - 2 = 0$ so that $m = 1, -2$

$\therefore$ C.F. = $\phi_1(y + x) + \phi_2(y - 2x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{Now } P.I. = \frac{1}{D^2 + DD' - 2D'^2} (2x + y)^{1/2}$$

$$= \frac{1}{2^2 + 2.1 - 2.1^2} \iint v^{1/2} dv dv, \text{ where } v = 2x + y$$

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$$= \frac{1}{4} \int \frac{2}{3} v^{3/2} dv = \frac{1}{4} \cdot \frac{2}{3} \cdot \frac{2}{5} v^{5/2} = \frac{1}{15} (2x+y)^{5/2}$$

Hence the required general solution is $y = C.F. + P.I.$

$$\text{or } z = \phi_1(y+x) + \phi_2(y-2x) + (1/15)(2x+y)^{5/2}.$$

Example: $(D^2 - 3DD' + 2D'^2)z = e^{2x-y} + e^{x+y} + \cos(x+2y)$

Solution: Here auxiliary equation is $m^2 - 3m + 2 = 0$ so that $m = 1, 2$.

$$\therefore C.F. = \phi_1(y+x) + \phi_2(y+2x), \phi_1, \phi_2 \text{ being arbitrary functions.} \quad \dots(1)$$

Now, $P.I.$ corresponds to e^{2x-y}

$$= \frac{1}{D^2 - 3DD' + 2D'^2} e^{2x-y} = \frac{1}{2^2 - 3 \cdot 2(-1) + 2(-1)^2} \int e^v dv, \quad \text{where } v = 2x - y$$

$$\frac{1}{12} \int e^v dv = (1/12)e^v = (1/12)e^{2x+2y} \quad \dots(2)$$

Again, $P.I.$ corresponding to e^{x+y}

$$\begin{aligned} &= \frac{1}{D^2 - 3DD' + 2D'^2} e^{x+y} = \frac{1}{D-D'} \left\{ \frac{1}{D-2D'} e^{x+y} \right\} \\ &= \frac{1}{D-D'} \left\{ \frac{1}{1-2.1} \int e^v dv \right\}, \text{ where } v = x+y, \text{ using formula (i)} \\ &= -\frac{1}{D-D'} e^v = -\frac{1}{(D-D')^1} e^{x+y} = -\frac{x}{1!1!} e^{x+y} = -x e^{x+y} \quad \dots(3) \end{aligned}$$

Finally, $P.I.$ corresponding to $\cos(x+2y)$

$$= \frac{1}{D^2 - 3DD' + 2D'^2} \cos(x+2y) = \frac{1}{1^2 - 3 \cdot 1 \cdot 2 + 2 \cdot 2^2} \iint \cos v dv dv, \text{ where } v = x+2y \quad \dots(4)$$

$$= \frac{1}{3} \int \sin v dv = -\frac{1}{3} \cos v = -\frac{1}{3} \cos(x+2y).$$

From (1), (2), (3) and (4) the required solution is $z = C.F. + P.I.$

$$\text{or } z = \phi_1(y+x) + \phi_2(y+2x) + \frac{1}{12} e^{2x-y} - x e^{x+y} - \frac{1}{3} \cos(x+2y).$$

Method II. When $f(x, y)$ is of The Form $x^m y^n$ or A Rational Integral Algebraic Function of x and y .

If $n < m$, $\frac{1}{f(D, D')}$ should be expanded in powers of $\frac{D'}{D}$ whereas if

$m < n$, $\frac{1}{f(D, D')}$ should be expanded in powers of $\frac{D}{D'}$.

Example: Solve $(D^2 - a^2 D^2)z = x$ or $\frac{\partial^2 z}{\partial x^2} - a^2 \frac{\partial^2 z}{\partial y^2} = x$

Solution: Here auxiliary equation is so that $m = a, -a$

$\therefore C.F. = \phi_1(y + ax) + \phi_2(y - ax)$, ϕ_1, ϕ_2 being arbitrary functions ... (1)

$$\begin{aligned} \text{Now, } P.I. &= \frac{1}{D^2 - a^2 D^2} = \frac{1}{D^2 [1 - a^2 (D^2/D^2)]} x = \frac{1}{D^2} \left(1 - a^2 \frac{D^2}{D^2} \right)^{-1} \\ &= \frac{1}{D^2} \left(1 + \frac{a^2 D^2}{D^2} + \dots \right) x = \frac{1}{D^2} x = \frac{x^3}{6} \end{aligned} \quad \dots (2)$$

Alternatively, we can compute P.I. as follows:

$$\begin{aligned} P.I. &= \frac{1}{D^2 - a^2 D^2} x = \frac{1}{-a^2 D^2 [1 - (D^2/a^2 D^2)]} x = -\frac{1}{a^2 D^2} \left(1 - \frac{D^2}{a^2 D^2} \right)^{-1} x \\ &= -\frac{1}{a^2 D^2} \left(1 + \frac{D^2}{a^2 D^2} + \dots \right) x = -\frac{1}{a^2 D^2} x = -\frac{1}{a^2} \cdot \frac{xy^2}{2} \end{aligned} \quad \dots (3)$$

Hence the required general solution is $z = C.F. + P.I.$

$z = \phi_1(y + ax) + \phi_2(y - ax) + x^3/6$, using (1) and (2).

or $z = \phi_1(y + ax) + \phi_2(y - ax) - (xy^2)/2$, using (1) and (3)

A General Method of Finding the Particular Integral of Linear Homogeneous Equation with Constant Coefficients

$$P.I. = \frac{1}{(D - m_1 D')(D - m_2 D') \dots (D - m_n D')} f(x, y) \quad \dots (1)$$

We shall use one of the following formulas:

$$\text{Formula I: } \frac{1}{D - mD'} f(x, y) = \int f(x, c - mx) dx \text{ where } c = y + mx \quad \dots (2)$$

$$\text{Formula II: } \frac{1}{D + mD'} f(x, y) = \int f(x, c + mx) dx, \text{ where } c = y - mx \quad \dots (3)$$

Hence in order to evaluate P.I. (1), we apply (2) or (3) depending on the factor $D - mD'$ and $D + mD'$. Note that result (3) can be obtained from (2) by replacing m by $-m$.

Example: Solve $\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = \sin x$.

Solution: Re-writing, the given equation is $(D + D')z = \sin x$... (1)

$\therefore C.F. = \phi(y - x)$, Where ϕ is an arbitrary function

$$\text{and } P.I. = \frac{1}{D + D'} \sin x = \int \sin x dx = -\cos x$$

Hence the required solution is $z = C.F. + P.I. = \phi(y - x) - \cos x$

Non-Homogeneous Linear Partial Differential Equations with Constant Coefficients:

A linear partial differential equation which is not homogeneous is called a non-homogeneous linear equation.

Consider a linear equation with constant coefficients

$$F(D, D')z = f(x, y) \quad \dots(2)$$

If (1) is a homogeneous equation, then $F(D, D')$ can always be resolved into linear factors. On the other hand if (1) is not homogeneous then $F(D, D')$ cannot always be resolved into linear factors.

Hence, we classify non-homogeneous linear equations into the following types:

- (i). When $F(D, D')$ can be resolved into factors each of which is first degree in D' . For example consider the equation,
 $(D^2 - D'^2 + 2D + 1)z = x^2 y^3$ i.e. $(D + D' + 1)(D - D' + 1)z = x^2 y^3$
- (ii). When $F(D, D')$ cannot be resolved into linear form $(aD + bD' + c)$ For example, consider the equation
 $(2D^4 - 3D^2 D' + D'^2)z = 0$ i.e. $(2D^2 - D')(D^2 - D')z = 0$.

Working rule for finding C.F. of non-homogeneous equations which are reducible into linear partial differential equation with constant coefficients, namely, $F(D, D')z = 0$.

Method I When $F(D, D')$ can be expressed as product of factors linear in D and D' . Then we apply the following rules:

- (i) Corresponding to each non-repeated factor $(bD - aD' - c)$, the part of C.F. is taken as $e^{(cx/b)}\phi(by + ax)$, if $b \neq 0$.
- (ii) Corresponding to a repeated factor $(bD - aD' - c)^r$, the part of C.F. is taken as (assuming that $b \neq 0$)
 $e^{(cx/b)}[\phi_1(by + ax) + x\phi_2(by + ax) + x^2\phi_3(by + ax) + \dots + x^{r-1}\phi_r(by + ax)]$.
- (iii) As a particular case of (i) with $c = 0$, corresponding to each non-repeated factor $(bD - aD')$, the part of C.F. is taken as $\phi(by + ax)$.
- (iv) As a particular case of (ii) with $c = 0$, corresponding to each repeated factor $(bD - aD')^r$, the part of C.F. is taken as
 $\phi_1(by + ax) + x\phi_2(by + ax) + x^2\phi_3(by + ax) + \dots + x^{r-1}\phi_r(by + ax)$.
- (v) As a particular case of (iii) with $a = 0, b = 1$, corresponding to each non-repeated factor D , the part of C.F. is taken as $\phi(y)$.
- (vi) As a particular case of (iv) with $a = 0, b = 1$, corresponding to each repeated factor D^r , the part of C.F. is
 $\phi_1(y) + x\phi_2(y) + x^2\phi_3(y) + \dots + x^{r-1}\phi_r(y)$.
- (vii) Corresponding to each non-repeated linear factor $(bD - aD' - c)$, the part of C.F. is taken as $e^{-(cy/a)}\phi(by + ax)$, if $a \neq 0$

(viii) Corresponding to each non-repeated factor $(aD' + c)$, the part of C.F. is $e^{-(cy)/a} \phi(ax)$, when $a \neq 0$.

(ix) Corresponding to repeated factor $(aD' + c)^r$ the part of C.F. is $e^{-(cy)/a} [\phi_1(ax) + y\phi_2(ax) + y^2\phi_3(ax) + \dots + y^{r-1}\phi_r(ax)]$

(x) As a particular case of (viii), with $c=0, a=1$, corresponding to repeated factor D' , the part of C.F. = $\phi_1(x)$.

(xi) An a particular case of (ix), with $c=0, a=1$, corresponding to repeated factor D'' , the part of C.F. is taken as $\phi_1(x) + y\phi_2(x) + y^2\phi_3(x) + \dots + y^{r-1}\phi_r(x)$.

Method II: When $F(D, D')$ Cannot Be Resolved Into Linear Factors In D and D'

In such case we apply trial solution as explained the following example.

Example: Solve $(2D^4 - 3D^2D' + D'^2)z = 0$

Solution: The given equation can be re-written as

$$(2D^2 - D')(D^2 - D')z = 0 \quad \dots(1)$$

$$\text{Consider } (2D^2 - D')z = 0 \quad \dots(2)$$

Let $z = Ae^{hx+ky}$ be a trial solution of (1). then we have

$$D^2z = Ah^2e^{hx+ky} \text{ and } D'z = Ake^{hx+ky}. \text{ Putting these values in (2), we get}$$

$$A(2h^2 - k)e^{hx+ky} = 0 \text{ so that } 2h^2 - k = 0 \text{ or } k = 2h^2.$$

$$\text{Hence the most general solution of (2) is } z = \sum Ae^{hx+2h^2y} \quad \dots(3)$$

$$\text{Next, consider } (D^2 - D')z = 0 \quad \dots(4)$$

Let $z = A'e^{h'x+k'y}$ be a trial solution of (1). Then we have

$$D^2z = A'h'^2e^{h'x+k'y} \text{ and } D'z = A'k'e^{h'x+k'y}. \text{ Putting these values in (4), we get}$$

$$A'(h'^2 - k')e^{h'x+k'y} = 0 \text{ So that } h'^2 - k' = 0 \text{ or } k' = h'^2$$

$$\text{Hence the most general solution of (3) is } z = \sum A'e^{h'x+h'^2y} \quad \dots(5)$$

From (3) and (5), the most general solution of (1) is

$$Z = \sum Ae^{hx+2h^2y} + \sum A'e^{h'x+h'^2y}, A, A', h, h', k, k' \text{ being arbitrary constants.}$$

Methods of Finding P.I. of Non-Homogeneous Equations with Constant Coefficients.

$$F(D, D')z = f(x, y)$$

Case I: When $f(x, y) = e^{ax+by}$ and $F(a, b) \neq 0$.

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$$\text{Then, } P.I. = \frac{1}{F(D, D')} e^{ax+by} = \frac{1}{F(a, b)} e^{ax+by}$$

Thus in this case we replace D by a and D' by b .

Case II: When $f(x, y) = \sin(ax + by)$ or $\cos(ax + by)$.

$$\text{Then, } P.I. = \frac{1}{F(D, D')} \sin(ax + by) [\text{or } \cos(ax + by)],$$

Which is evaluated by putting $D^2 = -a^2$, $D'^2 = -b^2$ and $DD' = -ab$, provided the denominator is non-zero.

Case III: When $f(x, y) = x^m y^n$

$$\text{Then } P.I. = \frac{1}{F(D, D')} x^m y^n = [F(D, D')]^{-1} x^m y^n,$$

Case IV: When $f(x, y) = V e^{ax+by}$, when V is a function of x and y

$$\text{The } P.I. = \frac{1}{F(D, D')} V e^{ax+by} = e^{ax+by} \frac{1}{F(D+a, D'+b)} V$$

Example: Solve $(D^2 - D'^2 + D - D')z = e^{2x+3y}$

Solution: The given equation can be re-written as

$$[(D - D')(D + D') + (D - D')]z = e^{2x+3y} \text{ or } (D - D')(D + D' + 1)z = e^{2x+3y}$$

$\therefore$ C.F. = $\phi_1(y+x) + e^{-x}\phi_2(y-x)$, ϕ_1, ϕ_2 being arbitrary functions.

$$\text{and } P.I. = \frac{1}{(D - D')(D + D' + 1)} e^{2x+3y} = \frac{1}{(2-3)(2+3+1)} e^{2x+3y} = -\frac{1}{6} e^{2x+3y}$$

Hence the required general solution is $z = \text{C.F.} + \text{P.I.}$

$$\text{i.e. } z = \phi_1(y+x) + e^{-x}\phi_2(y-x) - \frac{1}{6} e^{2x+3y}$$

Example: Find a particular integral of the differential equation:

$$(D^2 - D')z = e^{x+y} + 5 \cos(x+2y)$$

Solution: P.I. corresponding to e^{x+y}

$$= \frac{1}{D^2 - D'} e^{x+y} = e^{x+y} \frac{1}{(D+1)^2 - (D'+1)} \cdot 1 = e^{x+y} \frac{1}{D^2 + 2D - D'}$$

$$= e^{x+y} \frac{1}{2D} \left[1 + \left(\frac{D}{2} - \frac{D'}{2D} \right) \right]^{-1} = e^{x+y} \frac{1}{2D} \{ 1 + \dots \} = \frac{1}{2} x e^{x+y}$$

P.I. Corresponding to $5 \cos(x+2y)$

$$= 5 \frac{1}{D^2 - D'} \cos(x+2y) = 5 \frac{1}{-1^2 - D'} \cos(x+2y) = -\frac{5}{D'+1} \cos(x+2y)$$

$$= -\frac{5(D'-1)}{D^2 - 1} \cos(x+2y) = -5 \frac{1}{-2^2 - 1} (D'-1) \cos(x+2y)$$

$$= (D'-1) \cos(x+2y) = D' \cos(x+2y) - \cos(x+2y) = -2 \sin(x+2y) - \cos(x+2y)$$

$$\therefore \text{Required } P.I. = \frac{1}{2} x e^{x+y} - 2 \sin(x+2y) - \cos(x+2y)$$

4.4 Partial Differential Equations with Variable Coefficients Reducible to Equations with Constant Coefficients

We propose to discuss the method of solving the so-called Euler-Cauchy type partial differential equations of the form

$$a_0 x^n \frac{\partial^n z}{\partial x^n} + a_1 x^{n-1} y \frac{\partial^n z}{\partial x^{n-1} \partial y} + a_2 x^{n-2} y^2 \frac{\partial^n z}{\partial x^{n-2} \partial y^2} + \dots + a_n y^n \frac{\partial^n z}{\partial y^n} + \dots = f(x, y) \quad \dots(1)$$

having variable coefficients in particular form (namely the term $\frac{\partial^n z}{\partial x^n}$ is

multiplied by x^n , $\frac{\partial^n z}{\partial x^{n-1} \partial y}$ is multiplied by y^n , $\frac{\partial^n z}{\partial x^{n-2} \partial y^2}$, $r=1, 2, \dots, n-1$ is

multiplied by $x^r y^{n-r}$ and so on. If $D \equiv \frac{\partial}{\partial x}$ and $D' \equiv \frac{\partial}{\partial y}$, then (1) can be re-written as

$$(a_0 x^n D^n + a_1 x^{n-1} y D^{n-1} D' + a_2 x^{n-2} y^2 D^{n-2} D'^2 + \dots + a_n y^n D'^n + \dots) z = f(x, y) \quad \dots(2)$$

Examples of such equations are:

$$(x^2 D^2 - y^2 D'^2) z = xy; x^2 D^2 - y^2 D'^2 + xD - yD' = \log x$$

4.4.1. Method of Reducing Euler-Cauchy Type Equation to A Linear Partial Differential Equation with Constant Coefficients

Consider Euler-Cauchy type equation

$$(a_0 x^n D^n + a_1 x^{n-1} y D^{n-1} D' + a_2 x^{n-2} y^2 D^{n-2} D'^2 + \dots + a_n y^n D'^n + \dots) z = f(x, y) \quad \dots(1)$$

Where $D \equiv \frac{\partial}{\partial x}$ and $D' \equiv \frac{\partial}{\partial y}$. Define two new variables u and v by

$$x = e^u \text{ and } y = e^v \text{ so that } u = \log x \text{ and } v = \log y \quad \dots(2)$$

$$\text{Let } D_1 \equiv \frac{\partial}{\partial u} \text{ and } D'_1 \equiv \frac{\partial}{\partial v} \quad \dots(3)$$

Now, $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} = \frac{1}{x} \frac{\partial z}{\partial u}$, using (2)

$$\therefore x \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \text{ so that } x \frac{\partial}{\partial x} = \frac{\partial}{\partial u} \text{ or } xD = D_1 \quad \dots(4)$$

$$\text{Again, } x \frac{\partial}{\partial x} \left(x^{n-1} \frac{\partial^{n-1} z}{\partial x^{n-1}} \right) = x^n \frac{\partial^n z}{\partial x^n} + (n-1) x^{n-1} \frac{\partial^{n-1} z}{\partial x^{n-1}}$$

$$\therefore x^n \frac{\partial^n z}{\partial x^n} = \left(x \frac{\partial}{\partial x} - n + 1 \right) x^{n-1} \frac{\partial^{n-1} z}{\partial x^{n-1}}$$

$$\text{So that } x^n D^n = (D_1 - n + 1) x^{n-1} D^{n-1} \quad \dots(5)$$

Putting $n=2, 3, \dots$ in (5), we have

$$x^2 D^2 = (D_1 - 1) xD \text{ or } x^2 D^2 = D_1 (D_1 - 1), \text{ using (4)} \quad \dots(6)$$

$$x^3 D^3 = (D_1 - 2) x^2 D^2 \text{ or } x^3 D^3 = D_1 (D_1 - 1) (D_1 - 2), \text{ using (6)} \quad \dots(7)$$

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and so on. Similarly, we have

$$yD' = D_1', \quad y^2D'^2 = D_1'(D_1' - 1) \quad y^3D'^3 = D_1'(D_1' - 1)(D_1' - 2) \quad \dots(8)$$

and so on. Also, we have $xyDD' = D_1D_1'$(9)

$$\text{and } x^m y^n D^m D'^n = D_1(D_1 - 1) \dots (D_1 - m + 1) D_1' (D_1' - 1) \dots (D_1' - n + 1). \quad \dots(10)$$

Using the substitutions (2) and results (4), (5), (6), (7), (8), (9) and (10), the given equation (1) reduces to an equation having constant coefficients and now it can easily be solved by the methods already discussed for homogeneous and non-homogeneous linear equations with constant coefficients. Finally, with help of (2), the solution is obtained in terms of old variables x and y .

4.4.2. Working Rule For Solving Euler-Cauchy Type Partial Differential Equation

$$(a_0 x^n D^n + a_1 x^{n-1} y D^{n-1} D' + a_2 x^{n-2} y^2 D^{n-2} D'^2 + \dots + a_n y^n D'^n + \dots) z = f(x, y) \quad \dots(1)$$

Step-1. Introduce two new variables u and v :

$$x = e^u \text{ and } y = e^v, \text{ i.e., } u = \log x \text{ and } v = \log y \quad \dots(2)$$

Step-2. We have $D \equiv \frac{\partial}{\partial x}$ and $D' \equiv \frac{\partial}{\partial y}$. Also let $D_1 \equiv \frac{\partial}{\partial u}$, $D_1' \equiv \frac{\partial}{\partial v}$

Step-3. Use the following results in (1)

$$\left. \begin{aligned} xD &= D_1, yD' = D_1', x^2D^2 = D_1(D_1 - 1), y^2D'^2 = D_1'(D_1' - 1) \\ x^3D^3 &= D_1(D_1 - 1)(D_1 - 2), y^3D'^3 = D_1'(D_1' - 1)(D_1' - 2) \text{ and so on} \end{aligned} \right\} \quad \dots(3)$$

Step-4. Using (2) and (3) in (1), we obtain the following linear partial differential (homogeneous or non-homogeneous)

$$(b_0 D_1^n + b_1 D_1^{n-1} D_1' + b_2 D_1^{n-2} D_1'^2 + \dots + b_n D_1'^n + \dots) z = g(u, v) \quad \dots(4)$$

Step-5. If (4) is homogeneous linear partial differential equation, then it is solved with help of previous method. Again, if (4) is non-homogeneous linear partial differential equation, then it is solved with help of previous methods.

Step-6. Using $u = \log x$ and $v = \log y$ in the solution obtained in step 5, we finally obtain the required solution in terms of the original variables x and y

Example: Solve $x^2 r - 3xy s + 2y^2 t + px + 2qy = x + 2y$

Solution: The given equation can be re-written as

$$x^2 \left(\frac{\partial^2 z}{\partial x^2} \right) - 3xy \left(\frac{\partial^2 z}{\partial x \partial y} \right) + 2y^2 \left(\frac{\partial^2 z}{\partial y^2} \right) + x \left(\frac{\partial z}{\partial x} \right) + 2y \left(\frac{\partial z}{\partial y} \right) = x + 2y$$

$$\text{Or } (x^2 D^2 - 3xy DD' + 2y^2 D'^2 + xD + 2yD') z = x + 2y. \quad \dots(1)$$

$$\text{Let } x = e^u, \quad y = e^v, \text{ so that } u = \log x, \quad v = \log y. \quad \dots(2)$$

$$\text{Also, let } D \equiv \frac{\partial}{\partial x}, \quad D' \equiv \frac{\partial}{\partial y}, \quad D_1 \equiv \frac{\partial}{\partial u} \text{ and } D_1' \equiv \frac{\partial}{\partial v}.$$

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$$\therefore (1) \text{ Becomes } [D_1(D_1 - 1) - 3D_1D'_1 + 2D'_1(D'_1 - 1) + D_1 + 2D'_1]z = e^u + 2e^v.$$

$$\text{Or } (D_1^2 - 3D_1D'_1 + 2D_1'^2)z = e^u + 2e^v \text{ or } (D_1 - D'_1)(D_1 - 2D'_1)z = e^u + 2e^v.$$

$$\therefore \text{C.F.} = \phi(v + u) + \phi_2(v + 2u) = \phi_1(\log y + \log x) + \phi_2(\log y + 2\log x)$$

Or C.F. $\phi_1(\log xy) + \phi_2(\log x^2 y) = f_1(xy) + f_2(x^2 y)$ where f_1 and f_2 are arbitrary functions.

Also, P.I. =

$$\begin{aligned} \frac{1}{(D_1 - D'_1)(D_1 - 2D'_1)}(e^u + 2e^v) &= \frac{1}{(D_1 - D'_1)(D_1 - 2D'_1)}e^{1u+0v} + 2\frac{1}{(D_1 - D'_1)(D_1 - 2D'_1)}e^{0u+1v} \\ &= \frac{1}{(1-0)(1-0)}e^u + 2\frac{1}{(0-1)(0-2)}e^v = x + y, \text{ using (2)} \end{aligned}$$

Hence the required general solution is $z = \text{C.F.} + \text{P.I.}$ or
 $z = f_1(xy) + f_2(x^2 y) + x + y.$

Example: Solve $x^2 r - y^2 t + px - qy = \log x.$

$$\text{or } (x^2 D^2 - y^2 D'^2 + xD - yD')z = \log x.$$

Solution: Let $x = e^u$, $y = e^v$ so that $u = \log x$, $v = \log y$ (1)

$$\text{Also, let } D \equiv \frac{\partial}{\partial x}, D' \equiv \frac{\partial}{\partial y}, D_1 \equiv \frac{\partial}{\partial u} \text{ and } D'_1 \equiv \frac{\partial}{\partial v}$$

Then the given equation $(x^2 D^2 - y^2 D'^2 + xD - yD')z = \log x$ becomes

$$[D_1(D_1 - 1) - D'_1(D'_1 - 1) + D_1 - D'_1]z = u \text{ or } (D_1^2 - D_1'^2)z = u$$

$$\text{or } \therefore \text{C.F.} = \phi_1(v - u) + \phi_2(v + u) = \phi_1(\log y - \log x) + \phi_2(\log y + \log x)$$

or C.F. $= \phi_1(\log xy) + \phi_2\left(\log \frac{y}{x}\right) = f_1(xy) + f_2\left(\frac{y}{x}\right)$ where f_1 and f_2 are arbitrary functions. and

$$\begin{aligned} \text{P.I.} &= \frac{1}{D_1^2 - D_1'^2}u = \frac{1}{D_1^2\left(1 - \frac{D_1'^2}{D_1^2}\right)}u = \frac{1}{D_1^2}\left(1 - \frac{D_1'^2}{D_1^2}\right)^{-1}u = \frac{1}{D_1^2}\left(1 + \frac{D_1'^2}{D_1^2} + \dots\right)u \\ &= \frac{1}{D_1^2}u = \frac{1}{6}u^3 = \frac{1}{6}(\log x)^3, \text{ by (1).} \end{aligned}$$

$\therefore$ Required solution is $z = f_1(xy) + f_2\left(\frac{y}{x}\right) + \left(\frac{1}{6}\right) \times (\log x)^3$, f_1, f_2 being arbitrary functions.

Example: Solve $(x^2 D^2 - 4y^2 D'^2 - 4yD' - 1)z = x^2 y^2 \log y.$

Solution: Let $x = e^u$, $y = e^v$ so that $u = \log x$, $v = \log y$ (1)

Also the given equation reduces to

$$[D_1(D_1 - 1) - 4D_1'(D'_1 - 1) - 4D'_1 - 1]z = e^{2u}e^{2v}v$$

$$\text{or } (D_1^2 - D_1 - 4D_1'^2 - 1)z = e^{2u+2v}v. \dots (2)$$

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Here $(D_1^2 - D_1 - 4D_1'^2 - 1)$ cannot be resolved into linear factors in D_1 and D_1' . To find C.F. corresponding to it, we consider the equation:

$$(D_1^2 - D_1 - 4D_1'^2 - 1)z = 0. \quad \dots(3)$$

Let a trial solution of (3) be $z = Ae^{hu+kv}$ and

$$D_1'^2 z = Ak^2 e^{hu+kv}, D_1 z = Ahe^{hu+kv}, D_1^2 z = Ah^2 e^{hu+kv} \quad \dots(4)$$

$$\text{Then (3)} \Rightarrow A(h^2 - h - 4k^2 - 1)e^{hu+kv} = 0 \Rightarrow h^2 - h - 4k^2 - 1 = 0. \quad \dots(5)$$

$$\text{C.F. of (2)} = \sum Ae^{hu+kv} = \sum A(e^u)^h (e^v)^k = \sum Ax^h y^k$$

P.I. of (2) (2)

$$\begin{aligned} &= \frac{1}{D_1^2 - D_1 - 4D_1'^2 - 1} e^{2u+2v} v = e^{2u+2v} \frac{1}{(D_1+2)^2 - (D_1+2) - 4(D_1+2)^2 - 1} v \\ &= e^{2u+2v} \frac{1}{D_1^2 + 3D_1 - 4D_1'^2 - 16D_1' - 15} v \\ &= e^{2u+2v} \frac{1}{(-15) \times \left[1 + \left(\frac{16}{15}\right) \times D_1' + \left(\frac{4}{15}\right) \times D_1'^2 - \left(\frac{1}{5}\right) \times D_1 - \left(\frac{1}{5}\right) \times D_1 - \left(\frac{1}{15}\right) \times D_1^2 \right]} v \\ &= e^{2u+2v} \frac{1}{(-15)} \left[1 + \left\{ \frac{16}{15} D_1' + \frac{4}{15} D_1'^2 - \frac{1}{5} D_1 - \frac{1}{15} D_1^2 \right\} \right]^{-1} v \\ &= \frac{e^{2u+2v}}{(-15)} \left(1 - \frac{16}{15} D_1' + \dots \right) v = \frac{e^{2u+2v}}{(-15)} \left(v - \frac{16}{15} v \right) = \frac{(e^u)^2 \times (e^v)^2 (16 - 15v)}{225} \\ &= \left(\frac{1}{225} \right) \times x^2 y^2 (16 - 15 \log y), \text{ using (1).} \end{aligned}$$

$$\text{The required general solution is } z = \sum Ax^h y^k + \left(\frac{1}{225} \right) \times x^2 y^2 (16 - 15 \log y)$$

where $h^2 - h - 4k^2 - 1 = 0$, and A , h and k are arbitrary constants.

CHAPTER-5

CLASSIFICATION OF SECOND ORDER PARTIAL DIFFERENTIAL EQUATIONS, CHARACTERISTIC CURVES AND REDUCTION TO CANONICAL FORMS

Consider equation of the type $Rr + Ss + Tt + f(x, y, z, p, q) = 0$... (1)

Where R, S, T are continuous functions of x and y possessing continuous partial derivatives of as high an order as necessary. Laplace transformation on (1) consists of changing the independent variables x, y to new set of independent variables u, v where

$$u = u(x, y) \text{ and } v = v(x, y) \quad \dots (2)$$

are to be chosen so that the resulting equation in independent variables u, v is transformed into one of three canonical forms which are easily integrable.

From (2), we have

$$p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x}, \quad q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} \quad \dots (3)$$

$$\Rightarrow \frac{\partial}{\partial x} = \frac{\partial u}{\partial x} \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial}{\partial v}, \quad \frac{\partial}{\partial y} = \frac{\partial u}{\partial y} \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial}{\partial v} \quad \dots (4)$$

$$\therefore r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial u}{\partial x} \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial x} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial z}{\partial v} \right)$$

$$= \frac{\partial u}{\partial x} \frac{\partial}{\partial u} \left(\frac{\partial u}{\partial x} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial z}{\partial v} \right) + \frac{\partial v}{\partial x} \frac{\partial}{\partial v} \left(\frac{\partial u}{\partial x} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial z}{\partial v} \right)$$

$$= \frac{\partial^2 z}{\partial u^2} \left(\frac{\partial u}{\partial x} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial^2 z}{\partial v^2} \left(\frac{\partial v}{\partial x} \right)^2 + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial x^2} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial x^2}$$

$$s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial u}{\partial x} \frac{\partial}{\partial u} + \frac{\partial v}{\partial x} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right) \text{ by (3) and (4)}$$

$$= \frac{\partial^2 z}{\partial u^2} \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial^2 z}{\partial u \partial v} \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) + \frac{\partial^2 z}{\partial v^2} \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial y \partial x} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial y \partial x}$$

$$\text{and } t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial u}{\partial y} \frac{\partial}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial}{\partial v} \right) \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right), \text{ by (3) and (4)}$$

$$= \frac{\partial u}{\partial y} \frac{\partial}{\partial u} \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right) + \frac{\partial v}{\partial y} \frac{\partial}{\partial v} \left(\frac{\partial u}{\partial y} \frac{\partial z}{\partial u} + \frac{\partial v}{\partial y} \frac{\partial z}{\partial v} \right)$$

$$= \frac{\partial^2 z}{\partial u^2} \left(\frac{\partial u}{\partial y} \right)^2 + 2 \frac{\partial^2 z}{\partial u \partial v} \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + \frac{\partial^2 z}{\partial v^2} \left(\frac{\partial v}{\partial y} \right)^2 + \frac{\partial z}{\partial u} \frac{\partial^2 u}{\partial y^2} + \frac{\partial z}{\partial v} \frac{\partial^2 v}{\partial y^2}$$

Putting these values of p, q, r, s, t in (1) simplifying, we get

$$A \frac{\partial^2 z}{\partial u^2} + 2B \frac{\partial^2 z}{\partial u \partial v} + C \frac{\partial^2 z}{\partial v^2} + F\left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right) = 0, \quad \dots(5)$$

$$\text{Where } A = R\left(\frac{\partial u}{\partial x}\right)^2 + S \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + T\left(\frac{\partial u}{\partial y}\right)^2, \quad \dots(6)$$

$$B = R \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{1}{2} S \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right) + T \frac{\partial u}{\partial y} \frac{\partial v}{\partial y}, \quad \dots(7)$$

$$C = R\left(\frac{\partial v}{\partial x}\right)^2 + S \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + T\left(\frac{\partial v}{\partial y}\right)^2 \quad \dots(8)$$

and $F\left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v}\right)$ is the transformed form of $f(x, y, z, p, q)$.

Now we shall find out u and v so that (5) reduces to simplest possible form. The method of evaluation of desired values of u and v becomes easy when the discriminant $S^2 - 4RT$ of the quadratic equation

$$R\lambda^2 + S\lambda + T = 0 \quad \dots(9)$$

is everywhere either positive, negative or zero, and now we shall present these three cases separately.

Case I: Let $S^2 - 4RT > 0$. When this condition is satisfied, then the roots λ_1, λ_2 of the equation (9) are real and distinct. The coefficients of $\frac{\partial^2 z}{\partial u^2}$ and $\frac{\partial^2 z}{\partial v^2}$ in the equation (5) will vanish if we choose u and v such that

$$\frac{\partial u}{\partial x} = \lambda_1 \left(\frac{\partial u}{\partial y} \right) \quad \dots(10)$$

$$\text{and } \frac{\partial v}{\partial x} = \lambda_1 \left(\frac{\partial v}{\partial y} \right) \quad \dots(11)$$

$$\text{Since } \lambda_1 \text{ is a root of (9), we have } R\lambda_1^2 + S\lambda_1 + T = 0 \quad \dots(12)$$

$$\text{Using (10), (6) gives } A = (R\lambda_1^2 + S\lambda_1 + T) \left(\frac{\partial u}{\partial y} \right)^2 = 0, \text{ by (12)} \quad \dots(13)$$

$$\text{Again, since } \lambda_2 \text{ is a root of (9), we have } R\lambda_2^2 + S\lambda_2 + T = 0 \quad \dots(14)$$

$$\text{Using (11), (8) gives } C = (R\lambda_2^2 + S\lambda_2 + T) \left(\frac{\partial v}{\partial y} \right)^2 = 0 \text{ by (14)} \quad \dots(15)$$

$$\text{Re-writing (10), we have } \left(\frac{\partial u}{\partial x} \right) - \lambda_1 \left(\frac{\partial u}{\partial y} \right) = 0. \quad \dots(16)$$

$$\frac{dx}{1} = \frac{dy}{-\lambda_1} = \frac{du}{0} \quad \dots(17)$$

Taking third fraction of (17), $du = 0$ so that $u = c_1$, c_1 being an arbitrary constant $\dots(18)$

Taking first and second fractions of (17), $\left(\frac{dy}{dx}\right) + \lambda_1 = 0$... (19)

Let the solution of (19) be $f_1(x, y) = c_2$, c_2 being an arbitrary constant ... (20)

From (18) and (20), the general solution of (16) [i.e. (10)] is

$$u = f_1(x, y) \quad \dots (21)$$

Similarly, the general solution of (11) can be taken as

$$v = f_2(x, y) \quad \dots (22)$$

We can easily verify that

$$AC - B^2 = \frac{1}{4}(4RT - S^2) \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2$$

or $B^2 = \frac{1}{4}(S^2 - 4RT) \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right)^2$ as $A = C = 0$... (23)

Since $S^2 - 4RT > 0$, (23) shows that $B^2 > 0$. Hence we may divide both sides of (5) by B^2 . Then nothing that $A = C = 0$, (5) transforms of the form

$$\frac{\partial^2 z}{\partial u \partial v} = \phi \left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right) \quad \dots (24)$$

which is the canonical form of (1) in this case.

Case II : Let $S^2 - 4RT = 0$. When this condition is satisfied, the roots λ_1, λ_2 of (9) are real and equal. We now take u exactly as in case I and take v to be an y function of x, y which is independent of u . We have, as in case I, $A = 0$. Also, since $S^2 - 4RT = 0$ (23) shows that $B^2 = 0$ so that $B = 0$.

Moreover in this case $C \neq 0$, otherwise v would be a function of u and consequently v would not be independent of u as already assumed.

Putting $A = 0, B = 0$ and dividing by C , (5) transforms to the form

$$\frac{\partial^2 z}{\partial v^2} = \phi \left(u, v, z, \frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} \right) \quad \dots (25)$$

Which is the canonical form of (1) in this case.

Case III: Let $S^2 - 4RT < 0$. when this condition is satisfied, the roots λ_1, λ_2 of (9) are complex. Hence is case III is formally the same as case I. Therefore, proceeding as in case I, we find that (1) reduces to (24) but that variables u, v instead of being real are now complex conjugates. To obtain a real canonical form we make further transformation $u = \alpha + i\beta$ and $v = \alpha - i\beta$ so that

$$\alpha = \left(\frac{1}{2} \right) (u + v), \beta = \frac{i(v - u)}{2} \quad \dots (26)$$

$$\text{Now, } \frac{\partial z}{\partial u} = \frac{\partial z}{\partial \alpha} \frac{\partial \alpha}{\partial u} + \frac{\partial z}{\partial \beta} \frac{\partial \beta}{\partial u} = \frac{1}{2} \left(\frac{\partial z}{\partial \alpha} - i \frac{\partial z}{\partial \beta} \right), \text{ by (26)} \quad \dots (27)$$

$$\text{and } \frac{\partial z}{\partial v} = \frac{\partial z}{\partial \alpha} \frac{\partial \alpha}{\partial v} + \frac{\partial z}{\partial \beta} \frac{\partial \beta}{\partial v} = \frac{1}{2} \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right), \text{ by (26)} \quad \dots (28)$$

$$\therefore \frac{\partial^2 z}{\partial u \partial v} = \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial v} \right) = \frac{1}{2} \left(\frac{\partial}{\partial \alpha} - i \frac{\partial}{\partial \beta} \right) \cdot \frac{1}{2} \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right), \text{ by (27) and (28)}$$

$$= \frac{1}{4} \left[\frac{\partial}{\partial \alpha} \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) - i \frac{\partial}{\partial \beta} \left(\frac{\partial z}{\partial \alpha} + i \frac{\partial z}{\partial \beta} \right) \right]$$

$$= \frac{1}{4} \left(\frac{\partial^2 z}{\partial \alpha^2} + i \frac{\partial^2 z}{\partial \alpha \partial \beta} - i \frac{\partial^2 z}{\partial \beta \partial \alpha} + \frac{\partial^2 z}{\partial \beta^2} \right)$$

$$\text{or } \frac{\partial^2 z}{\partial u \partial v} = \frac{1}{4} \left(\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} \right), \text{ as } \frac{\partial^2 z}{\partial \alpha \partial \beta} = \frac{\partial^2 z}{\partial \beta \partial \alpha} \quad \dots (29)$$

Putting $u = \alpha + i\beta$, $v = \alpha - i\beta$ and using (27), (28) and (29), (24) reduces to

$$\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = \psi \left(\alpha, \beta, z, \frac{\partial z}{\partial \alpha}, \frac{\partial z}{\partial \beta} \right) \text{ which is the canonical form of (1) in this case.}$$

Remark: Second order partial differential equations of the type (1) are classified by their canonical forms. An equation of the form (1) is said to be

(i) Hyperbolic if $S^2 - 4RT > 0$

(ii) Parabolic if $S^2 - 4RT = 0$

(iii) Elliptic if $S^2 - 4RT < 0$.

Example: Reduce $\frac{\partial^2 z}{\partial x^2} = x^2 \left(\frac{\partial^2 z}{\partial y^2} \right)$ to canonical form.

Solution: Re-writing, the given equation becomes $r - x^2 t = 0$(1)

Comparing (1) with $Rr + Ss + Tt + f(x, y, z, p, q) = 0$, we have

$$R = 1, S = 0, T = -x^2$$

Now, the λ -quadratic $R\lambda^2 + S\lambda + T = 0$ gives

$$\lambda^2 - x^2 = 0 \text{ so that } \lambda = \pm x.$$

$\therefore$ Here $\lambda_1 = x$ and $\lambda_2 = -x$ (Real and distinct roots)

Hence $\frac{dy}{dx} + \lambda_1 = 0$ and $\frac{dy}{dx} + \lambda_2 = 0$ become

$$\frac{dy}{dx} + x = 0 \text{ and } \frac{dy}{dx} - x = 0$$

Integrating these, $y + (1/2)x^2 = c_1$ and $y - (1/2)x^2 = c_2$.

Hence in order to reduce (i) to canonical form, we change x, y to u, v by taking.

$$u = y + (1/2)x^2 \text{ and } v = y - (1/2)x^2. \quad \dots (2)$$

$$P = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = x \frac{\partial z}{\partial u} - x \frac{\partial z}{\partial v}, \text{ using (2)}$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \text{ using (2)}$$

$$\begin{aligned} r &= \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} \left\{ x \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \right\} = x \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) + 1 \cdot \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \\ &= x \left[\frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial u}{\partial x} + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \frac{\partial v}{\partial x} \right] + \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \\ &= x^2 \left(\frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) + \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \end{aligned}$$

$$\text{and } t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2}$$

Putting these values of r and t in (1), we get

$$x^2 \left(\frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) + \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} - x^2 \left(\frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \right) = 0$$

$$\text{or } \frac{\partial^2 z}{\partial u \partial v} = \frac{1}{4x^2} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right) \quad \text{or } \frac{\partial^2 z}{\partial u \partial v} = \frac{1}{4(u-v)} \left(\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} \right), \text{ by (2)}$$

Which is the required canonical form of the given equation.

Example: Reduce the equation $\frac{\partial^2 z}{\partial x^2} + 2 \left(\frac{\partial^2 z}{\partial x \partial y} \right) + \frac{\partial^2 z}{\partial y^2} = 0$ to canonical form and hence solve it.

Solution: Re-writing the given equation, we get $r + 2s + t = 0$... (1)

Comparing (1) with $Rr + Ss + Tt + f(x, y, z, p, q) = 0$, here $R = 1, S = 2, T = 1$ so that $S^2 - 4RT = 0$, showing that (1) is parabolic.

The λ -quadratic equation reduces to $\lambda^2 + 2\lambda + 1 = 0$ so that $\lambda = -1, -1$ (equal roots).

The corresponding characteristic equation is $\left(\frac{dy}{dx} \right) - 1 = 0$ or $dx - dy = 0$

Integrating, $x - y = c$, c being an arbitrary constant.

Choose $u = x - y$ and $v = x + y$, ... (2)

Where we have chosen $v = x + y$ in such a manner that u and v are independent functions as verified below.

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} = 1.1 + 1.1 = 2 \neq 0.$$

$$\text{Now, } p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = \frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \text{ using (2)} \quad \dots (3)$$

$$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = -\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v}, \text{ using (2)} \quad \dots (4)$$

$$\text{From (3) and (4), } \frac{\partial}{\partial x} = \frac{\partial}{\partial u} + \frac{\partial}{\partial v} \quad \text{and} \quad \frac{\partial}{\partial y} = -\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \quad \dots (5)$$

$$\therefore r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right), \text{ by (3) and (5)}$$

$$= \frac{\partial}{\partial u} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = \frac{\partial^2 z}{\partial u^2} + 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \quad \dots(6)$$

$$t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \left(-\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right), \text{ by (4) and (5)}$$

$$= -\frac{\partial}{\partial u} \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = \frac{\partial^2 z}{\partial u^2} - 2 \frac{\partial^2 z}{\partial u \partial v} + \frac{\partial^2 z}{\partial v^2} \quad \dots(7)$$

$$\text{and } s = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) \text{ by (4) and (5)}$$

$$= \frac{\partial}{\partial u} \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) + \frac{\partial}{\partial v} \left(-\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} \right) = -\frac{\partial^2 z}{\partial u^2} + \frac{\partial^2 z}{\partial v^2} \quad \dots(8)$$

Using (6), (7) and (8) in (1), the required canonical form is

$$\frac{\partial^2 z}{\partial v^2} = 0 \text{ or } \frac{\partial}{\partial v} \left(\frac{\partial z}{\partial v} \right) = 0 \quad \dots(9)$$

To Find The Required Solution, integrating (9) partially w. r. t. 'v',

$$\text{we get } \frac{\partial z}{\partial v} = \phi(u), \phi \text{ being an arbitrary function.} \quad \dots(10)$$

$$\text{Integrating (10) partially w. r. t. 'v', } z = \int \phi(u) dv + \psi(u) = v\phi(u) + \psi(u)$$

or $z = (x+y)\phi(x-y) + \psi(x-y)$, which is the desired solution, ϕ, ψ being arbitrary functions.

Example: Reduce the following partial differential equations to canonical forms:

$$\frac{\partial^2 z}{\partial x^2} + x^2 \left(\frac{\partial^2 z}{\partial y^2} \right) = 0 \text{ or } r + x^2 t = 0$$

$$\text{Solution: Re-writing the given equations, we get } r + x^2 t = 0 \quad \dots(1)$$

Comparing (1) with $Rr + Ss + Tt + f(x, y, z, p, q) = 0$, here $R=1, S=0, T=x^2$ so that $S^2 - 4RT = -4x^2 < 0, x \neq 0$, showing that (1) is elliptic.

The λ -quadratic $R\lambda^2 + S\lambda + T = 0$ reduces to $\lambda^2 + x^2 = 0$ giving $\lambda = ix, -ix$.

The corresponding characteristic equation are given by

$$\frac{dy}{dx} + ix = 0 \text{ and } \frac{dy}{dx} - ix = 0$$

$$\text{Integrating, } y + i \left(\frac{x^2}{2} \right) = c_1 \text{ and } y - i \left(\frac{x^2}{2} \right) = c_2.$$

$$\text{Choose } u = y + i \left(\frac{x^2}{2} \right) = \alpha + i\beta \text{ and } v = y - i \left(\frac{x^2}{2} \right) = \alpha - i\beta,$$

Where $\alpha = y$ and $\beta = \frac{x^2}{2}$... (2)

are now two new independent variables.

Now, $p = \frac{\partial z}{\partial x} = \frac{\partial z}{\partial \alpha} \frac{\partial \alpha}{\partial x} + \frac{\partial z}{\partial \beta} \frac{\partial \beta}{\partial x} = x \frac{\partial z}{\partial \beta}$, by (2) ... (3)

$q = \frac{\partial z}{\partial y} = \frac{\partial z}{\partial \alpha} \frac{\partial \alpha}{\partial y} + \frac{\partial z}{\partial \beta} \frac{\partial \beta}{\partial y} = \frac{\partial z}{\partial \alpha}$, by (2) ... (4)

$r = \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial x} \left(x \frac{\partial z}{\partial \beta} \right) = \frac{\partial z}{\partial \beta} + x \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial \beta} \right)$, by (3)

$= \frac{\partial z}{\partial \beta} + x \left[\frac{\partial}{\partial \alpha} \left(\frac{\partial z}{\partial \beta} \right) \frac{\partial \alpha}{\partial x} + \frac{\partial}{\partial \beta} \left(\frac{\partial z}{\partial \beta} \right) \frac{\partial \beta}{\partial x} \right] = \frac{\partial z}{\partial \beta} + x^2 \frac{\partial^2 z}{\partial \beta^2}$... (5)

and $t = \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial \alpha} \left(\frac{\partial z}{\partial \alpha} \right) = \frac{\partial^2 z}{\partial \alpha^2}$, by (4) ... (6)

Using (5) and (6) in (1) the required canonical form is

$\frac{\partial z}{\partial \beta} + x^2 \frac{\partial^2 z}{\partial \beta^2} + x^2 \frac{\partial^2 z}{\partial \alpha^2} = 0$ or $\frac{\partial^2 z}{\partial \alpha^2} + \frac{\partial^2 z}{\partial \beta^2} = \frac{1}{2\beta} \frac{\partial z}{\partial \beta}$, as $\beta = \frac{x^2}{2}$

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CHAPTER-6

HEAT, WAVE AND LAPLACE EQUATION

6.1. Introduction

In physical problems, we always seek a solution of the differential equations which satisfies some specified conditions known as the boundary conditions. The differential equation together with these boundary conditions, constitute a boundary value problem.

In problems involving ordinary differential equations, we may first find the general solution and then determine the arbitrary constants from the initial values. But the same process is not applicable to problems involving partial differential equations for the general solution of a partial differential equation contains arbitrary functions which are difficult to adjust so as to satisfy the given boundary conditions. Most of the boundary value problems involving linear partial differential equations can be solved by the following method.

A solution which breaks up into a product of functions each of which contains only one of the variables. The following explains this method.

6.2. Method of Separation of Variables

Example: Solve by the method of separation of variables:

$$\frac{\partial^2 z}{\partial x^2} - 2 \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$$

Suppose that the given equation has solution of the form

$$z(x, y) = X(x) Y(y) \quad \dots(i)$$

Where X is a function of x alone and Y that of y alone.

Substituting this value of z in the given equation, we have

$$X''Y - 2X'Y + XY' = 0 \text{ Where } X' = \frac{dX}{dx}, Y' = \frac{dY}{dy} \text{ etc.}$$

$$\text{Separating the variables, we get } \frac{X'' - 2X'}{X} = -\frac{Y'}{Y} \quad \dots(ii)$$

Since x and y are independent variables, therefore, (ii) can only be true if each side is equal to the same constant, a (say).

$$\therefore \frac{X'' - 2X'}{X} = a, \text{ i.e. } X'' - 2X' - aX = 0 \quad \dots(iii)$$

$$\text{and } -\frac{Y'}{Y} = a, \text{ i.e. } Y' + aY = 0 \quad \dots(iv)$$

To solve the ordinary linear equation (iii), the auxiliary equation is

$$m^2 - 2m - a = 0, \text{ where } m = 1 \pm \sqrt{1+a}$$

$$\therefore \text{The solution of (iii) is } X = c_1 e^{[1+\sqrt{1+a}]x} + c_2 e^{[1-\sqrt{1+a}]x}$$

and the solution of (iv) is $Y = c_3 e^{-ay}$.

Substituting these values of x and y in (i) we get

$$z = \left\{ c_1 e^{\left[1+\sqrt{1+a}\right]x} + c_2 e^{\left[1-\sqrt{1+a}\right]x} \right\} \cdot c_3 e^{-ay}$$

$$\text{i.e. } z = \left\{ k_1 e^{\left[1+\sqrt{1+a}\right]x} + k_2 e^{\left[1-\sqrt{1+a}\right]x} \right\} e^{-ay}$$

Which is the required complete solution.

Example: Using the method of separation of variables, solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} + u$

where $u(x, 0) = 6e^{-3x}$.

Solution: Assume the solution $u(x, t) = X(x)T(t)$

Substituting in the given equation, we have

$$XT = 2XT' + XT \quad \text{or} \quad (X' - X)T = 2XT'$$

$$\text{or } \frac{X' - X}{2X} = \frac{T'}{T} = k \quad (\text{say})$$

$$\therefore X' - X - 2kX = 0 \quad \text{Or } \frac{X'}{X} = 1 + 2k \quad \dots(i)$$

$$\text{And } \frac{T'}{T} = k \quad \dots(ii)$$

$$\text{Solving (i), } \log X = (1+2k)x + \log c \quad \text{or } X = ce^{(1+2k)x}$$

$$\text{From (ii) } \log T = kt + \log c' \quad \text{or } T = c'e^{kt}$$

$$\text{Thus } u(x, t) = XT = cc'e^{(1+2k)x} e^{kt} \quad \dots(iii)$$

$$\text{Now } 6e^{-3x} = u(x, 0) = cc'e^{(1+2k)x}$$

$$\therefore cc' = 6 \quad \text{and } 1+2k = -3 \quad \text{or } k = -2$$

Substituting these values in (iii), we get

$$u = 6e^{-3x} e^{-2t} \quad \text{i.e. } u = 6e^{-(3x+2t)} \quad \text{which is the required solution.}$$

6.3. Some Important Equations

$$1. \text{ Heat (Diffusion) Equation: } \nabla^2 u = \left(\frac{1}{k}\right) \left(\frac{\partial u}{\partial t}\right).$$

$$\text{One-dimensional: } \frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{k}\right) \left(\frac{\partial u}{\partial t}\right)$$

$$\text{Two-dimensional: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{k}\right) \left(\frac{\partial u}{\partial t}\right)$$

$$\text{Three-dimensional: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \left(\frac{1}{k}\right) \left(\frac{\partial u}{\partial t}\right)$$

Note: Sometimes we write c^2 instead of k in heat equation.

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Remark: 1. ∇^2 is called the Laplacian operator and is defined as

$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$. A function u which satisfies Laplace's equation is called the harmonic function.

2. Wave Equation: $\nabla^2 u = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$

One-dimensional: $\frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial^2 u}{\partial t^2}\right)$

Two-dimensional: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial^2 u}{\partial t^2}\right)$

Three-dimensional: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial^2 u}{\partial t^2}\right)$

3. Laplace's (or harmonic) Equation: $\nabla^2 u = 0$

Two-dimensional: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Three-dimensional: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$

6.4. Solution of the Heat Equation

Assume that a solution of $\frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial u}{\partial t}\right)$... (1)

is of the form $u(x, t) = X(x) \cdot T(t)$

Where X is a function of x alone and T is a function of t only.

Substituting this in (1), we get

$XT' = c^2 X''T$, i.e. $X''/X = T'/c^2 T$... (2)

Clearly the left side of (2) is a function of x only and the right side is a function of t alone. Since x and t are independent variables, (2) can hold good if each side is equal to a constant k (say). Then (2) leads to the ordinary differential equations.

$\frac{d^2 X}{dx^2} - kX = 0$... (3)

and $\frac{dT}{dt} - kc^2 T = 0$... (4)

Solving (3) and (4) we get

(i) When k is positive and $= p^2$, say:

$X = c_1 e^{px} + c_2 e^{-px}$, $T = c_3 e^{c^2 p^2 t}$;

(ii) When k is negative and $= -p^2$, say:

$$X = c_4 \cos px + c_5 \sin px, T = c_6 e^{-c^2 p^2 t};$$

(iii) When k is zero:

$$X = c_7 x + c_8, T = c_9$$

Thus the various possible solution of the heat equation (1) are

$$u = (c_1 e^{px} + c_2 e^{-px}) c_3 e^{c^2 p^2 t} \quad \dots(5)$$

$$u = (c_4 \cos px + c_5 \sin px) c_6 e^{-c^2 p^2 t} \quad \dots(6)$$

$$u = (c_7 x + c_8) c_9 \quad \dots(7)$$

Of the three solutions, we have to choose that solution which is consistent with the physical nature of the problem. As we are dealing with problems on heat conduction, it must be a transient solution, i.e., u is to decrease with the increase of time t . Accordingly, the solution given by (6), i.e. of the form

$$u = (C_1 \cos px + C_2 \sin px) C_3 e^{-c^2 p^2 t} \quad \dots(8)$$

is the only suitable solution of the heat equation.

Example: General solution of heat equation when both the ends of a bar are kept at temperature zero and the initial temperature is prescribed.

If both the ends of a bar of length a are at temperature zero and the initial temperature is to be prescribed function $f(x)$ in the bar, then find the temperature at a subsequent time t .

or The faces $x=0$ and $x=a$ of an infinite slab are maintained at zero temperature. Given that the temperature $u(x, t) = f(x)$ at $t=0$. Determine the temperature at a subsequent time t .

Solution: Here the temperature $u(x, t)$ in the given solid is governed by the

$$\text{one dimensional heat equation } k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t} \quad \dots(1)$$

Since the ends $x=0$ and $x=a$ are kept at zero temperature, the boundary conditions are

$$u(0, t) = 0 \text{ and } u(a, t) = 0, \text{ for all } t \quad \dots(2)$$

$$\text{The initial condition is given by } u(x, 0) = f(x), 0 < x < a \quad \dots(3)$$

$$\text{Suppose that (1) has solutions of the form } u(x, t) = X(x) T(t) \quad \dots(4)$$

Where X is a function of x alone and T that of t alone.

Substituting this value of u in (1), we get

$$k X'' T = X T' \text{ or } \frac{X''}{X} = \frac{T'}{kT} \quad \dots(5)$$

Since x and t are independent variables, (5) can only be true if each side is equal to the same constant, say μ . Hence (5) leads to the following equations:

$$X'' - \mu X = 0 \quad \dots(6)$$

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$$\text{and } T' = \mu k T. \quad \dots(7)$$

$$\text{Using (2), (4) gives } X(0) T(t) = 0 \quad \text{and } X(a) T(t) = 0 \quad \dots(8)$$

Since $T(t) = 0$ leads to $u = 0$, so suppose that $T(t) \neq 0$.

$$\therefore \text{Form (8), } X(0) = 0 \text{ and } X(a) = 0 \quad \dots(9)$$

We now solve (6) under B.C. (9). Three cases arise.

$$\text{Case I. Let } \mu = 0. \text{ Then solution of (6) is } X(x) = Ax + B \quad \dots(10)$$

Using B.C. (9), (10) gives $0 = B$, $0 = Aa + B$ so that $A = B = 0$.

Hence $X(x) \equiv 0$ so that $u \equiv 0$, which does not satisfy (3). So we reject $\mu = 0$,

Case II. Let $\mu = \lambda^2$, $\lambda \neq 0$. Then solution of (6) is

$$X(x) = Ae^{\lambda x} + Be^{-\lambda x} \quad \dots(11)$$

Using B.C. (9), (11) gives $0 = A + B$ and

$$0 = Ae^{a\lambda} + Be^{-a\lambda} \quad \dots(12)$$

Solving (12), $A = B = 0$ so that $X(x) \equiv 0$ and hence $u \equiv 0$,

Which does not satisfy (3). So we also reject $\mu = \lambda^2$.

Case III. Let $\mu = -\lambda^2$, $\lambda \neq 0$. Then solution of (6) is

$$X(x) = A \cos \lambda x + B \sin \lambda x \quad \dots(13)$$

Using B.C. (9), (13) gives $0 = A$ and $0 = A \cos \lambda a + B \sin \lambda a$

So $\sin \lambda a = 0$. we have taken $B \neq 0$, since otherwise $X \equiv 0$ so that $u \equiv 0$ which does not satisfy (3). Solving the trigonometric equation $\sin \lambda a = 0$, we have

$$\lambda a = n\pi \text{ so that } \lambda = \frac{n\pi}{a}, \text{ where } n = 1, 2, \dots \quad \dots(14)$$

Hence non-zero solutions $X_n(x)$ of (6) are given by

$$X_n(x) = B_n \sin\left(\frac{n\pi x}{a}\right) \quad \dots(15)$$

$$\text{Using (14), (7) reduces to } \frac{dT}{T} = -\frac{n^2 \pi^2 k}{a^2} dt, \text{ as } \mu = -\lambda^2 = -\frac{n^2 \pi^2}{a^2}$$

$$\text{or } \left(\frac{1}{T}\right) dT = -C_n^2 dt, \text{ as } C_n^2 = \frac{(n^2 \pi^2 k)}{a^2}$$

$$\text{Whose general solution is } T_n(t) = D_n e^{-C_n^2 t} \quad \dots(16)$$

$$\therefore u_n(x, t) = X_n(x) T_n(t) = E_n \sin\left(\frac{n\pi x}{a}\right) e^{-C_n^2 t} \quad \dots(17)$$

are solutions of (1), satisfying (2). Here $E_n (= B_n D_n)$ is another arbitrary constant. In order to obtain a solution also satisfying (3), we consider more general solution.

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right) e^{-C_n^2 t} \quad \dots(18)$$

Substituting $t = 0$ in (18) and using (3), we get $f(x) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right)$ which is Fourier sine series. So the constants E_n are given by

$$E_n = \frac{2}{a} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx, \quad n = 1, 2, 3$$

Hence (18) is the required solution where E_n is given by (19).

Working Rule For Solving Heat Equation When Both The Ends of A Bar of Length a are Kept At Temperature Zero and The Initial Temperature $f(x)$ is Prescribed.

Step-1. Proceed as previous and prove that the solution of the heat equation

$$k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t} \quad \dots(1)$$

subject to the boundary conditions $u(0, t) = u(a, t) = 0$, for all t $\dots(2)$

and the initial condition $u(x, 0) = f(x)$, $0 < x < a$ $\dots(3)$

is given by $u(x, t) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{n\pi x}{a}\right) e^{-C_n^2 t}$ $\dots(4)$

Where $E_n = \frac{2}{a} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$, $n = 1, 2, 3, \dots$ $\dots(5)$

and $C_n^2 = \frac{(n^2 \pi^2 k)}{a^2}$ $\dots(6)$

Step-II. Compare the given problem with (1), (2) and (3) and find particular values of k , a and $f(x)$.

Step-III. Substitute the particular values of k , a and $f(x)$ in (5) and (6) to get E_n and C_n^2

Step IV. Substitute the values of coefficients E_n and C_n^2 obtained in step III in (4) to arrive at the desired solution of the given boundary value problem.

Example: Solve the one-dimensional diffusion equation $\frac{\partial^2 u}{\partial x^2} = \left(\frac{1}{k}\right) \left(\frac{\partial u}{\partial t}\right)$ in the range $0 \leq x \leq 2\pi$, $t \geq 0$ subject to the boundary conditions: $u(x, 0) = \sin^3 x$ for $0 \leq x \leq 2\pi$ and $u(0, t) = u(2\pi, t) = 0$ for $t \geq 0$

Solution: Proceed upto equation (18) as in previous example by taking $a = 2\pi$ and $f(x) = \sin^3 x$. Then, equation (18) for the present problem reduces to

$$u(x, t) = \sum_{n=1}^{\infty} E_n \sin\left(\frac{nx}{2}\right) e^{-\frac{(n^2 k)}{4} t}, \quad n = 1, 2, 3, \dots \quad \dots(i)$$

Putting $t = 0$ in (i) and using given condition $u(x, 0) = \sin^3 x$, we get

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$$\sum_{n=1}^{\infty} E_n \sin\left(\frac{nx}{2}\right) = \sin^3 x = \frac{3}{4} \sin x - \frac{1}{4} \sin 3x$$

$$\left[\therefore \sin 3x = 3 \sin x - 4 \sin^3 x \Rightarrow \sin^3 x = \left(\frac{1}{4}\right) \times (3 \sin x - \sin 3x) \right]$$

$$\begin{aligned} \text{Or } E_1 \sin\left(\frac{x}{2}\right) + E_2 \sin x + E_3 \sin\left(\frac{3x}{2}\right) + E_4 \sin 2x + E_5 \sin\left(\frac{5x}{2}\right) \\ + E_6 \sin 3x + E_7 \sin\left(\frac{7x}{2}\right) + \dots = \left(\frac{3}{4}\right) \times \sin x - \left(\frac{1}{4}\right) \times \sin 3x \quad \dots(ii) \end{aligned}$$

Equating the coefficients of the like terms on both sides of (ii), we get

$E_2 = \frac{3}{4}$, $E_6 = -\frac{1}{4}$ and $E_n = 0$ when $n \neq 2$ or $n \neq 6$. Substituting these values in (i), the required solution is

$$u(x, t) = E_2 \sin x e^{-kt} + E_6 \sin 3x e^{-9kt} = \left(\frac{1}{4}\right) \times [3 \sin x e^{-kt} - \sin 3x e^{-9kt}]$$

General Solution of Heat Equation When Both The Ends of A Bar are Insulated and The Initial Temperature is Prescribed.

If both the ends of a bar of length a are insulated and the initial temperature $f(x)$ is prescribed, then to find the temperature at a subsequent time t .

Solution: Here the temperature $u(x, t)$ in the given bar is governed by one dimensional heat equation

$$k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t} \quad \dots(1)$$

Physical experiments show that the rate of heat flow is proportional to the gradient $\frac{\partial u}{\partial x}$ of the temperature $u(x, t)$. Hence if the ends $x = 0$ and $x = a$ of the bar are insulated, so that no heat can flow through the ends, we have

$$\text{the boundary conditions: } u_x(0, t) = u_x(a, t) = 0 \text{ for all } t \quad \dots(2)$$

$$\text{Also, the prescribed initial condition } \Rightarrow u(x, 0) = f(x) \text{ for all } x \quad \dots(3)$$

$$\text{Suppose that (1) has solutions of the form } u(x, t) = X(x)T(t), \quad \dots(4)$$

Where X is a function of x alone and T that of t alone.

Substituting this value of u in (1), we get

$$kX''T = XT' \text{ or } \frac{X''}{X} = \frac{T'}{kT} \quad \dots(5)$$

Since x and t are independent variables, (5) can be true if each side is equal to the same constant, say μ . Thus, (5) yields. $X'' - \mu X = 0$... (6)

$$\text{and } T' = \mu kT \quad \dots (7)$$

$$\text{Differentiating (4) partially w. r. t. 'x', we get } u_x(x, t) = X'(x)T(t) \quad \dots (8)$$

$$\text{Using (2), (8) gives } X'(0)T(t) = 0 \text{ and } X'(a)T(t) = 0 \quad \dots (9)$$

Since $T(t) = 0$ leads to $u = 0$, so assume that $T(t) \neq 0$. Hence, from (9), we get

$$X'(0) = 0 \text{ and } X'(a) = 0 \quad \dots (10)$$

Three cases arise:

$$\text{Case I. Let } \mu = 0. \text{ Then solution of (6) is } X(x) = Ax + B, \quad \dots (11)$$

$$\text{Which yields } X'(x) = A \quad \dots (12)$$

$$\text{Using B.C. (10), (12) gives } A = 0. \text{ Then (11) reduces to } X(x) = B.$$

Again, corresponding to $\mu = 0$, (7) yields

$$\frac{dT}{dt} = 0 \text{ So that } T = \text{constant} = \frac{E_0}{2B}, \text{ say}$$

$\therefore$ Corresponding to $\mu = 0$ a solution of the given boundary value problem from (4) is given by

$$u(x, t) = B \times \left(\frac{E_0}{2B} \right) = \frac{E_0}{2} \quad \dots (13)$$

Case II. Let $\mu = \lambda^2$, $\lambda \neq 0$ Then solution of (6) is

$$X(x) = Ae^{\lambda x} + Be^{-\lambda x} \quad \dots (14)$$

$$\text{Which yields } X'(x) = A\lambda e^{\lambda x} - B\lambda e^{-\lambda x} \quad \dots (15)$$

$$\text{Using B.C. (10), (15) gives } 0 = \lambda(A - B) \text{ and } 0 = \lambda(Ae^{a\lambda} - Be^{-a\lambda}).$$

$$\text{These equations yield } A = B = 0 \text{ so that } X(x) \equiv 0 \text{ and hence } u = 0,$$

Which does not satisfy (3). So we reject $\mu = \lambda^2$.

Case III. Let $\mu = -\lambda^2$, $\lambda \neq 0$. Then solution of (6) is

$$X(x) = A \cos \lambda x + B \sin \lambda x \quad \dots (16)$$

$$\text{Which yields } X'(x) = -A\lambda \sin \lambda x + B\lambda \cos \lambda x \quad \dots (16)'$$

Using B.C. (10), (16)' gives $0 = B\lambda$ and $0 = -A\lambda \sin \lambda a + B\lambda \cos \lambda a$. Thus, $B = 0$ and $A\lambda \sin \lambda = 0$. Since $\lambda \neq 0$ and $A \neq 0$ for a non-trivial solution, we must have

$$\sin \lambda a = 0 \text{ So that } \lambda a = n\pi$$

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$$\text{Thus, } \lambda = \frac{n\pi}{a}, \quad n=1,2,3,\dots \quad \dots(17)$$

Hence non-zero solutions $X_n(x)$ of (6) are given by

$$X_n(x) = A_n \cos\left(\frac{n\pi x}{a}\right), \text{ by (16) and (17)} \quad \dots(18)$$

Using (17), (7) reduces to

$$\frac{dT}{T} = -\frac{n^2 \pi^2 k}{a^2} dt, \text{ as } \mu = -\lambda^2 = -\frac{n^2 \pi^2}{a^2}$$

$$\text{or } \left(\frac{1}{T}\right) dT = -C_n^2 dt \quad \dots(19)$$

$$\text{Where } C_n^2 = \frac{(n^2 \pi^2 k)}{a^2} \quad \dots(20)$$

$$\text{Solving (19), } T_n(t) = D_n e^{-C_n^2 t} \quad \dots(21)$$

$$\therefore u_n(x, t) = X_n(X) T_n(t) = E_n \cos\left(\frac{n\pi x}{a}\right) e^{-C_n^2 t}, \text{ by (18) and (21)} \quad \dots(22)$$

are solutions of (1). For $n=1,2,3,\dots$, each one these satisfy the boundary conditions (2). Here $E_n (= A_n D_n)$ in another arbitrary constant.

Thus, (13) and (23) constitute a set of infinite solutions of (1). To obtain a solution also satisfying the initial condition (3), we consider a linear combination of these solutions. Hence the complete solution of (1) may be taken in the following form.

$$u(x, t) = \frac{E_0}{2} + \sum_{n=1}^{\infty} u_n(x, t) \text{ or } u(x, t) = \frac{E_0}{2} + \sum_{n=1}^{\infty} E_n \cos \frac{n\pi x}{a} e^{-C_n^2 t} \quad \dots(23)$$

Substituting $t=0$ in (23) and using (3), we have

$$f(x) = \frac{E_0}{2} + \sum_{n=1}^{\infty} E_n \cos \frac{n\pi x}{a}, \quad \dots(24)$$

Which is Fourier cosine series. So the constants E_0 and E_n are given by

$$E_0 = \frac{2}{a} \int_0^a f(x) dx \text{ and } E_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx, \quad n=1,2,3,\dots \quad \dots(25)$$

Hence (24) is the required solution where E_0 and E_n are given by (25)

Working Rule For Solving Heat Equation When Both The Ends of A Bar of Length a are Insulated and The Initial Temperature $f(x)$ is Prescribed.

Step-1. Solution of the heat equation

$$k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t} \quad \dots(1)$$

$$\text{Subject to the boundary conditions } u_x(0, t) = u_x(a, t) = 0 \text{ for all } t \quad \dots(2)$$

$$\text{and the initial condition } u(x, 0) = f(x), \text{ for all } x \quad \dots(3)$$

is given by $u(x,t) = \frac{E_0}{2} + \sum_{n=1}^{\infty} E_n \cos \frac{n\pi x}{a} e^{-C_n^2 t}$... (4)

Where $E_0 = \frac{2}{a} \int_0^a f(x) dx$, $E_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx$, $n=1, 2, 3, \dots$... (5)

and $C_n^2 = \frac{(n^2 \pi^2 k)}{a^2}$... (6)

Step-II. Compare the given problem with (1), (2) and (3) and find particular values of k, a and $f(x)$.

Step-III. Substitute the particular values of k, a and $f(x)$ in (5) and (6) and calculate E_0, E_1 and C_n^2 .

Step-IV. Substitute the values of coefficients E_0, E_n and C_n^2 obtained in step III in (4) to arrive at the desired solution of the given boundary value problem.

Example: Solve $k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t}$ for $0 < x < \pi, t > 0$, if $u_x(0,t) = u_x(\pi,t) = 0$ and $u(x,0) = \sin x$.

Solution: We can prove that the solution of heat equation

$k \left(\frac{\partial^2 u}{\partial x^2} \right) = \frac{\partial u}{\partial t}$... (1)

Subject to the boundary conditions $u_x(0,t) = u_x(\pi,t) = 0$ for all t ... (2)

and the initial condition $u(x,0) = f(x)$, $0 < x < \pi$... (3)

is given by $u(x,t) = \frac{E_0}{2} + \sum_{n=1}^{\infty} E_n \cos \frac{n\pi x}{a} e^{-C_n^2 t}$... (4)

Where $E_0 = \frac{2}{a} \int_0^a f(x) dx$, $E_n = \frac{2}{a} \int_0^a f(x) \cos \frac{n\pi x}{a} dx$, $n=1, 2, 3, \dots$... (5)

and $C_n^2 = \frac{(n^2 \pi^2 k)}{a^2}$... (6)

Comparing the given boundary value problem with the boundary value problem given by (1), (2) and (3), we have $k=k$, $a=\pi$ and $f(x)=\sin x$. Hence, from (5), we get

$E_0 = \frac{2}{\pi} \int_0^{\pi} \sin x dx = \frac{2}{\pi} [-\cos x]_0^{\pi} = \frac{2}{\pi} \times 2 = \frac{4}{\pi}$... (7)

and $E_n = \frac{2}{\pi} \int_0^{\pi} \sin x \cos nx dx = \frac{1}{\pi} \int_0^{\pi} \{ \sin(n+1)x - \sin(n-1)x \} dx$

$= \frac{1}{\pi} \left[-\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1} \right]_0^{\pi} = \frac{1}{\pi} \left[\frac{(-1)^{n+1}}{(n+1)} + \frac{(-1)^{n-1}}{n-1} + \frac{1}{n+1} - \frac{1}{n-1} \right]$

$$\text{Thus, } E_n = \frac{1}{\pi} \left[\frac{1 - (-1)^{n+1}}{n+1} + \frac{(-1)^{n-1} - 1}{n-1} \right] \quad \dots(8)$$

If $n = 2m - 1$ (odd) with $m = 1, 2, 3, \dots$, then $E_n = 0$, by (8)

If $n = 2m$ (even) with $m = 1, 2, 3, \dots$, then from (8), we get

$$E_{2m} = \frac{2}{m} \left(\frac{1}{2m+1} - \frac{1}{2m-1} \right) = -\frac{4}{\pi(4m^2 - 1)}, \quad m = 1, 2, 3 \quad \dots(9)$$

$$\text{Also, from (6), } C_{2m}^2 = \frac{\{(2m)^2 \times \pi^2 \times k\}}{\pi^2} = 4m^2 \pi^2$$

Substituting the values of E_0 and E_{2m} given by (7) and (9) in (4), the required solution is

$$u(x, t) = \frac{2}{\pi} - \frac{4}{\pi} \sum \frac{\cos 2mx}{4m^2 - 1} e^{-4m^2 kt}$$

6.5. Solution of The Wave Equation

$$\text{Assume that a solution of } \frac{\partial^2 y}{\partial x^2} = \left(\frac{1}{c^2} \right) \left(\frac{\partial^2 y}{\partial t^2} \right) \quad \dots(1)$$

is of the form

$y(x, t) = X(x)T(t)$ where X is a function of x and T is a function of t only.

$$\text{Then } \frac{\partial^2 y}{\partial t^2} = X \cdot T'' \text{ and } \frac{\partial^2 y}{\partial x^2} = X'' \cdot T$$

$$\text{Substituting these in (1), we get } XT'' = c^2 X''T \text{ i.e. } \frac{X''}{X} = \frac{1}{c^2} \frac{T''}{T} \quad \dots(2)$$

Clearly the left side of (2) is a function of x only and the right side is a function of t only. Since x and t are independent variables (2) can hold good if each side is equal to a constant k (say). Then (2) leads to the ordinary differential equations:

$$\frac{d^2 X}{dx^2} - kX = 0 \quad \dots(3)$$

$$\text{and } \frac{d^2 T}{dt^2} - kc^2 T = 0 \quad \dots(4)$$

Solving (3) and (4), we get

$$(i) \text{ When } k \text{ is positive and } = p^2, \text{ say } X = c_1 e^{px} + c_2 e^{-px}; T = c_3 e^{cpt} + c_4 e^{-cpt}.$$

$$(ii) \text{ When } k \text{ is negative and } = -p^2 \text{ say. } X = c_5 \cos px + c_6 \sin px; T = c_7 \cos cpt + c_8 \sin cpt.$$

$$(iii) \text{ When } k \text{ is zero } X = c_9 x + c_{10}; T = c_{11} t + c_{12}.$$

Thus the various possible solutions of wave equation (1) are

$$y = (c_1 e^{px} + c_2 e^{-px})(c_3 e^{cpt} + c_4 e^{-cpt}) \quad \dots(5)$$

$$y = (c_5 \cos px + c_6 \sin px)(c_7 \cos cpt + c_8 \sin cpt) \quad \dots(6)$$

$$y = (c_9 x + c_{10})(c_{11} t + c_{12}) \quad \dots(7)$$

Of these three solutions, we have to choose that solution which is consistent with the physical nature of the problem. As we will be dealing with problems on vibrations, y must be a periodic function of x and t . Hence their solution must involve trigonometric terms. According the solution given by (6), i.e. of the form

$$y = (C_1 \cos px + C_2 \sin px)(C_3 \cos cpt + C_4 \sin cpt) \quad \dots(8)$$

is the only suitable solution of the wave equation?

Example: A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $y = a \sin(\pi x / l)$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by

$$y(x, t) = a \sin(\pi x / l) \cos(\pi c t / l)$$

Solution: The vibration of the string is given by $\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2} \quad \dots(i)$

As the end points of the string are fixed, for all time,

$$y(0, t) = 0 \quad \dots(ii)$$

$$\text{and } y(l, t) = 0 \quad \dots(iii)$$

Since the initial transverse velocity of any point of the string is zero,

$$\text{Therefore, } \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0 \quad \dots(iv)$$

$$\text{Also } y(x, 0) = a \sin(\pi x / l) \quad \dots(v)$$

Now we have to solve (i) subject to the boundary conditions (ii) and (iii) and initial conditions (iv) and (v). Since the vibration of the string is periodic, therefore, the solution of (i) is of the form

$$y(x, t) = (C_1 \cos px + C_2 \sin px)(C_3 \cos cpt + C_4 \sin cpt) \quad \dots(vi)$$

$$\text{By (ii), } y(0, t) = C_1 (C_3 \cos cpt + C_4 \sin cpt) = 0$$

$$\text{For this to be true for all time, } C_1 = 0 \quad \dots(vii)$$

$$\text{and } \frac{\partial y}{\partial t} = C_2 \sin px \{ C_3 (-cp \sin cpt) + C_4 (cp \cos cpt) \}$$

$$\therefore \text{By (iv), } \left(\frac{\partial y}{\partial t} \right)_{t=0} = C_2 \sin px (C_4 cp) = 0, \text{ whence } C_2 C_4 cp \sin px = 0$$

If $C_2 = 0$, (vii) will lead to the trivial solution $y(x, t) = 0$

∴ The only possibility is that $C_4 = 0$

Thus (vii) becomes $y(x, t) = C_2 C_3 \sin px \cos cpt$... (viii)

∴ By (iii), $y(l, t) = C_2 C_3 \sin pl \cos cpt = 0$ for all t .

Since C_2 and $C_3 \neq 0$, we have $\sin pl = 0$. ∴ $pl = n\pi$, i.e. $p = n\pi/l$, where n is an integer.

Hence (i) reduces to $y(x, t) = C_2 C_3 \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}$.

[These are the solutions of (i) satisfying the boundary conditions. These functions are called the eigen function corresponding to the eigen values $l_n = n\pi/l$ of the vibrating string. The set of values $l_1, l_2, l_3, \dots$ is called is spectrum].

Finally, imposing the last condition (v), we have $y(x, 0) = C_2 C_3 \sin \frac{n\pi x}{l} = a \sin \frac{\pi x}{l}$ which will be satisfied by taking $C_2 C_3 = a$ and $n = 1$

Hence the required solution is $y(x, t) = a \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l}$

Working Rule for Solving One-Dimensional Wave Equation Satisfying the Given Boundary and Initial Conditions

Step-I. The given solution of the wave equation $\frac{\partial^2 y}{\partial x^2} = \left(\frac{1}{c^2}\right) \times \left(\frac{\partial^2 y}{\partial t^2}\right)$... (1)

subject to the boundary conditions $y(0, t) = y(a, t) = 0$ for all t ... (2)

and the initial conditions $y(x, 0) = f(x)$, $y_t(x, 0) = g(x)$ $0 \leq x \leq a$... (3)

is given by $y(x, t) = \sum_{n=1}^{\infty} \left(E_n \cos \frac{n\pi ct}{a} + F_n \sin \frac{n\pi ct}{a} \right) \sin \frac{n\pi x}{a}$... (4)

where $E_n = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$ and $F_n = \frac{2}{n\pi c} \int_0^a g(x) \sin \frac{n\pi x}{a} dx$... (5)

Step-II. Compare the given problem with (1), (2) and (3) and find particular values of c , a , $f(x)$ and $g(x)$.

Step-III. Substitute the particular values of a , $f(x)$ and $g(x)$ in (5) and compute E_n and F_n .

Step-IV. Substitute the values of E_n and F_n in (4) to arrive at the desired solution of the given boundary value problem.

Working rule for solving One-Dimensional Wave equation when both the ends of the string of length l are fixed and they initial velocity of the string is zero.

First of all, as explained in working rule, prove that the solution of one-

dimensional wave equation $\frac{\partial^2 y}{\partial x^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial^2 y}{\partial t^2}\right)$... (i)

subject to boundary conditions $y(0,t) = y(l,t) = 0$ for all t ... (ii)

and $\left. \begin{aligned} \text{initial velocity} &= \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0, 0 < x < l \\ \text{initial deflection} &= y(x,0) = f(x), 0 < x < l \end{aligned} \right\}$... (iii)

is $y(x,t) = \sum_{n=1}^{\infty} E_n \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}$... (iv)

Where $E_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$ (v)

Example: A string of length l has its ends $x=0$ and $x=l$ fixed. It is released from rest in the position $y = \frac{4\lambda x(1-x)}{l^2}$. Find an expression for the displacement of the string at any subsequent time.

The displacement function $y(x,t)$ is the solution of the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \left(\frac{1}{c^2} \right) \left(\frac{\partial^2 y}{\partial t^2} \right) \quad \dots (1)$$

subject to boundary conditions : $y(0,t) = y(l,t) = 0$ for all $t \geq 0$ (2)

and initial conditions, namely,

Initial velocity $= \left(\frac{\partial y}{\partial t} \right)_{t=0} = 0$ for $0 \leq x \leq l$ 3(a)

and initial displacement $= y(x,0) = f(x) = \frac{\{4\pi x(l-x)\}}{l}$ 3(b)

Comparing the given problem given by (1), (2), 3(a) and 3(b) with the boundary value problem given by (i), (ii) and (iii), the solution of (1) satisfying the above boundary and initial conditions is given by

$y(x,t) = \sum_{n=1}^{\infty} E_n \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}$, by (iv) ... (4)

Where $E_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$, by (v) ... (5)

Substituting the value of $f(x)$ given by 3(a) in (5), we get

$$\begin{aligned} E_n &= \left(\frac{2}{l} \right) \times \left(\frac{4\lambda}{l^2} \right) \int_0^l (lx - x^2) \sin \frac{n\pi x}{l} dx. \\ &= \frac{8\lambda}{l^3} \left[(lx - x^2) \left(-\frac{l}{n\pi} \cos \frac{n\pi x}{l} \right) - (l - 2x) \left(-\frac{l^2}{n^2 \pi^2} \sin \frac{n\pi x}{l} \right) + (-2) \left(\frac{l^3}{n^3 \pi^2} \cos \frac{n\pi x}{l} \right) \right]_0^l \end{aligned}$$

[Using the chain rule of integrating by parts]

$$= \begin{cases} 0, & \text{if } n = 2m (\text{even}) \text{ with } m = 1, 2, 3, \dots \\ (32\lambda) / (2m-1)^3 \pi^3, & \text{if } n = 2m-1 (\text{odd}) \text{ with } m = 1, 2, 3, \dots \end{cases}$$

Put Your Own Notes

Substituting the value of E_n in (4), the required expression is

$$y(x, t) = \frac{32\lambda}{\pi^3} \sum_{m=1}^{\infty} \frac{1}{(2m-1)^3} \sin \frac{(2m-1)\pi x}{l} \cos \frac{(2m-1)\pi ct}{l}$$

6.6. Solution of Laplace's Equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \dots(1)$$

Let $u(x, y) = X(x)Y(y)$ be a solution of (1).

Substituting it in (1), we get $\frac{d^2 X}{dx^2} Y + X \frac{\partial^2 Y}{\partial y^2} = 0$

$$\text{or separating the variables, } \frac{1}{X} \frac{d^2 X}{dx^2} = -\frac{1}{Y} \frac{d^2 Y}{dy^2} \quad \dots(2)$$

Since x and y are independent variables, (2) can hold good only if each side of (2) is equal to a constant k (say). Then (2) leads to the ordinary differential equations

$$\frac{d^2 X}{dx^2} - kX = 0 \text{ and } \frac{d^2 Y}{dy^2} + kY = 0.$$

Solving these equations, we get

- (i) When k is positive and is equal to p^2 , say
 $X = c_1 e^{px} + c_2 e^{-px}$, $Y = c_3 \cos py + c_4 \sin py$
- (ii) When k is negative, and is equal to $-p^2$, say
 $X = c_5 \cos px + c_6 \sin px$, $Y = c_7 e^{py} + c_8 e^{-py}$
- (iii) When k is zero; $X = c_9 x + c_{10}$, $Y = c_{11} y + c_{12}$.

Thus the various possible solutions of (1) are
 $u = (c_1 e^{px} + c_2 e^{-px})(c_3 \cos py + c_4 \sin py) \quad \dots(3)$

$$u = (c_5 \cos px + c_6 \sin px)(c_7 e^{py} + c_8 e^{-py}) \quad \dots(4)$$

$$u = (c_9 x + c_{10})(c_{11} y + c_{12}) \quad \dots(5)$$

Of these we take that solution which is consistent with the given boundary conditions.

6.6.1 General Solution of Two-Dimensional Laplace's Equation

$$\text{Consider } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0. \quad \dots(1)$$

$$\text{Suppose that (1) has solutions of the form } u(x, y) = X(x)Y(y) \quad \dots(2)$$

Where X and Y are functions of x and y respectively.

$$\text{From (2) } \frac{\partial^2 u}{\partial x^2} = X''Y \text{ and } \frac{\partial^2 u}{\partial y^2} = XY'' \quad \dots(3)$$

Since the L.H.S. of (3) depends only on x and the R.H.S depends only on y , both sides of (3) must be equal to same constant, say μ . This leads to two ordinary differential equations

$$X'' - \mu X = 0 \text{ and } Y'' + \mu Y = 0, \quad \dots(4)$$

whose solutions depends only on the value of μ . Three cases arise:

Case I. When $\mu = 0$. Then (4) reduces to $X'' = 0$ and $Y'' = 0$.

Solving these $X = A_1x + B_1$ and $Y = C_1y + D_1$

$$\text{Then a solution of (1) is } u(x, y) = (A_1x + B_1)(C_1y + D_1). \quad \dots(5)$$

Case II. When $\mu = \lambda^2$, i.e., positive. Here $\lambda \neq 0$. Then (4) reduces to

$$X'' - \lambda^2 X = 0 \text{ and } Y'' + \lambda^2 Y = 0.$$

Solving these, $X = A_2e^{\lambda x} + B_2e^{-\lambda x}$ and $Y = C_2 \cos \lambda y + D_2 \sin \lambda y$.

or, alternatively, $X = A_2 \cosh \lambda x + B_2 \sinh \lambda x$, $Y = C_2 \cos \lambda y + D_2 \sin \lambda y$

$$\text{Then a solution of (1) is } u(x, y) = (A_2e^{\lambda x} + B_2e^{-\lambda x})(C_2 \cos \lambda y + D_2 \sin \lambda y) \quad \dots(6)$$

$$\text{or alternatively, } u(x, y) = (A_2 \cosh \lambda x + B_2 \sinh \lambda x)(C_2 \cos \lambda y + D_2 \sin \lambda y) \quad \dots(6)'$$

Case III. When $\mu = -\lambda^2$, i.e., negative. Here $\lambda \neq 0$. Then (4) reduces to

$$X'' + \lambda^2 X = 0 \text{ and } Y'' - \lambda^2 Y = 0.$$

Solving these, $X = A_3 \cos \lambda x + B_3 \sin \lambda x$ and $Y = C_3e^{\lambda y} + D_3e^{-\lambda y}$.

or alternatively, $X = A_3 \cos \lambda x + B_3 \sin \lambda x$ and $Y = C_3 \cosh \lambda y + D_3 \sinh \lambda y$

Then a solution of (1) is

$$u(x, y) = (A_3 \cos \lambda x + B_3 \sin \lambda x)(C_3e^{\lambda y} + D_3e^{-\lambda y}). \quad \dots(6)$$

$$\text{or alternatively, } u(x, y) = (A_3 \cos \lambda x + B_3 \sin \lambda x)(C_3 \cosh \lambda y + D_3 \sinh \lambda y) \quad \dots(7)'$$

Out of the above mentioned three types of solutions (5), (6) and (7), we must select an appropriate solution which suits the physical nature of the problem and given boundary conditions.

6.6.2 Dirichlet Problem in A Rectangle

1st Case . The Dirichlet problem in a rectangle is defined as follows:

$$\text{Laplace's equation: } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad 0 \leq x \leq a, \quad 0 \leq y \leq b \quad \dots(1)$$

$$\text{Boundary conditions: } u(0, y) = 0, \quad u(a, y) = 0 \quad 0 \leq y \leq b \quad \dots(2)$$

$$u(x, b) = 0, \quad 0 \leq x \leq a \quad \dots 3(a) \text{ and } u(x, 0) = f(x), \quad 0 \leq x \leq a \quad \dots 3(b)$$

We shall solve (1) subject to the boundary conditions (2), 3(a) and 3(b).

Suppose (1) has a solution of the form

$$u(x, y) = X(x)Y(y) \quad \dots(4)$$

Substituting this value of u in (1), we get

$$X''Y + XY'' = 0 \text{ or } \frac{X''}{X} = -\frac{Y''}{Y}. \quad \dots(5)$$

Since x and y are independent, variables each side of (5) must be equal to the same constant, say μ ,

Then (5) gives $X'' - \mu X = 0$... (6)

and $Y'' + \mu Y = 0$... (7)

Using (2), (4) gives $X(0)Y(y) = 0$ and $X(a)Y(y) = 0$

giving $X(0) = 0$ and $X(a) = 0$... (8)

Where we have taken $Y(y) \neq 0$, since otherwise $u = 0$ which does not satisfy 3(b).

We now solve (6) under B.C. (8). Three cases arise:

Case I. Let $\mu = 0$. Then solution of (6) is $X(x) = Ax + B$... (9)

Using B.C. (8), (9) gives $0 = B$ and $0 = Aa + B$. These give $A = B = 0$ so that $X(x) \equiv 0$. This leads to $u \equiv 0$ which does not satisfy 3(b). So we reject $\mu = 0$.

Case II. Let $\mu = \lambda^2$, $\lambda \neq 0$. Then solution of (6) is $X(x) = Ae^{\lambda x} + Be^{-\lambda x}$... (10)

Using B.C. (8), (10) gives $0 = A + B$ and $0 = Ae^{a\lambda} + Be^{-a\lambda}$... (11)

Solving (11), $A = B = 0$ so that $X(x) = 0$ and hence $u \equiv 0$. So we reject $\mu = \lambda^2$.

Case III. Let $\mu = -\lambda^2$, $\lambda \neq 0$. Solution of (6) is

$X(x) = A \cos \lambda x + B \sin \lambda x$... (12)

Using B.C. (8), (12) gives $0 = A$ and $0 = -A \cos \lambda a + B \sin \lambda a$ so that

$A = 0$ and $\sin \lambda a = 0$,

Where we have taken $B \neq 0$, since otherwise $X(x) = 0$ and hence $u = 0$ which does not satisfy 3(b).

Now, $\sin \lambda a = 0 \Rightarrow \lambda a = n\pi \Rightarrow \lambda = \frac{n\pi}{a}$, $n = 1, 2, 3, \dots$... (13)

Hence non-zero solution $X_n(x)$ of (6) are given by

$X_n(x) = B_n \sin\left(\frac{n\pi x}{a}\right)$... (14)

Using $\mu = -\lambda^2 = -\frac{n^2\pi^2}{a^2}$ (7) reduces to $Y'' - \left(\frac{n^2\pi^2}{a^2}\right)Y = 0$... (15)

Whose solution is $Y_n(y) = C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}}$... (16)

Using 3(a), (4) gives $0 = X(x)Y(b)$ so that $Y(b) = 0$, where we have taken $X(x) \neq 0$, since otherwise $u = 0$ which does not satisfy 3(b).

Again, $Y(b) = 0 \Rightarrow Y_n(b) = 0$... (17)

Putting $y = b$ in (16) and using (17), we have

$0 = C_n e^{n\pi b/a} + D_n e^{-n\pi b/a}$ so that $C_n = -\frac{D_n e^{-n\pi b/a}}{e^{n\pi b/a}}$... (18)

Putting this value of C_n given by (18) in (16), we get

$$Y_n(y) = D_n \frac{\left(e^{-\frac{n\pi y}{a}} e^{\frac{n\pi b}{a}} - e^{-\frac{n\pi y}{a}} e^{-\frac{n\pi b}{a}} \right)}{e^{\frac{n\pi b}{a}}} \quad \text{or} \quad Y_n(y) = D_n \frac{\left(e^{\frac{n\pi(b-y)}{a}} - e^{-\frac{n\pi(b-y)}{a}} \right)}{e^{\frac{n\pi b}{a}}}$$

$$\text{or } Y_n(y) = 2D_n e^{-\frac{n\pi b}{a}} \sinh \left\{ \frac{n\pi(b-y)}{a} \right\}, \quad \text{as } e^{\theta} - e^{-\theta} = 2 \sinh \theta$$

$$\therefore u_n(x, y) = X_n(x) Y_n(y) = F_n \sin \left(\frac{n\pi x}{a} \right) \sinh \left\{ \frac{n\pi(b-y)}{a} \right\}$$

are solutions of (1), satisfying (2) and 3(a). Here $F_n \left(= 2B_n D_n e^{-\frac{n\pi b}{a}} \right)$ are new arbitrary constants. In order to obtain a solution also satisfying 3(b), we consider more general solution, namely,

$$u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} F_n \sin \left(\frac{n\pi x}{a} \right) \sinh \left\{ \frac{n\pi(b-y)}{a} \right\} \quad \dots(19)$$

Putting $y=0$ in (20) and using 3(b), we have

$$f(x) = \sum_{n=1}^{\infty} \left\{ F_n \sinh \left(\frac{n\pi b}{a} \right) \right\} \sin \left(\frac{n\pi x}{a} \right),$$

Which is the half range Fourier sine series of $f(x)$ in $(0, a)$. Hence, we get

$$F_n \sinh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$$

$$\text{or } F_n = \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \int_0^a f(x) \sin \frac{n\pi x}{a} dx \quad \dots(20)$$

Example: Find the steady state temperature distribution in a rectangular plate of sides a and b insulated at the lateral surface and satisfying the boundary conditions

$$u(0, y) = u(a, y) = 0 \quad \text{for } 0 \leq y \leq b$$

$$\text{and } u(x, b) = 0 \quad \text{and } u(x, 0) = x(a-x) \quad \text{for } 0 \leq x \leq a$$

Solution: First proceed as in above article upto equation (20)

For the present problem, $u(x, 0) = f(x) = x(a-x)$, $0 \leq x \leq a$

$$\therefore \Rightarrow F_n = \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \int_0^a f(x) \sin \frac{n\pi x}{a} dx = \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \int_0^a (ax - x^2) \sin \frac{n\pi x}{a} dx \quad \dots(21)$$

$$= \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \left[(ax - x^2) \left(-\frac{a}{n\pi} \right) \cos \frac{n\pi x}{a} - (a-2x) \left(-\frac{a^2}{n^2 \pi^2} \right) \sin \frac{n\pi x}{a} + (-2) \left(\frac{a^3}{n^3 \pi^3} \right) \cos \frac{n\pi x}{a} \right]_0^a$$

[Using chain rule of integration by parts]

$$= \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \left[-\frac{2a^3(-1)^n}{n^3 \pi^3} + \frac{2a^3}{n^3 \pi^3} \right] = \frac{4a^3}{n^3 \pi^3} [1 - (-1)^n] \operatorname{cosech} \frac{n\pi b}{a}$$

$$= \begin{cases} 0, & \text{if } n = 2m \text{ (even) and } m = 1, 2, 3, \dots \\ \{8a^2 / \pi^3 (2m-1)^3\} \operatorname{cosech} \{(2m-1)\pi b / a\}, & \text{if } n = 2m-1 \text{ and } m = 1, 2, 3, \dots \end{cases}$$

Substituting the above value of F_n in (21), the required steady temperature $u(x, y)$ is given by

$$u(x, y) = \frac{8a^2}{\pi^3} \sum_{m=1}^{\infty} \frac{1}{(2m-1)^3} \sin \frac{(2m-1)\pi x}{a} \sinh \frac{(2m-1)(b-y)\pi}{a} \operatorname{cosech} \frac{(2m-1)\pi b}{a}$$

2nd Case

The steady state temperature distribution in a rectangular plate of sides a and b insulated at the lateral surface and satisfying the boundary conditions.

$$u(0, y) = u(a, y) = 0 \text{ for } 0 \leq y \leq b \text{ and } u(x, 0) = 0 \text{ and } u(x, b) = f(x) \text{ for } 0 \leq x \leq a$$

Solution: The heat flow in a body for a two-dimensional case is governed by

$$\text{the equation } \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \left(\frac{1}{c^2}\right) \left(\frac{\partial u}{\partial t}\right). \text{ When the steady state condition}$$

prevails, temperature u is independent of t so that $\frac{\partial u}{\partial t} = 0$. Then, the above equation reduces to Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \dots(1)$$

$$\text{Given that } u(0, y) = u(a, y) = 0 \text{ for } 0 \leq y \leq b \quad \dots(2)$$

$$\text{Also, given that } u(x, 0) = 0 \text{ for } 0 \leq x \leq a \quad \dots(3(a))$$

$$\text{and } u(x, b) = f(x) \text{ for } 0 \leq x \leq a \quad \dots(3(b))$$

Suppose (1) has a solution of the form

$$u(x, y) = X(x)Y(y) \quad \dots(4)$$

Substituting this value of u in (1), we get

$$X''Y + XY'' = 0 \text{ or } \frac{X''}{X} = -\frac{Y''}{Y} \quad \dots(5)$$

Since x and y are independent variables, each side of (5) must be equal to the constant, say μ . Then (5) gives $X'' - \mu X = 0$ $\dots(6)$

$$\text{and } Y'' + \mu Y = 0 \quad \dots(7)$$

Using (2), (4) gives $X(0)Y(y) = 0$ and $X(a)Y(y) = 0$

$$\text{giving } X(0) = 0 \text{ and } X(a) = 0, \quad \dots(8)$$

Where we have taken $Y(y) \neq 0$, since otherwise $u \equiv 0$ which does not satisfy (3(b)).

We now solve (6) under B.C. (8). Three cases arise:

$$\text{Case I. Let } \mu = 0. \text{ solution of (6) is } X(x) = Ax + B \quad \dots(9)$$

Using B.C. (8),(9) gives $0 = B$ and $0 = Aa + B$. these give $A = B = 0$ so that $X(x) = 0$. This leads to $u \equiv 0$ which does not satisfy (3)b. So we reject $\mu = 0$.

Case II. Let $\mu = \lambda^2, \lambda \neq 0$ Then solution of (6) is

$$X(x) = Ae^{\lambda x} + Be^{-\lambda x} \quad \dots(10)$$

$$\text{Using B.C. (8), (10) gives } A + B = 0 \text{ and } 0 = e^{a\lambda} + Be^{-a\lambda} \quad \dots(11)$$

Solving (11), $A = B = 0$ so that $X(x) = 0$ and hence $u \equiv 0$.

So we reject $\mu = \lambda^2$.

Case III. Let $\mu = -\lambda^2, \lambda \neq 0$. Then solution of (6) is

$$X(x) = A \cos \lambda x + B \sin \lambda x \quad \dots(12)$$

Using B.C. (8),(12) gives $0 = A$ and $0 = A \cos \lambda a + B \sin \lambda a$

so that $A = 0$ and $\sin \lambda a = 0$

Where we have taken $B \neq 0$, since otherwise $X(x) \equiv 0$ and hence $u \equiv 0$ which does not satisfy 3(b).

Now, $\sin \lambda a = 0$ gives $\lambda a = n\pi, n = 1, 2, 3$

$$\therefore \lambda = \frac{n\pi}{a}, n = 1, 2, 3, \dots \quad \dots(13)$$

Hence non-zero solutions $X_n(x)$ of (6) are

$$\text{given by } X_n(x) = B_n \sin\left(\frac{n\pi x}{a}\right) \quad \dots(14)$$

$$\text{Using } \mu = -\lambda^2 = -\frac{n^2\pi^2}{a^2}, (7) \text{ becomes } Y'' - \left(\frac{n^2\pi^2}{a^2}\right)Y = 0 \quad \dots(15)$$

$$\text{Whose general solution is } Y_n(y) = C_n e^{\frac{n\pi y}{a}} + D_n e^{-\frac{n\pi y}{a}} \quad \dots(16)$$

Using 3(a), (4) gives $0 = X(x) Y(0)$ so that $Y(0) = 0$, where we have taken $X(x) \neq 0$, for otherwise we shall get $u = 0$ which does not satisfy 3(b).

But $Y(0) = 0$

$$\Rightarrow Y_n(y) = 0 \quad \dots(17)$$

Putting $y = 0$ in (16) and using (17), we have

$$0 = C_n + D_n \text{ so that } D_n = -C_n.$$

Then (6) reduces to

$$Y_n(y) = C_n \left(e^{\frac{n\pi y}{a}} - e^{-\frac{n\pi y}{a}} \right) = 2 \sinh\left(\frac{n\pi y}{a}\right),$$

$$\text{as } e^\theta - e^{-\theta} = 2 \sinh \theta. \quad \dots(18)$$

$$\therefore u_n(x, y) = X_n(x) Y_n(y) = E_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right) \quad \dots(19)$$

are solutions of (1), satisfying (2) and 3(a), we now consider more general solution given by

$$\text{i.e., } u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} E_n \sin \frac{n\pi x}{a} \sinh \frac{n\pi y}{a} \quad \dots(20)$$

Putting $y = b$ in (20) and using 3(b), we get

$$f(x) = \sum_{n=1}^{\infty} \left(E_n \sinh \frac{n\pi b}{a} \right) \sin \frac{n\pi x}{a},$$

Which is the half range Fourier sine series of $f(x)$ in $(0, a)$.

Hence, we have

$$E_n \sinh \frac{n\pi b}{a} = \frac{2}{a} \int_0^a f(x) \sin \frac{n\pi x}{a} dx$$

$$\Rightarrow E_n = \frac{2}{a \sinh \left(\frac{n\pi b}{a} \right)} \int_0^a f(x) \sin \frac{n\pi x}{a} dx \quad \dots(21)$$

Hence, (20) is the required solution wherein E_n is given by (21).

3rd Case

The Laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the following boundary conditions: $u(x, 0) = u(x, b) = 0$ for $0 \leq x \leq a$, $u(0, y) = 0$ and $u(a, y) = f(y)$ for $0 \leq y \leq b$.

Solution: Given Laplace's equation is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (1)

Also given $u(x, 0) = u(x, b) = 0$ for $0 \leq x \leq a$, ... (2)

$u(0, y) = 0$ for $0 \leq y \leq b$ (3(a))

and $u(a, y) = f(y)$ for $0 \leq y \leq b$ (3(b))

Let a solution of (1) be of the form $u(x, y) = X(x)Y(y)$ (4)

Using (4), (1) reduces to

$$X''Y + XY'' = 0 \text{ or } \left(\frac{1}{X} \right) X'' = - \left(\frac{1}{Y} \right) Y'' \quad \dots(5)$$

Since the L.H.S. of (5) depends only on x and the R.H.S. depends only on y , each side of (5) must be equal to the same constant, say μ . Then, (5) leads to

$$X'' - \mu X = 0 \quad \dots(6)$$

$$\text{and } Y'' + \mu Y = 0 \quad \dots(7)$$

Using (2), (4) gives $X(x)Y(0) = 0$ and $X(x)Y(b) = 0$

So that $Y(0) = 0$ and $Y(b) = 0$... (8)

where we have taken $X(x) \neq 0$, since otherwise $u = 0$ does not satisfy (3b).

We now solve (7) under boundary conditions (8). Three cases arise:

Case I. Let $\mu = 0$. Then, solution of (7) is $Y(y) = Ay + B$ (9)

Using B.C. (8), (9) gives $0 = B$ and $0 = Ab + B$. These give $A = B = 0$ so that $Y(y) \equiv 0$. This leads to $u = 0$ which does not satisfy 3(b). So we reject $\mu = 0$.

Case II. Let $\mu = -\lambda^2$ (negative). Here $\lambda \neq 0$. Then solution of (7) is

$$Y(y) = Ae^{\lambda y} + Be^{-\lambda y} \quad \dots(10)$$

$$\text{Using B.C. (8), (10) gives } 0 = A + B \text{ and } 0 = Ae^{b\lambda} + Be^{-b\lambda}. \quad \dots(11)$$

(11) $\Rightarrow A = B = 0$ so that $X(x) = 0$. This leads to $u = 0$ which does not satisfy 3(b). So we reject $\mu = -\lambda^2$.

Case III. Let $\mu = \lambda^2$ (positive). Here $\lambda \neq 0$. Solution of (7) is

$$Y(y) = A \cos \lambda y + B \sin \lambda y. \quad \dots(12)$$

Using B.C. (8), (12) gives $0 = A$ and $0 = A \cos \lambda b + B \sin \lambda b$ so that $A = 0$ and $\sin \lambda b = 0$, where we have taken $B \neq 0$, since otherwise we shall get $Y(y) \equiv 0$. This leads to $u = 0$ which does not satisfy 3(b).

$$\text{Now, } \sin \lambda b = 0 \Rightarrow \lambda b = n\pi \Rightarrow \lambda = \frac{n\pi}{b}, \quad n = 1, 2, 3, \dots$$

Hence, non-zero solutions $Y_n(y)$ of (7) are given by

$$Y_n(y) = B_n \sin\left(\frac{n\pi y}{b}\right), \quad n = 1, 2, 3, \dots \quad \dots(13)$$

Also, then $\mu = \lambda^2 = \frac{n^2 \pi^2}{b^2}$ hence, (6) reduces to

$$X'' - \left(\frac{n^2 \pi^2}{b^2}\right) X = 0, \quad n = 1, 2, 3, \dots \quad \dots(14)$$

$$\text{Whose solutions are } X_n(x) = C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}}. \quad \dots(15)$$

Using 3(a), (4) gives $0 = X(0)Y(y)$ so that $X(0) = 0$, where we have taken $Y(y) \neq 0$, since otherwise $u = 0$ which does not satisfy 3(b).

Now $X(0) = 0 \Rightarrow X_n(0) = 0$. Putting $x = 0$ in (15) and using $X_n(0) = 0$, we get

$$0 = C_n + D_n \text{ so that } D_n = -C_n. \text{ Then (15) reduces to}$$

$$X_n(x) = C_n \left(e^{\frac{n\pi x}{b}} - e^{-\frac{n\pi x}{b}} \right) = 2C_n \sinh\left(\frac{n\pi x}{b}\right), \text{ as } e^\theta - e^{-\theta} = 2 \sinh \theta \quad \dots(16)$$

From (4), (13) and (16), we see that a solution $u_n(x, y)$ of (1) is given by

$$u_n(x, y) = X_n(x)Y_n(y) = E_n \sinh\left(\frac{n\pi x}{b}\right) \sin\left(\frac{n\pi y}{b}\right), \quad n = 1, 2, 3, \dots \quad \dots(17)$$

Where $E_n (= B_n C_n)$ are new arbitrary constants. In order to obtain a solution also satisfying B.C. (3b), we consider more general solution

$$u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} E_n \sinh\left(\frac{n\pi x}{b}\right) \sin\left(\frac{n\pi y}{b}\right) \quad \dots(18)$$

Putting $x = a$ in (18) and using B.C. (3b), we have

$$f(y) = \sum_{n=1}^{\infty} E_n \sinh \frac{n\pi a}{b} \sin \frac{n\pi y}{b}$$

which is the half range Fourier sine series of $f(y)$ in $(0, b)$. Hence

$$E_n \sinh \frac{n\pi a}{b} = \frac{2}{b} \int_0^b f(y) \sin \frac{n\pi y}{b} dy$$

$$\text{Or } E_n = \frac{2}{b \sinh \left(\frac{n\pi a}{b} \right)} \int_0^b f(y) \sin \frac{n\pi y}{b} dy \quad \dots(19)$$

Hence, (18) is the required solution wherein E_n is given by (19).

4th Case

The laplace's equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ subject to the following boundary

condition $u(x, 0) = u(x, b) = 0$ for $0 \leq x \leq a$

$u(0, y) = f(y)$ and $u(a, y) = 0$ for $0 \leq y \leq b$

Solution: Given laplace's equation is $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \dots(1)$

Also given $u(x, 0) = u(x, b) = 0$ for $0 \leq x \leq a \quad \dots(2)$

$u(0, y) = f(y)$ for $0 \leq y \leq b \quad \dots 3(a)$

and $u(a, y) = 0$ for $0 \leq y \leq b \quad \dots 3(b)$

Let a solution of (1) be of the form

$$u(x, y) = X(x)Y(y) \quad \dots(4)$$

Using (4), (1) reduces to

$$X''Y + XY'' = 0 \text{ or } \frac{X''}{X} = -\frac{Y''}{Y} \quad \dots(5)$$

Since the L.H.S of (5) depends only on x and the R.H.S depends only y , each side of (5) must be equal to the same constant, say μ . Then, (5) leads to

$$X'' - \mu X = 0 \quad \dots(6)$$

$$\text{And } Y'' + \mu Y = 0 \quad \dots(7)$$

Using (2), (4) gives $X(x)Y(0) = 0$ and $X(x)Y(b) = 0$

$$\text{So that } Y(0) = 0 \text{ and } Y(b) = 0 \quad \dots(8)$$

Where we have taken $X(x) \neq 0$, since otherwise $u = 0$ does not satisfy (3b)

We now solve (7) under boundary conditions (8). Three cases arise.

Case I: Let $\mu = 0$. Then, solution of (7) is $Y(y) = Ay + B \quad \dots(9)$

Using B.C (8), (9) gives $0 = B$ and $Ab + B = 0$

This give $A = B = 0$ so that $Y(y) = 0$

This leads to $u = 0$ which does not satisfy $3(b)$. So we reject $\mu = 0$

Case II: Let $\mu = -\lambda^2$ (negative). Here $\lambda \neq 0$. Then solution of (7) is

$$Y(y) = Ae^{\lambda y} + Be^{-\lambda y} \quad \dots(10)$$

$$\text{Using B.C(8), (10) gives } 0 = A + B \text{ and } 0 = Ae^{\lambda b} + Be^{-\lambda b} \quad \dots(11)$$

(11) $\Rightarrow A = B = 0$ so that $X(x) = 0$. This leads to $u = 0$ which does not satisfy $3(b)$. so we reject $\mu = -\lambda^2$

Case III: Let $\mu = \lambda^2$ (positive). Here $\lambda \neq 0$. Then solution of (7) is

$$Y(y) = A \cos \lambda y + B \sin \lambda y \quad \dots(12)$$

Using B.C (8), (12) gives $0 = A$ and $0 = B \sin \lambda b$ so that $\sin \lambda b = 0$, where we have taken $B \neq 0$, since otherwise we shall get $Y(y) \equiv 0$.

This leads to $u = 0$ which does not satisfy $3(b)$.

Now, $\sin \lambda b = 0 \Rightarrow \lambda b = n\pi$

$$\Rightarrow \lambda = \frac{n\pi}{b}, \quad n = 1, 2, \dots$$

Hence, non zero solution $Y_n(y)$ of (7) are given by

$$y_n(y) = B_n \sin\left(\frac{n\pi y}{b}\right); \quad n = 1, 2, 3, \dots \quad \dots(13)$$

Also then $\mu = \lambda^2 = \frac{n^2 \pi^2}{b^2}$. Hence (6) reduces to

$$X'' - \left(\frac{n^2 \pi^2}{b^2}\right) X = 0; \quad n = 1, 2, \dots \quad \dots(14)$$

$$\text{whose solutions are } X_n(x) = C_n e^{\frac{n\pi x}{b}} + D_n e^{-\frac{n\pi x}{b}} \quad \dots(15)$$

Using $3(b)$, (4) gives $X(a)Y(y) = 0$ so that $X(a) = 0$ where we have taken $Y(y) \neq 0$, since otherwise $u = 0$ which does not satisfy $3(a)$.

Now, $X(a) = 0 \Rightarrow X_n(a) = 0$. Putting $x = a$ in (15) and using $X_n(a) = 0$, we get

$$C_n e^{\frac{n\pi a}{b}} + D_n e^{-\frac{n\pi a}{b}} = 0$$

$$\therefore C_n = \frac{-D_n e^{-\frac{n\pi a}{b}}}{e^{\frac{n\pi a}{b}}}$$

Then (15) reduces to

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$$X_n(x) = \frac{D_n}{e^{\frac{n\pi a}{b}}} \left\{ e^{\frac{n\pi}{b}(a-x)} - e^{-\frac{n\pi}{b}(a-x)} \right\}$$

$$= 2D_n e^{-\frac{n\pi a}{b}} \sinh \left(\frac{n\pi}{b}(a-x) \right), \text{ as } e^\theta - e^{-\theta} = 2 \sinh \theta \quad \dots(16)$$

From (4), (13) and (16) we see that a solution $u_n(x, y)$ of (1) is given by

$$u_n(x, y) = X_n(x)Y_n(y) = E_n \sinh \left(\frac{n\pi}{b}(a-x) \right) \sin \left(\frac{n\pi y}{b} \right) \quad n=1, 2, \dots \quad \dots(17)$$

Where $E_n = \left(2B_n D_n e^{-\frac{n\pi a}{b}} \right)$ are new arbitrary constants. In order to obtain

a solution also satisfying B.C 3(a), we consider more general solution

$$u(x, y) = \sum_{n=1}^{\infty} u_n(x, y) = \sum_{n=1}^{\infty} E_n \sinh \left(\frac{n\pi}{b}(a-x) \right) \sin \left(\frac{n\pi y}{b} \right) \quad \dots(18)$$

Putting $x=0$ in (18) and using B.C 3(a), we have

$$f(y) = \sum_{n=1}^{\infty} E_n \sinh \left(\frac{n\pi a}{b} \right) \sin \left(\frac{n\pi y}{b} \right)$$

Which is the half range fourier sine series of $f(y)$ in $(0, b)$. Hence

$$E_n \sinh \left(\frac{n\pi a}{b} \right) = \frac{2}{b} \int_0^b f(y) \sin \left(\frac{n\pi y}{b} \right) dy$$

$$E_n = \frac{2}{b \sinh \left(\frac{n\pi a}{b} \right)} \int_0^b f(y) \sin \left(\frac{n\pi y}{b} \right) dy \quad \dots(19)$$

Hence (18) is the required solution where E_n is given by (19).

CHAPTER-7

GREEN'S FUNCTION

7.1. Construction of Green's Function

Consider a differential equation of order n :

$$L(u) \equiv P_0(x)u^n + P_1(x)u^{n-1} + P_2(x)u^{n-2} + \dots + P_n(x)u = 0, \quad \dots(1)$$

Where the function $P_0(x), P_1(x), P_2(x), \dots, P_n(x)$ are continuous on $[a, b]$, $P_0(x) \neq 0$ on $[a, b]$ and the boundary conditions

$$V_k(u) = a_k u(a) + a_k^1 u'(a) + a_k^2 u''(a) + \dots + a_k^{n-1} u^{(n-1)}(a) + b_k u(b) + b_k^1 u'(b) + b_k^2 u''(b) + \dots + b_k^{n-1} u^{(n-1)}(b), \quad (k=1, 2, \dots, n) \quad \dots(2)$$

Where the linear forms $V_1, V_2, \dots, V_n$ in

$u(a), u'(a), \dots, u^{(n-1)}(a), u(b), u'(b), \dots, u^{(n-1)}(b)$ are linearly independent.

The homogeneous boundary value problem (1), (2) contains only a trivial solution $u(x) \equiv 0$.

Green's function of the boundary value problem (1), (2) is the function $G(x, \xi)$ constructed in x for any point $\xi, a < x < b$ satisfying the following $u(x) \equiv 0$.

- (i). $G(x, \xi)$ is continuous in x for fixed ξ and has continuous derivatives with regard to x upto order $(n-2)$ inclusive for $a \leq x \leq b$.
- (ii). Its $(n-1)$ th derivative with regard to x at the point $x = \xi$ has a discontinuity of first kind, the jump being equal to $-\frac{1}{[P_0(x)]_{x=\xi}}$,

$$\text{i.e. } \left\{ \frac{\partial^{n-1}}{\partial x^{n-1}} G(x, \xi) \right\}_{x=\xi+0} - \left\{ \frac{\partial^{n-1}}{\partial x^{n-1}} G(x, \xi) \right\}_{x=\xi-0} = -\frac{1}{P_0(\xi)}, \quad \dots(3)$$

Where $G|_{x=\xi+0}$ defines the limit $G(x, \xi)$ as $x \rightarrow \xi$ from the right and $G|_{x=\xi-0}$ defines the limit of $G(x, \xi)$ as $x \rightarrow \xi$ from the left.

- (iii). In each of the intervals $[a, \xi)$ and $(\xi, b]$ the function $G(x, \xi)$, considered as a function of x , is a solution of the equation (1):

$$L(G) = 0 \quad \dots(4)$$

- (iv). The function $G(x, \xi)$ satisfies the boundary conditions (2)

$$V_k(G) = 0, \quad k = 1, 2, 3, \dots, n \quad \dots(5)$$

If the boundary value problem (1), (2) contains only the trivial solution $u(x) \equiv 0$ then the operator L contains one and only one Green's function $G(x, \xi)$.

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Since $a_k = b_k - c_k$, then we have

$$(b_1 - c_1)A_k(u_1) + (b_2 - c_2)A_k(u_2) + \dots + (b_n - c_n)A_k(u_n) + b_1B_k(u_1) + b_2B_k(u_2) + \dots + b_nB_k(u_n) = 0$$

$$\Rightarrow b_1V_k(u_1) + b_2V_k(u_2) + \dots + b_nV_k(u_n)$$

$$= c_1A_k(u_1) + c_2A_k(u_2) + \dots + c_nA_k(u_n), \quad \dots(8)$$

Which is a linear system in the quantities $b_1, b_2, \dots, b_n$. The determinant of the system is different from zero:

$$\begin{vmatrix} V_1(u_1) & V_1(u_2) & \dots & V_1(u_n) \\ V_2(u_1) & V_2(u_2) & \dots & V_2(u_n) \\ \dots & \dots & \dots & \dots \\ V_n(u_1) & V_n(u_2) & \dots & V_n(u_n) \end{vmatrix} \neq 0$$

The system of the equations (8) contain a unique solution in $b_1(\xi), b_2(\xi), \dots, b_n(\xi)$ and since

$a_k(\xi) = b_k(\xi) - c_k(\xi)$, it follows that the quantities $a_k(\xi)$ are defined uniquely.

(i). If the boundary value problem (1), (2) is self adjoint, then Green's function is symmetric i.e.,

$$G(x, \xi) = G(\xi, x)$$

The converse is true as well.

(ii). If at one of the extremities of an interval $[a, b]$ the coefficient of the derivative vanishes e.g., $P_0(a) = 0$, then the natural boundary condition for the boundedness of the solution $x = a$ is imposed, and at the other extremity the ordinary condition is specified.

- **Particular Case:** We shall construct the Green's function $G(x, \xi)$ for a given number ξ , for the second differential equation

$$L(u) + \phi(x) = 0 \quad \dots(1)$$

$$\text{Where } L \equiv \frac{d}{dx} \left(P \frac{d}{dx} \right) + q \quad \dots(2)$$

Together with the homogeneous boundary conditions of the form

$$\alpha u + \beta \frac{du}{dx} = 0 \quad \dots(3)$$

The Green's function $G(x, \xi)$ constructed for any point ξ , $a < x < b$ contains the following properties.

- $G_1(\xi) = G_2(\xi)$; it follows that the function $G(x, \xi)$ is continuous in x for fixed ξ , in particular, continuous at the point $x = \xi$.
- The derivatives of G (which are of finite magnitude) are continuous at every point within the range of except at $x = \xi$ where it is continuous so that

$$G'_2(\xi) - G'_1(\xi) = -\frac{1}{p(\xi)}$$

- (iii). The function G_1 and G_2 satisfy the homogeneous conditions at the end point $x = a$ and $x = b$ respectively.
- (iv). The functions G_1 and G_2 satisfy the homogeneous equations $LG = 0$ in their defined intervals except at $z = \xi$, i.e.,

$$LG_1 = 0, x < \xi$$

$$LG_2 = 0, x > \xi$$

Consider the Green's function $G(x, \xi)$ exists, then the solution of the given differential equation can be transformed to the relation

$$u(x) = \int_a^b G(x, \xi) \phi(\xi) d\xi \quad \dots(4)$$

Consider two linearly independent solutions of the homogeneous equation $L(u) = 0$. Let $u = u_1(x)$ and $u = u_2(x)$ be the non-trivial solutions of the equation, which satisfy the homogeneous conditions at $x = a$ and $x = b$ respectively.

Consider the Green's functions for the problem from the conditions III and IV, in the form

$$G(x, \xi) = \begin{cases} C_1 u_1(x), & x < \xi, \\ C_2 u_2(x), & x > \xi, \end{cases} \quad \dots(5)$$

Where the constant C_1 and C_2 are chosen in a manner that the conditions I and II are fulfilled.

Thus, we have

$$C_2 u_2(\xi) - C_1 u_1(\xi) = 0$$

$$C_2 u_2'(\xi) - C_1 u_1'(\xi) = -\frac{1}{p(\xi)} \quad \dots(6)$$

The determinant of the system (6) is the Wronskian $W[u_1(\xi), u_2(\xi)]$ evaluated at the point $x = \xi$ for linearly independent solutions $u_1(x)$ and $u_2(x)$ and hence it is different from zero $W(\xi) \neq 0$.

$$W[u_1(\xi), u_2(\xi)] = \begin{vmatrix} u_1(\xi) & u_2(\xi) \\ u_1'(\xi) & u_2'(\xi) \end{vmatrix} = u_1(\xi)u_2'(\xi) - u_2(\xi)u_1'(\xi) \quad \dots(7)$$

* The conditions that $u_1(x)$ and $u_2(x)$ satisfy the equation $L(u) = 0$ are

$$(pu_1') + qu_1 = 0, \quad \dots(i)$$

$$\text{and } (pu_2') + qu_2 = 0 \quad \dots(ii)$$

Multiplying (i) by u_2 and (ii) by u_1 , and subtracting, we have

$$u_2(pu_1') - u_1(pu_2') = 0 \Rightarrow [p(u_1u_2' - u_2u_1')] = 0 \Rightarrow u_1u_2' - u_2u_1' = (C/p).$$

$$\text{i.e., } u_1(\xi)u_2'(\xi) - u_2(\xi)u_1'(\xi) = \frac{C}{p(\xi)} \quad \dots(8)$$

From the system (6), we have

$$C_1 = -\frac{1}{C}u_2(\xi), C_2 = -\frac{1}{C}u_1(\xi)$$

Thus the relation (5) reduces to

$$G(x, \xi) = \begin{cases} -\frac{1}{C}u_1(x)u_2(\xi), & x < \xi \\ -\frac{1}{C}u_1(\xi)u_2(x), & x > \xi \end{cases} \quad \dots(9)$$

This result breaks down iff C vanishes, so that u_1 and u_2 are linearly dependent, and hence are each multiples of a certain non-trivial function $U(x)$. In this case, the function $U(x)$ satisfies the equation $L(u) = 0$ together with the end conditions at $x = a$, $x = b$.

Converse: The integral equation

$$u(x) = \int_a^b G(x, \xi)\phi(\xi)d\xi, \quad \dots(10)$$

Where $G(x, \xi)$ are defined by the relation (9), satisfy the differential equation

$$L(u) + \phi(x) = 0 \quad \dots(11)$$

Together with the prescribed boundary condition.

We know that

$$u(x) = -\frac{1}{C} \left[\int_a^x u_1(\xi)u_2(x)\phi(\xi)d\xi + \int_x^b u_1(x)u_2(\xi)\phi(\xi)d\xi \right] \quad \dots(12)$$

$$u'(x) = -\frac{1}{C} \left[\int_a^x u_2'(x)u_1(\xi)\phi(\xi)d\xi + \int_x^b u_1'(x)u_2(\xi)\phi(\xi)d\xi \right] \quad \dots(13)$$

$$\begin{aligned} u''(x) &= -\frac{1}{C} \left[\int_a^x u_2''(x)u_1(\xi)\phi(\xi)d\xi + \int_x^b u_1''(x)u_2(\xi)\phi(\xi)d\xi \right] \\ &= -\frac{1}{C} [u_2'(x)u_1(x) - u_1'(x)u_2(x)]\phi(x) \quad \dots(14) \end{aligned}$$

Since $L(u) = p(x)u''(x) + p(x)u'(x) + q(x)u(x)$.

Thus

$$\begin{aligned} Lu(x) &= -\frac{1}{C} \left[\int_a^x \{Lu_2(x)\}u_1(\xi)\phi(\xi)d\xi + \int_x^b \{Lu_1(x)\}u_2(\xi)\phi(\xi)d\xi \right] \\ &\quad -\frac{1}{C} \left[p(x) \cdot \frac{C}{p(x)} \phi(x) \right] \end{aligned}$$

Again, $u_1(x)$ and $u_2(x)$ satisfy $L(u) = 0$, hence the first two terms vanish identically.

$$\text{So } Lu(x) = -\phi(x) \Rightarrow Lu(x) + \phi(x) = 0$$

Therefore, a function $u(x)$ satisfying (10) also satisfies the differential equation (11).

Again from (12) and (13), we have

$$u(a) = -\frac{u_1(a)}{C} \int_a^b u_2(\xi) \phi(\xi) d\xi$$

$$u'(a) = -\frac{u_1'(b)}{C} \int_a^b u_2(\xi) \phi(\xi) d\xi,$$

Which shows that the function u defined by (11) satisfies the same homogeneous condition at $x = a$ as the function u_1 .

Example: Find the Green's function for the boundary value problem

$$\frac{d^2 u}{dx^2} - u(x) = 0$$

with the conditions $u(0) = u(1) = 0$

Solution: The differential equation is given as

$$\frac{d^2 u}{dx^2} - u(x) = 0 \quad \dots(1)$$

$$\text{with } u(0) = u(1) = 0 \quad \dots(2)$$

The general solution of the differential equation is given as

$$u(x) = Ae^x + Be^{-x}$$

$$u(0) = 0 \Rightarrow A + B = 0$$

$$u(1) = 0 \Rightarrow Ae + Be^{-1} = 0$$

$$\Rightarrow A = B = 0$$

These conditions have only a trivial solution $u(x) \equiv 0$, hence we can construct a unique Green's function.

The fundamental system of solutions for the differential equation (1) is given as

$$u_1(x) = \sinh x \text{ Satisfies the boundary condition } u(0) = 0$$

$$u_2(x) = \sinh(x-1) \text{ Satisfies the boundary condition } u(1) = 0$$

The solution are linearly independent. The value of Wronskian for $\sinh x$ and $\sinh(x-1)$ at the point $x = \xi$;

$$W(\xi) = \begin{vmatrix} \sinh \xi & \sinh(\xi-1) \\ \cosh \xi & \cosh(\xi-1) \end{vmatrix}$$

$$\Rightarrow W(\xi) = \sinh \xi \cosh(\xi-1) - \cosh \xi \sinh(\xi-1) = \sinh 1$$

Here $p(x) = 1$. From (9), we obtain the Green Function of the form:

$$G(x, \xi) = \begin{cases} -\frac{\sinh x \sinh(\xi-1)}{\sinh 1}, & 0 \leq x \leq \xi \\ -\frac{\sinh \xi \sinh(x-1)}{\sinh 1}, & \xi \leq x \leq 1 \end{cases}$$

Example: If $u(x)$ has continuous first and second derivatives and satisfies the boundary value problem $\frac{d^2u}{dx^2} + \lambda u = 0$ with $u(0) = u(l) = 0$ then $u(x)$ is continuous and satisfies the homogeneous linear integral equation $u(x) = \int_0^l G(x, \xi) u(\xi) d\xi$.

Solution: The differential equation may be written as

$$\frac{d^2u}{dx^2} + \lambda u = 0 \Rightarrow \frac{d^2u}{dx^2} = -\lambda u \quad \dots(1)$$

By integrating with respect to x over the interval $(0, x)$ two times, we obtain

$$\frac{du}{dx} = -\lambda \int_0^x u(\xi) d\xi + C$$

$$\text{or } u(x) = -\lambda \int_0^x (x - \xi) u(\xi) d\xi + Cx + D \quad \dots(2)$$

Where C and D are the integration constants, to be determined by the boundary conditions.

$$u(0) = 0 \Rightarrow D = 0$$

$$u(l) = 0 \Rightarrow -\lambda \int_0^l (l - \xi) u(\xi) d\xi + Cl = 0$$

$$\Rightarrow C = \frac{\lambda}{l} \int_0^l (l - \xi) u(\xi) d\xi$$

Substituting the value of the constants C and D in (2), we have

$$u(x) = -\lambda \int_0^x (x - \xi) u(\xi) d\xi + \frac{\lambda}{l} \int_0^l x(l - \xi) u(\xi) d\xi$$

$$\text{or } u(x) = -\lambda \int_0^x (x - \xi) u(\xi) d\xi + \frac{\lambda}{l} \int_0^x x(l - \xi) u(\xi) d\xi + \frac{\lambda}{l} \int_x^l x(l - \xi) u(\xi) d\xi$$

$$\text{or } u(x) = \lambda \int_0^x \frac{\xi}{l} (l - x) u(\xi) d\xi + \lambda \int_x^l \frac{x}{l} (l - \xi) u(\xi) d\xi$$

$$\text{or } u(x) = \lambda \int_0^l G(x, \xi) u(\xi) d\xi$$

$$\text{where } G(x, \xi) = \begin{cases} \frac{\xi}{l} (l - x), & x > \xi \\ \frac{x}{l} (l - \xi), & x < \xi \end{cases}$$

Example: Construct the Green's function for the boundary value problem

$$\frac{d^2u}{dx^2} = 0 \text{ with the conditions } u(0) = u'(1) \text{ and } u'(0) = u(1).$$

Solution: The differential equation is given by

$$\frac{d^2u}{dx^2} = 0 \quad \dots(1)$$

with the conditions

$$u(0) = u'(1) \quad \dots(2)$$

$$u'(0) = u(1) \quad \dots(3)$$

The general solution of the given differential equation is

$$u(x) = Ax + B \quad \dots(4)$$

The boundary value problem contains only a trivial solution $u(x) \neq 0$, hence we can construct a unique Green's function for it.

Consider the unknown Green's function $G(x, \xi)$ must have the representation on the interval $[0, \xi]$ and $(\xi, 1]$

$$G(x, \xi) = \begin{cases} a_1x + a_2 & , 0 \leq x < \xi \\ b_1x + b_2 & , \xi < x \leq 1 \end{cases} \quad \dots(5)$$

Where a_1, a_2, b_1, b_2 are unknown functions of ξ . The Green's function $G(x, \xi)$ must satisfy the following properties.

I. $G(x, \xi)$ is continuous at $x = \xi$, i.e.,

$$a_1\xi + a_2 = b_1\xi + b_2 \Rightarrow (b_1 - a_1)\xi + (b_2 - a_2) = 0 \Rightarrow C_1\xi + C_2 = 0 \quad \dots(6)$$

II. The derivatives of G has a discontinuity of magnitude $\frac{1}{p_0(\xi)}$ at the point $x = \xi$, where $p_0(\xi) =$ coefficient of highest order derivative in the given differential equation ($=1$).

$$\left(\frac{\partial G}{\partial x}\right)_{x=\xi+0} - \left(\frac{\partial G}{\partial x}\right)_{x=\xi-0} = -1$$

$$\Rightarrow b_1 - a_1 = -1 \Rightarrow C_1 = -1 \text{ and } C_2 = \xi \quad \dots(7)$$

III. The Green's function $G(x, \xi)$ must satisfy the prescribed boundary conditions

$$G(0, \xi) = G'(1, \xi) \Rightarrow a_2 = b_1 \quad \dots(8)$$

$$G'(0, \xi) = G(1, \xi) \Rightarrow a_1 = b_1 + b_2 \quad \dots(9)$$

$$\text{Consider } C_k = b_k - a_k \Rightarrow C_1 = b_1 - a_1 \text{ and } C_2 = b_2 - a_2 \quad \dots(10)$$

From the equations (7) to (10), we have

$$b_2 - a_2 = \xi, \quad b_1 - a_1 = -1$$

$$b_1 + b_2 = a_1, \quad a_2 = b_1$$

By solving these equations, we have

$$a_1 = -\xi + 2, \quad a_2 = -\xi + 1, \quad b_1 = -\xi + 1, \quad b_2 = +1$$

The required Green's function $G(x, \xi)$ is obtained:

$$G(x, \xi) = \begin{cases} (-\xi + 2)x + (-\xi + 1) & 0 \leq x < \xi \\ (-\xi + 1)x + 1 & \xi < x \leq 1 \end{cases}$$

ASSIGNMENT (SHEET-01)

1. Consider the initial value problem

$$\frac{\partial u}{\partial x} + 2 \frac{\partial u}{\partial y} = 0, \quad u(0, y) = 4e^{-2y}.$$

Then the value of $u(1, 1)$ is

- (a.) $4e^{-2}$
- (b.) $4e^2$
- (c.) $2e^{-4}$
- (d.) $4e^4$

2. Let $a, b \in \mathbb{R}$ be such that $a^2 + b^2 \neq 0$. Then

the Cauchy problem $a \frac{\partial u}{\partial x} + b \frac{\partial u}{\partial y} = 1; \quad x, y \in \mathbb{R}$

$u(x, y) = x$ on

$$ax + by = 1$$

- (a.) has more than one solution if either a or b is zero
- (b.) has no solution
- (c.) has a unique solution
- (d.) has infinitely many solutions

3. Which of the following are complete integrals of the partial differential equation

$$pqx + yq^2 = 1?$$

- (a.) $z = \frac{x}{a} + \frac{ay}{x} + b$
- (b.) $z = \frac{x}{b} + \frac{ay}{x} + b$
- (c.) $z^2 = 4(ax + y) + b$
- (d.) $(z - b)^2 = 4(ax + y)$

4. For an arbitrary continuously differentiable function f , which of the following is a general solution of $z(px - qy) = y^2 - x^2$

- (a.) $x^2 + y^2 + z^2 = f(xy)$
- (b.) $(x + y)^2 + z^2 = f(xy)$
- (c.) $x^2 + y^2 + z^2 = f(y - x)$
- (d.) $x^2 + y^2 + z^2 = f((x + y)^2 + z^2)$

5. The Charpit's equation for the PDE

$$up^2 + q^2 + x + y = 0, \quad p = \frac{\partial u}{\partial x}, \quad q = \frac{\partial u}{\partial y} \text{ are given by}$$

- (a.) $\frac{dx}{-1-p^3} = \frac{dy}{-1-qp^2} = \frac{du}{2p^2u+2q^2} = \frac{dp}{2pu} = \frac{dq}{2q}$
- (b.) $\frac{dx}{2pu} = \frac{dy}{2q} = \frac{du}{2p^2u+2q^2} = \frac{dp}{-1-p^3} = \frac{dq}{-1-qp^2}$
- (c.) $\frac{dx}{up^2} = \frac{dy}{q^2} = \frac{du}{0} = \frac{dp}{x} = \frac{dq}{y}$
- (d.) $\frac{dx}{2q} = \frac{dy}{2pu} = \frac{du}{x+y} = \frac{dp}{p^2} = \frac{dq}{qp^2}$

6. Consider the Cauchy problem of finding

$$u = u(x, t) \text{ such that } \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \text{ for}$$

$$x \in \mathbb{R}, t > 0, \quad u(x, 0) = u_0(x), \quad x \in \mathbb{R} \text{ which}$$

choice(s) of the following function for u_0 yield a C^1 solution $u(x, t)$ for all $x \in \mathbb{R}$ and $t > 0$.

- (a.) $u_0(x) = \frac{1}{1+x^2}$
- (b.) $u_0(x) = x$
- (c.) $u_0(x) = 1+x^2$
- (d.) $u_0(x) = 1+2x$

7. Let $u = u(x, t)$ be the solution of the Cauchy

$$\text{problem } \frac{\partial u}{\partial t} + \left(\frac{\partial u}{\partial x} \right)^2 = 1 \quad x \in \mathbb{R}, t > 0$$

$$u(x, 0) = -x^2 \quad x \in \mathbb{R}$$

then

- (a.) $u(x, t)$ exists for all $x \in \mathbb{R}$ and $t > 0$.
- (b.) $|u(x, t)| \rightarrow \infty$ as $t \rightarrow t^*$ for some $t^* > 0$ and $x \neq 0$
- (c.) $u(x, t) \leq 0$ for all $x \in \mathbb{R}$ and for all $t < 1/4$
- (d.) $u(x, t) > 0$ for some $x \in \mathbb{R}$ and $0 < t < 1/4$

8. Let $x = x(s), y = y(s), u = u(s), s \in R$, be the characteristic curve of the PDE

$$\left(\frac{\partial u}{\partial x}\right)\left(\frac{\partial u}{\partial y}\right) - u = 0 \text{ passing through a given}$$

curve $x = 0, y = \tau, u = \tau^2, \tau \in R$. Then the characteristics are given by

- (a.) $x = 3\tau(e^s - 1), y = \frac{\tau}{2}(e^s + 1), u = \tau^2 e^{-2s}$
(b.) $x = 2\tau(e^s - 1), y = \tau(2e^{2s} - 1), u = \frac{\tau^2}{2}(1 + e^{-2s})$
(c.) $x = 2\tau(e^s - 1), y = \frac{\tau}{2}(e^s + 1), u = \tau^2 e^{2s}$
(d.) $x = \tau(e^s - 1), y = -2\tau(e^s - \frac{3}{2}), u = \tau^2(2e^{-2s} - 1)$

9. The initial value problem $\frac{\partial u}{\partial t} + x \frac{\partial u}{\partial x} = x, 0 \leq x \leq 1, t > 0$ and $u(x, 0) = 2x$ has

- (a.) A unique solution $u(x, t)$ which $\rightarrow \infty$ as $t \rightarrow \infty$
(b.) More than one solution
(c.) A solution which remains bounded as $t \rightarrow \infty$
(d.) No solution

10. Let $xyu = c_1$ and $x^2 + y^2 - 2u = c_2$, where c_1 and c_2 are arbitrary constants, be the first integrals of the PDE

$$x(u + y^2) \frac{\partial u}{\partial x} - y(u + x^2) \frac{\partial u}{\partial y} = (x^2 - y^2)u.$$

Then the solution of the PDE with $x + y = 0, u = 1$ is given by

- (a.) $x^3 + y^3 + 2xyu^2 + 2x^2u = 0$
(b.) $x^3 + yx^2 + (x^2 + xy)u = 0$
(c.) $x^2 + y^2 + 2(xy - 1)u + 2 = 0$
(d.) $x^2 - y^2 - u(x + y - 2) - 2 = 0$

11. Let $z = z(x, y)$ be a solution of $\frac{\partial z}{\partial x} \frac{\partial z}{\partial y} = 1$ passing through $(0, 0, 0)$. Then $z(0, 1)$ is

- (a.) 0.
(b.) 1.
(c.) 2.
(d.) 4.

12. The solution of the Cauchy problem for the first order PDE $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = z$, on $D = \{(x, y, z) | x^2 + y^2 \neq 0, z > 0\}$, with the initial condition $x^2 + y^2 = 1, z = 1$ is

- (a.) $z = x^2 + y^2$
(b.) $z = (x^2 + y^2)^2$
(c.) $z = (2 - (x^2 + y^2))^{1/2}$
(d.) $z = (x^2 + y^2)^{1/2}$

13. Consider the first order PDE $p + q = pq$ where $p \equiv \frac{\partial z}{\partial x}, q \equiv \frac{\partial z}{\partial y}$. Then which of the following are correct?

- (a.) The Charpit's equations for the above PDE reduce to

$$\frac{dx}{1-q} = \frac{dy}{1-p} = \frac{dz}{-pq} = \frac{dp}{p+q} = \frac{dq}{0}$$

- (b.) A solution of the Charpit's equation is $q = b$, where b is a constant.
(c.) The corresponding value of p is $p = \frac{b}{b-1}$

- (d.) A solution of the equation is $z = \frac{b}{b-1}x + by + a$, where a and b are constants.

14. The Cauchy problem

$$xu_x + yu_y = 0$$

$$u(x, y) = x, \text{ on } x^2 + y^2 = 1 \text{ has}$$

- (a.) A solution for all $x \in R, y \in R$
(b.) An unique solution in $\{(x, y) \in R^2 : (x, y) \neq (0, 0)\}$
(c.) A bounded solution in $\{(x, y) \in R^2 : (x, y) \neq (0, 0)\}$
(d.) An unique solution in $\{(x, y) \in R^2 : (x, y) \neq (0, 0)\}$ but the solution is unbounded.

15. Let $u = u(x, y)$ be the complete integral of the PDE $\frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial y} = xy$ passing through the points $(0, 0, 1)$ and $(0, 1, \frac{1}{2})$ in the $x-y-u$ space. Then the value of $u(x, y)$ evaluated at $(-1, 1)$ is
- 0
 - 1
 - 2
 - 3
16. The differential equation $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$ satisfying the initial condition $y = xg(x), u = f(x)$ with
- $f(x) = 2x, g(x) = 1$, has no solution
 - $f(x) = 2x^2, g(x) = 1$, has infinite number of solutions
 - $f(x) = x^3, g(x) = x$, has a unique solution
 - $f(x) = x^4, g(x) = x$, has a unique solution.
17. A general solution of the PDE $uu_x + yu_y = x$ is of the form
- $f\left(u^2 - x^2, \frac{y}{x+u}\right) = 0$, where $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is C^1 and $\nabla f \neq (0, 0)$ at every point
 - $u^2 = g\left(\frac{y}{x+u}\right) + x^2, g \in C^1(\mathbb{R})$
 - $f(u^2 + x^2) = 0, f \in C^1(\mathbb{R})$
 - $f(x+y) = 0, f \in C^1(\mathbb{R})$
18. The partial differential equation $(u_x)^4 u_{xx} + u_{yy} + (u_y)^2 = 0$ is
- Linear and of order 4
 - Quasi-linear and of order 2
 - Quasi-linear and of order 4
 - Linear and of order 2.
19. The solution of the Cauchy problem $u_x + u_y = 0, u(x, 0) = e^x$ is
- $u(x, y) = e^{x+y}$
 - $u(x, y) = e^x e^{2y}$
 - $u(x, y) = e^{x-y}$
 - $u(x, y) = \frac{1}{2}[e^{x+y} + e^{x-y}]$
20. The solution of the PDE $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$ with initial data $u(x, 1) = x(1-x)$ is
- $xy^2 - x^2y$
 - $xy^3 - x^3y$
 - $xy^2 - x^2y^2$
 - $xy - x^2y^2$
21. The complete integral of the non-linear first order partial differential equation $y(1+x^2)p^2 - qx^2 = 0$ is
- $z = \pm a\sqrt{1+x^2} + \frac{a^2y^2}{2} + c$
 - $z = \pm \frac{a}{\sqrt{1+x^2}} + a^2y + c$
 - $z = \pm \frac{2}{3}a(1+x^2)^{3/2} + \frac{a^2y^2}{2} + c$
 - $z = \pm a\sqrt{1+x^2} + a^2y^2 + c$
22. The characteristics of the partial differential equation $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = u$ are
- Exponential curves along which u remains constant
 - Exponential curves along which u varies linearly with x
 - Straight lines along which u remains constant
 - Straight lines along which u varies exponentially with x
23. For obtaining the complete integral of the non-linear partial differential equation $pq(x^4 + y^4) = (p+q)$, the required forms of $p = \frac{\partial z}{\partial x}, q = \frac{\partial z}{\partial y}$, involving an arbitrary constant a , are
- $p = (x^4 + a)^{-1}, q = (y^3 - a)^{-1}$
 - $p = (2x^4 + a)^{-1}, q = (3y^3 - a)^{-1}$
 - $p = (-2x^4 + a)^{-1}, q = (-3y^3 - a)^{-1}$
 - $p = (x^4 - a)^{-1}, q = (y^3 - a)^{-1}$

24. The general integral of $x(y-z)\frac{\partial z}{\partial x} + y(z-x)\frac{\partial z}{\partial y} = z(x-y)$ is

(a.) $F(2x+3y+z, xyz) = 0$
 (b.) $F(x+y+z, xyz) = 0$
 (c.) $F(x-y+z, xyz) = 0$
 (d.) $F(x+y-z, xyz) = 0$

25. The general solution of $2\frac{\partial z}{\partial x} + 3\frac{\partial z}{\partial y} = \cos(3x-2y)$ is

(a.) $2z = x \cos(3x-2y) + f(3x+2y)$
 (b.) $2z = x \sin(3x-2y) + f(3x+2y)$
 (c.) $2z = x \cos(3x-2y) + f(3x-2y)$
 (d.) $2z = x \sin(3x-2y) + f(3x-2y)$

26. The general integral of $z(xp-yq) = y^2 - x^2$ is

(a.) $z^2 = x^2 + y^2 + f(xy)$
 (b.) $z^2 = x^2 - y^2 + f(xy)$
 (c.) $z^2 = -x^2 - y^2 + f(xy)$
 (d.) $z^2 = y^2 - z^2 + f(xy)$

27. A singular solution of the partial differential equation $z + xp - x^2 yq^2 - x^3 pq = 0$ is

(a.) $z = \frac{x^2}{y}$
 (b.) $z = \frac{x}{y^2}$
 (c.) $z = \frac{y}{x^2}$
 (d.) $z = \frac{y^2}{x}$

28. Let $u(x, y) = 2f(y)\cos(x-2y)$, $(x, y) \in \mathbb{R}^2$, be a solution of the initial value problem

$2u_x + u_y = u$ $u(x, 0) = \cos(x)$. Then $f(1)$ is equal to

(a.) $\frac{1}{2}$
 (b.) $\frac{e}{2}$
 (c.) e
 (d.) $\frac{3e}{2}$

29. The integral surface of the first order partial differential equation

$$2y(z-3)\frac{\partial z}{\partial y} + (2x-z)\frac{\partial z}{\partial y} = y(2x-3)$$

passing through the curve $x^2 + y^2 = 2x, z = 0$ is

(a.) $x^2 + y^2 - z^2 - 2x + 4z = 0$
 (b.) $x^2 + y^2 - z^2 - 2x + 8z = 0$
 (c.) $x^2 + y^2 - z^2 - 2x + 16z = 0$
 (d.) $x^2 + y^2 + z^2 - 2x + 8z = 0$

30. The integral surface satisfying the partial differential equation $\frac{\partial z}{\partial x} + z^2 \frac{\partial z}{\partial y} = 0$ and passing through the straight line $x=1, y=z$ is

(a.) $(x-1)z + z^2 = y^2$
 (b.) $x^2 + y^2 - z^2 = 1$
 (c.) $(y-z)x + x^2 = 1$
 (d.) $(x-1)z^2 + z = y$

31. The integral surface for the Cauchy problem

$$\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 1,$$

which passes through the circle $z=0, x^2 + y^2 = 1$ is

(a.) $x^2 + y^2 + 2z^2 + 2zx - 2yz - 1 = 0$
 (b.) $x^2 + y^2 + 2z^2 + 2zx + 2yz - 1 = 0$
 (c.) $x^2 + y^2 + 2z^2 - 2zx - 2yz - 1 = 0$
 (d.) $x^2 + y^2 + 2z^2 + 2zx + 2yz + 1 = 0$

32. The integral surface satisfying the equation $y\frac{\partial z}{\partial x} + x\frac{\partial z}{\partial y} = x^2 + y^2$ and passing through the curve $x=1-t, y=1+t, z=1+t^2$ is

(a.) $z = xy + \frac{1}{2}(x^2 - y^2)^2$
 (b.) $z = xy + \frac{1}{4}(x^2 - y^2)^2$
 (c.) $z = xy + \frac{1}{8}(x^2 - y^2)^2$
 (d.) $z = xy + \frac{1}{16}(x^2 - y^2)^2$

33. The solution of $xu_x + yu_y = 0$ is of the form

- (a.) $f(y/x)$
- (b.) $f(x+y)$
- (c.) $f(x-y)$
- (d.) $f(xy)$

34. The characteristic curve of $2yu_x + (2x + y^2)u_y = 0$ passing through $(0,0)$ is

- (a.) $y^2 = 2(e^x + x - 1)$
- (b.) $y^2 = 2(e^x - x + 1)$
- (c.) $y^2 = 2(e^x - x - 1)$
- (d.) $y^2 = 2(e^x + x + 1)$

35. The initial value problem $u_x + u_y = 1, u(s, s) = \sin s, 0 \leq s \leq 1$ has

- (a.) Two solutions
- (b.) A unique solution
- (c.) No solution
- (d.) Infinitely many solutions

36. Let $u(x, y) = f(xe^y) + g(y^2 \cos(y))$

Where f and g are infinitely differentiable functions. Then the partial differential equation of minimum order satisfied by u is

- (a.) $u_{xy} + xu_{xx} = u_x$
- (b.) $u_{xy} + xu_{xx} = xu_x$
- (c.) $u_{xy} - xu_{xx} = u_x$
- (d.) $u_{xy} - xu_{xx} = xu_x$

37. Let $u(x, y)$ be the solution of the Cauchy problem

$$xu_x + u_y = 1, u(x, 0) = 2 \ln(x), x > 1$$

Then $u(e, 1) =$

- (a.) -1
- (b.) 0
- (c.) 1
- (d.) e

38. Consider the partial differential equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \text{ satisfying the initial condition } u(x, 0) = \alpha + \beta x.$$

If $u(x, t) = 1$ along the characteristic $x = t + 1$, then

- (a.) $\alpha = 1, \beta = 1$
- (b.) $\alpha = 2, \beta = 0$
- (c.) $\alpha = 0, \beta = 0$
- (d.) $\alpha = 0, \beta = 1$

39. Consider the partial differential equation $x \frac{\partial u}{\partial y} - y \frac{\partial u}{\partial x} = u$, the characteristic curves for

the above equation in the (x, y) plane are

- (a.) Straight line with slopes 1
- (b.) Straight lines with slopes -1
- (c.) Circles with centre at the origin
- (d.) Circles touching y axis and centered on x -axis

40. If $u(x, y)$ is a solution to the partial differential equation $x \frac{\partial u}{\partial y} - y \frac{\partial u}{\partial x} = u$ with

$$u\left(\frac{x}{4}, 0\right) = \sin\left(\frac{\pi x}{4}\right), \text{ then } u\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \text{ equals}$$

- (a.) $\frac{1}{\sqrt{2}} e^{\frac{\pi}{4}}$
- (b.) $\frac{\pi}{4} e^{\frac{\pi}{\sqrt{2}}}$
- (c.) $\frac{1}{\sqrt{2}} e^{\frac{1}{\sqrt{2}}}$
- (d.) $\frac{\pi}{4} e^{\frac{\pi}{4}}$

41. The integral surface of the partial differential equation $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$ satisfying the condition $u(1, y) = y$ is given by

- (a.) $u(x, y) = \frac{y}{x}$
- (b.) $u(x, y) = \frac{2y}{x+1}$
- (c.) $u(x, y) = \frac{y}{2-x}$
- (d.) $u(x, y) = y + x - 1$

42. The integral surface of the partial differential equation $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$ satisfying the condition $u(1, y) = y$ is given by

- (a.) $u(x, y) = \frac{y}{x}$
- (b.) $u(x, y) = \frac{2y}{x+1}$
- (c.) $u(x, y) = \frac{y}{2-x}$
- (d.) $u(x, y) = y + x - 1$

43. Complete integral for the partial differential equation $z = px + qy - \sin(pq)$ is
- $z = ax + by + \sin(ab)$
 - $z = ax + by - \sin(ab)$
 - $z = ax + y + \sin(b)$
 - $z = x + by - \sin(a)$
44. The Cauchy problem $u_x - u_y = 2$ with the Cauchy data on $\Gamma : (s, -s, 2s)$ has
- One solution
 - Two solutions
 - No solution
 - Infinite solutions
45. The general integral of the partial differential equation $(y + zx)z_x - (x + yz)z_y = x^2 - y^2$ is
- $F(x^2 + y^2 + z^2, xy + z) = 0$
 - $F(x^2 + y^2 - z^2, xy + z) = 0$
 - $F(x^2 - y^2 - z^2, xy + z) = 0$
 - $F(x^2 + y^2 + z^2, xy - z) = 0$
- Where F is an arbitrary function.
46. The partial differential equation of the family of surfaces $z = (x + y) + A(xy)$ is
- $xp - yq = 0$
 - $xp - yq = x - y$
 - $xp + yq = x + y$
 - $xp + yq = 0$
47. Using the transformation $u = \frac{w}{y}$ in the partial differential equation $xu_x = u + yu_y$, the transformed equation has a solution of the form $w =$
- $f(\frac{x}{y})$
 - $f(x + y)$
 - $f(x - y)$
 - $f(xy)$

48. A general solution of the equation $\frac{\partial u(x, y)}{\partial x} + u(x, y) = e^{-x}$ is
- $u(x, y) = e^{-x} f(y)$
 - $u(x, y) = e^{-x} f(y) + xe^{-x}$
 - $u(x, y) = e^x f(y) + xe^{-x}$
 - $u(x, y) = e^{-x} f(y) + xe^{-x}$
49. The integral curves of the equation $p + 2q = 5z + \cosh(2x - y)$ are given by the intersection of the surfaces
- $2x - y = c_1; [5z + \cosh(2x - y)]e^{-5x} = c_2$
 - $x - 2y = c_1; [5z + \cosh(x - 2y)]e^{-5x} = c_2$
 - $2x - y = c_1; [z + \cosh(2x - y)]e^{-x} = c_2$
 - $x - 2y = c_1; [z + \cosh(2x - y)]e^{-x} = c_2$
50. The complete integral of the partial differential equation $xp^3q^2 + yp^2q^3 + (p^3 + q^3) - zp^2q^2 = 0$ is $z =$
- $ax + by + (ab^{-2} + ba^{-2})$
 - $ax - by + (ab^{-2} - ba^{-2})$
 - $-ax + by + (ba^{-2} - ab^{-2})$
 - $ax + by - (ab^{-2} + ba^{-2})$
51. For arbitrary real valued smooth functions f and g , the function u defined as $u(x, t) = f(x + t) + g(x - t)$ is a general solution of
- $u_t + u_x = 0$
 - $u_t - u_{xx} = 0$
 - $u_{tt} + u_{xx} = 0$
 - $u_{tt} - u_{xx} = 0$

ASSIGNMENT (SHEET-02)

- The variables ξ and η which reduce the differential equation $\frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial^2 u}{\partial y^2} = 0$ to the canonical form are
 - $\xi = y^2 + \frac{1}{2}x, \eta = y^2 - \frac{1}{2}x$
 - $\xi = y + \frac{1}{2}x^2, \eta = y - \frac{1}{2}x^2$
 - $\xi = y + x^2, \eta = y - x^2$
 - $\xi = y + x^2, \eta = y^2 - x$
- If the partial differential equation $(x-1)^2 u_{xx} - (y-2)^2 u_{yy} + 2xu_x + 2yu_y + 2xyu = 0$ is parabolic in $S \subseteq R^2$ but not in $R^2 \setminus S$, then S is
 - $\{(x, y) \in R^2 : x=1 \text{ or } y=2\}$
 - $\{(x, y) \in R^2 : x=1 \text{ and } y=2\}$
 - $\{(x, y) \in R^2 : x=1\}$
 - $\{(x, y) \in R^2 : y=2\}$
- The characteristics curves of the equation $x^2 u_{xx} - y^2 u_{yy} = x^2 y^2 + x; x > 0, u = (x, y)$ are
 - Rectangular hyperbola
 - Parabola
 - Circle
 - Straight line
- If $f(x)$ and $g(y)$ are arbitrary functions, then the general solution of the partial differential equation $u \frac{\partial^2 u}{\partial x \partial y} - \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} = 0$ is given by
 - $u(x, y) = f(x) + g(y)$
 - $u(x, y) = f(x+y) + g(x-y)$
 - $u(x, y) = f(x)g(y)$
 - $u(x, y) = xg(y) + yf(x)$
- The number of characteristic curves of the PDE $(x^2 + 2y)u_{xx} + (y^3 - y + u)u_{yy} + x^2(y-1)u_{xy} + 3u_x + u = 0$ passing through the point $x=1, y=1$ is
 - 0
 - 1
 - 2
 - 3
- Which of the following is elliptic?
 - Laplace equation
 - Wave equation
 - Heat equation
 - $u_{xx} + 2u_{xy} - 4u_{yy} = 0$
- The equation $x^2(y-1)Z_{xx} - x(y^2-1)Z_{xy} + y(y^2-1)Z_{yy} + Z_x = 0$ is hyperbolic in the entire xy -plane except along
 - x -axis
 - y -axis
 - A line parallel to y -axis
 - A line parallel to x -axis
- The second order partial differential equation $\frac{(x-y)^2}{4} \frac{\partial^2 u}{\partial x^2} + (x-y) \sin(x^2 + y^2) \frac{\partial^2 u}{\partial x \partial y} + \cos^2(x^2 + y^2) \frac{\partial^2 u}{\partial y^2} + (x-y) \frac{\partial u}{\partial x} + \sin^2(x^2 + y^2) \frac{\partial u}{\partial y} + u = 0$ is
 - Elliptic in the region $\{(x, y) : x \neq y, x^2 + y^2 < \frac{\pi}{6}\}$
 - Hyperbolic in the region $\{(x, y) : x \neq y, \frac{\pi}{4} < x^2 + y^2 < \frac{3\pi}{4}\}$
 - Elliptic in the region $\{(x, y) : x \neq y, \frac{\pi}{4} < x^2 + y^2 < \frac{3\pi}{4}\}$
 - Hyperbolic in the region $\{(x, y) : x \neq y, x^2 + y^2 < \frac{\pi}{4}\}$
- The general solution of the partial differential equation $(D + D' - 2)(D - D')z = 0$, where $D \equiv \frac{\partial}{\partial x}$ and $D' \equiv \frac{\partial}{\partial y}$ is
 - $f_1(x+y) + e^{2x} f_2(x-y)$
 - $e^{2x} f_2(x+y) + f_2(x-y)$
 - $e^{-2x} f_1(x+y) + f_2(x-y)$
 - $f_1(x+y) + e^{2x} f_2(x-y)$

10. The partial differential equation $y \frac{\partial^2 u}{\partial x^2} + x \frac{\partial^2 u}{\partial y^2} = 0$ is hyperbolic in
- The second and fourth quadrants
 - The first and second quadrants
 - The second and third quadrants
 - The first and third quadrants
11. The general solution of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is of the form
- $u = f(x+iy) + g(x-iy)$
 - $u = f(x+y) + g(x-y)$
 - $u = cf(x-iy)$
 - $u = g(x+iy)$
12. The second order partial differential equation $u_{xx} + 4u_{xy} + (\cos x)u_{yy} + e^x u_x + e^y u_y = 0$ has
- Exactly one characteristic curve passing through every point
 - Exactly one characteristics curve passing through the point (0, 0)
 - Two distinct characteristic curves passing through every point
 - Three distinct characteristic curves passing through the point
13. The partial differential equation $u_{xx} + (x+y)u_{xy} + \left(\frac{x^2+y^2}{4}\right)u_{yy} + (\sin x)u_x = 0$, is
- Elliptic in $\{(x, y) : x \in \mathbf{R}, y \in \mathbf{R}\}$
 - Parabolic in $\{(x, y) : x \in \mathbf{R}, y > 0\}$
 - Hyperbolic in $\{(x, y) : xy > 0\}$
 - Parabolic in $\{(x, y) : xy < 0\}$
14. The complete integral of the PDE $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = xe^{x+y}$ involving arbitrary functions ϕ_1 and ϕ_2 is
- $\phi_1(y+x) + \phi_2(y+x) + \frac{1}{4}e^{x+y}$
 - $\phi_1(y+x) + x\phi_2(y+x) + \frac{(x-1)}{4}e^{x+y}$
 - $\phi_1(y-x) + \phi_2(y-x) + \frac{1}{4}e^{x+y}$
 - $\phi_1(y-x) + x\phi_2(y-x) + \frac{(x-1)}{4}e^{x+y}$
15. In the region $x > 0, y > 0$ the partial differential equation $(x^2 - y^2) \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + (x^2 + y^2) \frac{\partial^2 u}{\partial y^2} = 0$
- Changes type
 - Is elliptic
 - Is parabolic
 - Is hyperbolic
16. Pick the region in which the following differential equation is hyperbolic $yu_{xx} + 2xyu_{xy} = u_x + u_y$
- $xy \neq 1$
 - $xy \neq 0$
 - $xy > 1$
 - $xy > 0$
17. The expression $\frac{1}{D_x^2 - D_y^2} \sin(x-y)$ is equal to
- $-\frac{x}{2} \cos(x-y)$
 - $-\frac{x}{2} \sin(x-y) + \cos(x-y)$
 - $-\frac{x}{2} \cos(x-y) + \sin(x-y)$
 - $\frac{3x}{2} \sin(x-y)$
18. A general solution of the second order equation $4u_{xx} - u_{yy} = 0$ is of the form (where f and g are twice differentiable functions.)
- $f(x) + g(y)$
 - $f(x+2y) + g(x-2y)$
 - $f(x+4y) + g(x-4y)$
 - $f(4x+y) + g(4x-y)$
19. The general solution of the partial differential equation $\frac{\partial^2 z}{\partial x \partial y} = x+y$ is of the form
- $\frac{1}{2}xy(x+y) + F(x) + G(y)$
 - $\frac{1}{2}xy(x-y) + F(x) + G(y)$
 - $\frac{1}{2}xy(x-y) + F(x)G(y)$
 - $\frac{1}{2}xy(x+y) + F(x)G(y)$

20. The characteristics of the partial differential equation $36 \frac{\partial^2 z}{\partial x^2} - y^{14} \frac{\partial^2 z}{\partial y^2} - 7y^{13} \frac{\partial z}{\partial y} = 0$, are given by

(a.) $x + \frac{1}{y^6} = c_1, x - \frac{1}{y^6} = c_2$
(b.) $x + \frac{36}{y^6} = c_1, x - \frac{36}{y^6} = c_2$
(c.) $6x + \frac{7}{y^6} = c_1, 6x - \frac{7}{y^6} = c_2$
(d.) $6x + \frac{7}{y^8} = c_1, 6x - \frac{7}{y^8} = c_2$

21. The second order partial differential equation $(1 - \sqrt{xy}) \frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + (1 + \sqrt{xy}) \frac{\partial^2 u}{\partial y^2} = 0$ is

- (a.) Hyperbolic in the second and the fourth quadrants
(b.) Elliptic in the first and the third quadrants
(c.) Hyperbolic in the second and elliptic in the fourth quadrant
(d.) Hyperbolic in the first and the third quadrants

22. The partial differential equation $x \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y \frac{\partial^2 u}{\partial y^2} + x \frac{\partial u}{\partial y} + y \frac{\partial u}{\partial x} = 0$ is

- (a.) Elliptic in the region $x < 0, y < 0, xy > 1$
(b.) Elliptic in the region $x > 0, y > 0, xy > 1$
(c.) Parabolic in the region $x < 0, y < 0, xy > 1$
(d.) Hyperbolic in the region $x < 0, y < 0, xy > 1$

23. The partial differential equation $\frac{\partial^2 u}{\partial y^2} - y \frac{\partial^2 u}{\partial x^2} = 0$ has

- (a.) Two families of real characteristics curve for $y < 0$
(b.) No real characteristics for $y > 0$
(c.) Vertical lines as a family for characteristics curve for $y = 0$
(d.) Branches of quadratic curves as characteristics for $y \neq 0$

24. If $u(x, t)$ satisfy the partial differential equation $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$ then $u(x, t)$ can be of the form

(a.) $u(x, t) = f(e^{x-2t}) + g(x+2t)$
(b.) $u(x, t) = f(x^2 - 4t^2) + g(x^2 + 4t^2)$
(c.) $u(x, t) = f(2x - 4t) + g(x + 2t)$
(d.) $u(x, t) = f(2x - t) + g(2x + t)$

(Where f and g are non-trivial smooth function).

25. The second order partial differential equation $u_{xx} + xu_{yy} = 0$ is

- (a.) elliptic for $x > 0$
(b.) hyperbolic for $x > 0$
(c.) elliptic for $x < 0$
(d.) hyperbolic for $x < 0$

26. Let $P(x, y)$ be a particular integral of the partial differential equation

$\frac{\partial^2 z}{\partial x^2} - \frac{\partial z}{\partial y} = 2y - y^2$, Then $P(2, 3)$ equals

- (a.) 2.
(b.) 8.
(c.) 12.
(d.) 10.

27. The PDE is $\frac{\partial^2 u}{\partial x^2} + 2 \frac{\partial^2 u}{\partial x \partial y} + \frac{\partial^2 u}{\partial y^2} = 0$ is

- (a.) Parabolic and has characteristics $\xi(x, y) = x + 2y, \eta(x, y) = x - 2y$

- (b.) Reducible to the canonical form $\frac{\partial^2 u}{\partial \xi^2} = 0$, where $\xi(x, y) = x + 2y$

- (c.) Reducible to the canonical form $\frac{\partial^2 u}{\partial \eta^2} = 0$, where $\eta(x, y) = x + 2y$

- (d.) Parabolic and has the general solution $u = (x - y)f_1(x + y) + f_2(x - y)$ where f_1, f_2 are arbitrary functions.

28. Canonical form of $r - t = 0$ is/are

(a.) $\frac{\partial^2 z}{\partial V^2} = 0$
(b.) $\frac{\partial^2 z}{\partial U \partial V} = 0$
(c.) $\frac{\partial z}{\partial U} = 0$
(d.) $\frac{\partial z}{\partial V} = 0$

29. Partial Differential Equation

$$U_{xx} + U_{yy} + U_{zz} + U_{yz} + U_{zy} = 0$$

- (a.) Hyperbolic
- (b.) Parabolic
- (c.) Elliptic
- (d.) Laplace equation

30. The characteristics of the equation

$$x^2 \left(\frac{\partial^2 z}{\partial x^2} \right) - y^2 \left(\frac{\partial^2 z}{\partial y^2} \right) = 0 \text{ are}$$

- (a.) $x+y, x-y$
- (b.) $\frac{x}{y}, \frac{x}{y}$
- (c.) $xy, \frac{x}{y}$
- (d.) xy, x^2y^2

31. The second order partial differential equation.

$$(1-\sqrt{xy}) \frac{\partial^2 U}{\partial x^2} + 2 \frac{\partial^2 U}{\partial x \partial y} + (1+\sqrt{xy}) \frac{\partial^2 U}{\partial y^2} = 0 \text{ is}$$

- (a.) Hyperbolic in the second and the fourth quadrants.
- (b.) Elliptic in the first and the third quadrants.
- (c.) Hyperbolic in the second and elliptic in the fourth quadrant.
- (d.) Hyperbolic in the first and the third quadrants.

32. Find the integral surface passing through the given curve $p^2 + q^2 = 4z$ where $z = x^2 + 1$ and $y = 0$ then its general solution

- (a.) $z = x^2 + (y+1)^2$ if $b = a$
- (b.) $z = x^2 + (y-1)^2$ if $b = -a$
- (c.) $z = x^2 + (y-1)^2$ if $b = a$
- (d.) $z = x^2 + (y+1)^2$ if $b = -a$

33. The particular integral of the differential equation $(D^3 - D)y = e^x + e^{-x}$, $D \equiv \frac{d}{dx}$ is

- (a.) $\frac{1}{2}(e^x + e^{-x})$
- (b.) $\frac{1}{2}x(e^x + e^{-x})$
- (c.) $\frac{1}{2}x^2(e^x + e^{-x})$
- (d.) $\frac{1}{2}x^2(e^x - e^{-x})$

34. Solution of $\frac{2\partial^2 z}{\partial x^2} - \frac{5\partial^2 z}{\partial x \partial y} + \frac{2\partial^2 z}{\partial y^2} = 5 \sin(2x+y)$

is

- (a.) $z = \phi_1(y+2x) + \phi_2(2y+x) - \frac{5}{3} \cos(2x+y)$
- (b.) $z = \phi_1(y+2x) + \phi_2(2y+x) - \frac{5}{3} x \sin(2x+y)$
- (c.) $z = \phi_1(y+2x) + \phi_2(2y+3x) - \frac{5}{3} x \sin(2x+y)$
- (d.) $z = \phi_1(y+2x) + \phi_2(2y+x) - \frac{5}{3} x \cos(2x+y)$

35. The general solution of $(D + D' - 2)(D - D')z = 0$ is/are

- (a.) $z = e^{2x} \phi_1(x-y) + \phi_2(x+y)$
- (b.) $z = e^{2x} \phi_1(x+y) + \phi_2(x-y)$
- (c.) $z = e^{2y} \phi_1(x-y) + \phi_2(x+y)$
- (d.) $z = e^{2y} \phi_1(x+y) + \phi_2(x-y)$

36. In the (x, t) plane, the characteristics of the initial value problem $u_t + uu_x = 0$, with $u(x, 0) = x, 0 \leq x \leq 1$, are

- (a.) Parallel straight lines
- (b.) Straight lines which intersect at $(0, -1)$
- (c.) Non-intersecting parabolas
- (d.) Concentric circles with centre at the origin

37. Which of the following is NOT elliptic?

- (a.) Laplace equation
- (b.) $u_{xx} + 2u_{xy} + 4u_{yy} = 0$
- (c.) Harmonic function
- (d.) None of these

38. The Partial differential equation $y^3 u_{xx} - (x^2 - 1)u_{yy} = 0$ is

- (a.) Hyperbolic in $\{(x, y) : y < 0, |x| > 1\}$
- (b.) Parabolic in $\{(x, y) : y > 0, |x| > 1\}$
- (c.) Hyperbolic in $\{(x, y) : y > 0, |x| > 1\}$
- (d.) Hyperbolic for all pair (x, y)

39. Characteristic curves of $y^2r - x^2t = 0$ are

- (a.) $x^2 + y^2 = c_1, x^2 - y^2 = c_2$
- (b.) $x^2 + y^2 = c_1, x^3 - y^2 = c_2$
- (c.) $x^2 + y^2 = c_1, -x^2 + y^2 - y = c_2$
- (d.) None of the above

40. The characteristics curves of the equation $x^2u_{xx} - y^2u_{yy} = x^2y^2 + x$; $x > 0, u = u(x, y)$ are

- (a.) Rectangular hyperbola
- (b.) Parabola
- (c.) Circle
- (d.) Straight line

41. The partial differential equation $xU_{xx} + 2xyU_{xy} + yU_{yy} - xU_y - yU_x = 0$ is/are

- (a.) Hyperbolic in region $xy < 1$
- (b.) Hyperbolic in region $xy > 1$
- (c.) Elliptic in region $xy < 1$
- (d.) Elliptic in region $xy > 1$

42. A general solution of the equation $\frac{\partial u(x, y)}{\partial x} + u(x, y) = e^{-x}$ is

- (a.) $u(x, y) = e^{-x}f(y)$
- (b.) $u(x, y) = e^{-x}f(y) + xe^x$
- (c.) $u(x, y) = e^x f(y) + xe^{-x}$
- (d.) $u(x, y) = e^{-x}f(y) + xe^{-x}$

ASSIGNMENT SHEET -3

1. Given 4 points A, B, C and D (in that order) along a parallelogram whose sides have slopes $\pm \frac{1}{c}$ in the (x, t) plane and u is a solution of the wave equation $u_{tt} - c^2 u_{xx} = 0$. Suppose $u(A) = \frac{1}{2}$, $u(C) = \frac{1}{4}$, $u(B) = \frac{2}{3}$. Then $u(D)$ equals

- (a.) $\frac{1}{6}$
(b.) $\frac{1}{12}$
(c.) $\frac{1}{3}$
(d.) $\frac{7}{12}$

2. The function $u(x, t) = \begin{cases} \frac{1}{\sqrt{t}} e^{-\frac{x^2}{4t}} & t > 0, x \in \mathbb{R} \\ 0 & t \leq 0, x \in \mathbb{R} \end{cases}$

is a solution of the heat equation in

- (a.) $\{(x, t) : x \in \mathbb{R}, t \in \mathbb{R}\}$
(b.) $\{(x, t) : x \in \mathbb{R}, t > 0\}$ but not in the set $\{(x, t) : x \in \mathbb{R}, t < 0\}$
(c.) $\{(x, t) : x \in \mathbb{R}, t \in \mathbb{R}\} \setminus \{(0, 0)\}$
(d.) $\{(x, t) : x \in \mathbb{R}, t \in \mathbb{R}\}$

3. Consider the boundary value problem $u_{xx} + u_{yy} = 0$, $x \in (0, \pi)$, $y \in (0, \pi)$, $u(x, 0) = u(x, \pi) = u(0, y) = 0$. If an additional boundary condition $u_x(\pi, y) = \sin y$ is satisfied, then $u(x, \pi/2)$ is equal to

- (a.) $\frac{\pi}{2}(e^\pi - e^{-\pi})(e^x + e^{-x})$
(b.) $\frac{\pi(e^\pi + e^{-x})}{(e^\pi - e^{-\pi})}$
(c.) $\frac{(e^x - e^{-x})}{(e^\pi + e^{-\pi})}$
(d.) $\frac{\pi}{2}(e^\pi + e^{-\pi})(e^x + e^{-x})$

4. The diffusion equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$, $u = u(x, t)$, $u(0, t) = 0 = u(\pi, t)$, $u(x, 0) = \cos x \sin 5x$ admits the solution

- (a.) $\frac{e^{-36t}}{2} [\sin 6x + e^{20t} \sin 4x]$
(b.) $\frac{e^{-36t}}{2} [\sin 4x + e^{20t} \sin 6x]$
(c.) $\frac{e^{-20t}}{2} [\sin 3x + e^{15t} \sin 5x]$
(d.) $\frac{e^{-36t}}{2} [\sin 5x + e^{20t} \sin x]$

5. It is required to solve the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$, $0 < x < a$, $0 < y < b$, satisfying the boundary conditions $u(x, 0) = 0$, $u(x, b) = 0$, $u(0, y) = 0$ and $u(a, y) = f(y)$. If c_n 's are constants, then the equation and the homogeneous boundary conditions determine the fundamental set of solutions of the form

- (a.) $u(x, y) = \sum_{n=1}^{\infty} c_n \sin h \frac{n\pi x}{b} \sin \frac{n\pi y}{b}$
(b.) $u(x, y) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{b} \sin \frac{n\pi y}{b}$
(c.) $u(x, y) = \sum_{n=1}^{\infty} c_n \sin \frac{n\pi x}{b} \sin h \frac{n\pi y}{b}$
(d.) $u(x, y) = \sum_{n=1}^{\infty} c_n \sin h \frac{n\pi x}{b} \sin \frac{n\pi y}{b}$

6. Let $u(x, t)$ be the bounded of $\frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} = 0$

with $u(x, 0) = \frac{e^{2x} - 1}{e^{2x} + 1}$. Then $\lim_{t \rightarrow +\infty} u(1, t)$ equals

- (a.) $-1/2$
(b.) $1/2$
(c.) -1
(d.) 1

7. The vertical displacement $u(x, t)$ of an infinitely long elastic string is governed by the initial value problem $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$, $-\infty < x < \infty$, $t > 0$, $u(x, 0) = -x$ and $\frac{\partial u}{\partial t}(x, 0) = 0$. The value of $u(x, t)$ at $x = 2$ and $t = 2$ is equal to
(a.) 2
(b.) 4
(c.) -2
(d.) -4
8. Let $u(x, t)$ be the solution of $u_{tt} = u_{xx}$; $0 < x < 1$, $t > 0$, $u(x, 0) = x(1-x)$, $u_t(x, 0) = 0$. Then $u\left(\frac{1}{2}, \frac{1}{4}\right)$ is
(a.) $\frac{3}{16}$
(b.) $\frac{1}{4}$
(c.) $\frac{3}{4}$
(d.) $\frac{1}{16}$
9. Consider the Neumann problem $u_{xx} + u_{yy} = 0$, $0 < x < \pi$, $-1 < y < 1$ $u_x(0, y) = u_x(\pi, y) = 0$. The problem admits solution for
(a.) $\alpha = 0, \beta = 1$
(b.) $\alpha = -1, \beta = \frac{\pi}{2}$
(c.) $\alpha = 1, \beta = \frac{\pi}{2}$
(d.) $\alpha = 1, \beta = -\pi$
10. Consider the one dimensional heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $0 < x < 1$, $t > 0$ with the initial condition $u(x, 0) = 2 \cos^2 \pi x$ and the boundary conditions $\frac{\partial u}{\partial t}(0, t) = 0 = \frac{\partial u}{\partial t}(1, t)$. The temperature $u(x, t)$ is given by
(a.) $u(x, t) = 1 - e^{-4\pi^2 t} \cos 2\pi x$
(b.) $u(x, t) = 1 + e^{-4\pi^2 t} \cos 2\pi x$
(c.) $u(x, t) = 1 - e^{-4\pi^2 t} \sin 2\pi x$
(d.) $u(x, t) = 1 + e^{-4\pi^2 t} \sin 2\pi x$
11. For the diffusion problem $u_{xx} = u_t$ ($0 < x < \pi, t > 0$), $u(0, t) = 0$, $u(\pi, t) = 0$ and $u(x, 0) = 3 \sin 2x$, the solution is given by
(a.) $3e^{-t} \sin 2x$
(b.) $3e^{-4t} \sin 2x$
(c.) $3e^{-9t} \sin 2x$
(d.) $3e^{-2t} \sin 2x$
12. Let PQRS be a rectangle in the first quadrant whose adjacent sides PQ and QR have slopes 1 and -1 respectively. If $u(x, t)$ is a solution of $\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$ and $u(P) = 1$, $u(Q) = -1/2$, $u(R) = 1/2$, then $u(S)$ equals
(a.) 2
(b.) 1
(c.) $1/2$
(d.) $-1/2$
13. Consider the wave equation $\frac{\partial^2 u}{\partial t^2} = 4 \frac{\partial^2 u}{\partial x^2}$, $0 < x < \pi$, $t > 0$, with $u(0, t) = u(\pi, t) = 0$, $u(x, 0) = \sin x$ and $\frac{\partial u}{\partial t} = 0$ at $t = 0$. Then $u\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ is
(a.) 2
(b.) 1
(c.) 0
(d.) -1
14. Solve the heat equation $u_t - u_{xx} = 0$, $0 < x < \pi$, $0 < t < \infty$ $u(x, 0) = \sin x$, $0 < x < \pi$ $u(0, t) = u(\pi, t) = 0$, $t > 0$. Then u
(a.) Is unbounded in $(0, \pi) \times (0, \infty)$
(b.) Takes both positive and negative values in $(0, \pi) \times (0, \infty)$
(c.) Is negative in $(0, \pi) \times (0, \infty)$
(d.) $u(x, t) \leq e^{-t}$, for all $(x, t) \in (0, \pi) \times (0, \infty)$
15. Let $u(x, t)$ be the solution of the initial value problem $\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0$; $u(x, 0) = \sin x$; $\frac{\partial u}{\partial x}(x, 0) = 1$. Then $u(\pi, \pi/2)$ equals
(a.) $\pi/2$
(b.) $1/2$
(c.) -1
(d.) 1

16. Let $u(x, t)$ be the solution of the one dimensional wave equation
 $u_{tt} - 4u_{xx} = 0, -\infty < x < \infty, t > 0$
 $u(x, 0) = \begin{cases} 16 - x^2, & |x| \leq 4, \\ 0, & \text{otherwise,} \end{cases}$ and
 $u_t(x, 0) = \begin{cases} 1, & |x| \leq 2, \\ 0, & \text{otherwise,} \end{cases}$ For $1 < t < 3$, $u(2, t) =$
 (a.) $\frac{1}{2}[16 - (2 - 2t)^2] + \frac{1}{2}[1 - \min\{1, t - 1\}]$
 (b.) $\frac{1}{2}[32 - (2 - 2t)^2 - (2 + 2t)^2] + t$
 (c.) $\frac{1}{2}[32 - (2 - 2t)^2 - (2 + 2t)^2] + 1$
 (d.) $\frac{1}{2}[16 - (2 - 2t)^2] + \frac{1}{2}[1 - \max\{1 - t, -1\}]$
17. Let $u(x, t)$ be the solution to the wave equation
 $\frac{\partial^2 u}{\partial x^2}(x, t) = \frac{\partial^2 u}{\partial t^2}(x, t),$
 $u(x, 0) = \cos(5\pi x), \frac{\partial u}{\partial t}(x, 0) = 0$. Then, the value of $u(1, 1)$ is _____
18. A function $u(x, t)$ satisfies the wave equation
 $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, 0 < x < 1, t > 0$
 If $u\left(\frac{1}{2}, 0\right) = \frac{1}{4}, u\left(1, \frac{1}{2}\right) = 1$ and $u\left(0, \frac{1}{2}\right) = \frac{1}{2}$
 then $u\left(\frac{1}{2}, 1\right)$ is
 (a.) $\frac{7}{4}$
 (b.) $\frac{5}{4}$
 (c.) $\frac{4}{5}$
 (d.) $\frac{4}{7}$
19. Let $u(x, y)$ be a solution of Laplace's equation on $x^2 + y^2 \leq 1$. If
 $u(\cos \theta, \sin \theta) = \begin{cases} \sin \theta & \text{for } 0 \leq \theta \leq \pi \\ 0 & \text{for } \pi \leq \theta \leq 2\pi \end{cases}$ Then
 $u(0, 0)$ equals
 (a.) $1/\pi$
 (b.) $2/\pi$
 (c.) $1/(2\pi)$
 (d.) $\pi/2$

20. The solution of the Cauchy problem
 $u_{yy}(x, y) - u_{xx}(x, y) = 0;$
 $u(x, 0) = 0, u_y(x, 0) = x$ is $u(x, y) =$
 (a.) $\frac{x}{y}$
 (b.) $x.y$
 (c.) $xy + \frac{x}{y}$
 (d.) 0
21. If $u(x, t)$ is the D'Alembert's solution to the wave equation $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, x \in \mathbb{R}, t > 0$, with the condition $u(x, 0) = 0$ and $\frac{\partial u}{\partial t}(x, 0) = \cos x$, then $u\left(0, \frac{\pi}{4}\right)$ is _____
22. The function $u(r, \theta)$ satisfying the Laplace equation $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0, e < r < e^2$ subject to the conditions $u(e, \theta) = 1, u(e^2, \theta) = 0$ is
 (a.) $\ln\left(\frac{e}{r}\right)$
 (b.) $\ln\left(\frac{e}{r^2}\right)$
 (c.) $\ln\left(\frac{e^2}{r}\right)$
 (d.) $\sum_{n=1}^{\infty} \left(\frac{r - e^2}{e - e^2}\right) \sin n\theta$
23. The solution of Laplace's equation $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$ in the unit disc with boundary conditions $u(1, \theta) = 2 \cos^2 \theta$ is given by
 (a.) $1 + r^2 \cos \theta$
 (b.) $1 + \ln r + r \cos 2\theta$
 (c.) $2r^3 \cos^2 \theta$
 (d.) $1 - r^2 + 2r^2 \cos^2 \theta$

24. Consider the boundary value problem:
 $u_{xx} + u_{yy} = 0$ in $\Omega = \{(x, y); x^2 + y^2 < 1\}$ with
 $\frac{\partial u}{\partial n} = x^2 + y^2$ on the boundary of Ω ($\frac{\partial u}{\partial n}$ denotes the normal derivative of u). Then its solution $u(x, y)$
- (a.) Is unique and is identically zero
 (b.) Is unique up to a constant
 (c.) Does not exist
 (d.) Is Unique and non-zero
25. Let u be a solution of the initial value problem
 $\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0; u(x, 0) = x^2, \frac{\partial u}{\partial t}(x, 0) = 0$. Then $u(0, 1)$ equals
- (a.) 1
 (b.) 0
 (c.) 2
 (d.) $\frac{1}{2}$
26. Let $u(x, t), x \in \mathbb{R}, t \geq 0$, be the solution of the initial value problem $u_{tt} = u_{xx}$, $u(x, 0) = x$, $u_t(x, 0) = 1$. Then $u(2, 2)$ is equal to
27. The solution of the initial value problem $u_{tt} = 4u_{xx}, t > 0, -\infty < x < \infty$ satisfying the conditions $u(x, 0) = x, u_t(x, 0) = 0$ is
- (a.) x
 (b.) $\frac{x^2}{2}$
 (c.) $2x$
 (d.) $2t$
28. Which of the following, concerning the solution of the Neumann problem for Laplace's equation on a smooth bounded domain, is true?
- (a.) Solution is unique
 (b.) Solution is unique up to an additive constant
 (c.) Solution is unique up to a multiplicative constant
 (d.) No conclusion can be drawn about uniqueness
29. Let $G(x, y)$ be the Green's function of the boundary value problem $[(1+x)u']' + (\sin x)u = 0, x \in [0, 1], u(0) = u(1) = 0$. Then the function g defined by $g(x) = G\left(x, \frac{1}{2}\right), x \in [0, 1]$
- (a.) is continuous.
 (b.) is discontinuous at $x = \frac{1}{2}$.
 (c.) is differentiable.
 (d.) does not have the left derivative at $x = \frac{1}{2}$.
30. Consider the boundary value problem $u_{xx} + u_{yy} = 0, x \in (0, \pi), y \in (0, \pi), u(x, 0) = u(x, \pi) = u(0, y) = 0$. Any solution of this boundary value problem is of the form
- (a.) $\sum_{n \in \mathbb{N}} a_n \sinh nx \sin ny$
 (b.) $\sum_{n \in \mathbb{N}} a_n \cosh nx \sin ny$
 (c.) $\sum_{n \in \mathbb{N}} a_n \sinh nx \cosh ny$
 (d.) $\sum_{n \in \mathbb{N}} a_n \cosh nx \cos ny$
31. Let $u(x, t) = e^{i\omega x} v(t)$ with $v(0) = 1$ be a solution to $\frac{\partial u}{\partial t} = \frac{\partial^3 u}{\partial x^3}$ then
- (a.) $u(x, t) = e^{i\omega(x - \omega^2 t)}$
 (b.) $u(x, t) = e^{i\omega x - \omega^2 t}$
 (c.) $u(x, t) = e^{i\omega(x + \omega^2 t)}$
 (d.) $u(x, t) = e^{i\omega^3(x - t)}$
32. The partial differential equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + u$ can be transformed to $\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$ For
- (a.) $v = e^{-t}u$.
 (b.) $v = e^t u$.
 (c.) $v = tu$.
 (d.) $v = -tu$

33. Which of the following satisfies the heat equation without source terms and with diffusion constant in one space dimension?

(a.) $\sin\left(\frac{x^2}{4t}\right)$

(b.) $e^t \sin x$

(c.) $x^2 - t$

(d.) $\frac{e^{-x^2/4t}}{\sqrt{t}}$

34. Let $u(x, t)$ be the solution of $u_{tt} - u_{xx} = 1, x \in \mathbb{R}, t > 0$, with $u(x, 0) = 0, u_t(x, 0) = 0, x \in \mathbb{R}$.

Then $u(1/2, 1/2)$

(a.) $\frac{1}{8}$

(b.) $-\frac{1}{8}$

(c.) $\frac{1}{4}$

(d.) $-\frac{1}{4}$

35. If $u(x, t)$ satisfies the wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, x \in \mathbb{R}, t > 0$, with initial

conditions $u(x, 0) = \begin{cases} \sin \frac{\pi x}{c}, & 0 \leq x \leq c \\ 0 & \text{Elsewhere} \end{cases}$, and

$u_t(x, 0) = 0$ for all x , then for a given $t > 0$,

- (a.) There are values of x at which $u(x, t)$ is discontinuous
(b.) $u(x, t)$ is continuous but $u_x(x, t)$ is not continuous
(c.) $u(x, t), u_x(x, t)$ are continuous, but $u_{xx}(x, t)$ is not continuous
(d.) $u(x, t)$ is smooth for all x

36. The second order PDE $u_{yy} - yu_{xx} + x^3y = 0$ is

- (a.) Elliptic for all $x \in \mathbb{R}, y \in \mathbb{R}$
(b.) Parabolic for all $x \in \mathbb{R}, y \in \mathbb{R}$
(c.) Elliptic for all $x \in \mathbb{R}, y < 0$
(d.) Hyperbolic for all $x \in \mathbb{R}, y < 0$

37. Let a, b, c be continuous functions defined on $\mathbb{R}^2$. Let V_1, V_2, V_3 be nonempty subset of $\mathbb{R}^2$ such that $V_1 \cup V_2 \cup V_3 = \mathbb{R}^2$ and the PDE $a(x, y)u_{xx} + b(x, y)u_{xy} + c(x, y)u_{yy} = 0$ is elliptic in V_1 , parabolic in V_2 and hyperbolic in V_3 , then

(a.) V_1, V_2 and V_3 are open sets in $\mathbb{R}^2$

(b.) V_1 , and V_3 are open sets in $\mathbb{R}^2$

(c.) V_1 , and V_2 are open sets in $\mathbb{R}^2$

(d.) V_2 , and V_3 are open sets in $\mathbb{R}^2$

38. Consider the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, 0 < x < \pi, t > 0$, with the boundary conditions $u(0, t) = 0, u(\pi, t) = 0$ for $t > 0$, and the initial condition $u(x, 0) = \sin x$. Then $u\left(\frac{\pi}{2}, 1\right)$ is

39. Let $u(x, t)$ be the solution of the initial boundary value problem

$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, 0 < x < \infty, t > 0$,

$u(x, 0) = \cos\left(\frac{\pi x}{2}\right), 0 \leq x < \infty$,

$\frac{\partial u}{\partial t}(x, 0) = 0, t \geq 0$ Then

(a.) The value of $u(2, 2) = -1$

(b.) The value of $u(2, 2) = 1$

(c.) The value of $u\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{\sqrt{2}}$

(d.) The value of $u\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

40. Consider the Laplace equation in polar form:

$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0; 0 < r \leq a, 0 \leq \theta < 2\pi$

satisfying $u(a, \theta) = f(\theta)$, where f is a given function. Let σ be the separation constant that appears when one uses the method of separation of variables. Then for solution $u(r, \theta)$ to be bounded and also periodic in θ with period 2π ,

(a.) σ cannot be negative

(b.) σ can be zero, and in that case the solution is a constant

(c.) σ can be positive, and in the case it must be an integer

(d.) The fundamental set of solutions is $\{1, r^n \sin n\theta, r^n \cos n\theta\}$, where n is a positive integer.

41. The solution of the differential equation $\frac{d^2 y}{dx^2} = f(x)$; $x \in (0, 1)$ $y(0) = y(1) = 0$ is given by $y(x) = \int_0^1 G(x, \xi) f(\xi) d\xi$ where

(a.) $G(x, \xi) = \begin{cases} x(\xi-1); & x \leq \xi \\ \xi(x-1); & x > \xi \end{cases}$

(b.) $G(x, \xi) = \begin{cases} x^2(\xi-1); & x \leq \xi \\ \xi^2(x-1); & x > \xi \end{cases}$

(c.) $G(x, \xi) = \begin{cases} x(\xi^2-1); & x \leq \xi \\ \xi(x^2-1); & x > \xi \end{cases}$

(d.) $G(x, \xi) = \begin{cases} \sin x(\xi-1); & x \leq \xi \\ \sin \xi(x-1); & x > \xi \end{cases}$

42. Let u be a solution of the heat equation $u_t - u_{xx} = 0$ $0 < x < \pi$ and $t > 0$ $\left. \begin{array}{l} u(0, t) = u(\pi, t) = 0 \quad t > 0 \\ u(x, 0) = \sin x + \sin 2x, \quad 0 \leq x \leq \pi \end{array} \right\}$ Then

- (a.) $u(x, t) \rightarrow 0$ as $t \rightarrow \infty$ for all $x \in (0, \pi)$
- (b.) $t^2 u(x, t) \rightarrow 0$ as $t \rightarrow \infty$ for all $x \in (0, \pi)$
- (c.) $e^2 u(x, t)$ is a bounded function for $x \in (0, \pi), t > 0$
- (d.) $e^{2t} u(x, t) \rightarrow 0$ as $t \rightarrow \infty$ for all $x \in (0, \pi)$

43. Let $u(x, t)$ be the solution of the equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ which tends to zero as $t \rightarrow \infty$ and has the value $\cos(x)$ when $t = 0$ then

- (a.) $u = \sum_{n=1}^{\infty} a_n \sin(nx + b_n) e^{-n^2 t}$ where a_n, b_n are arbitrary constants
- (b.) $u = \sum_{n=1}^{\infty} a_n \sin(nx + b_n) e^{-n^2 t}$ where a_n, b_n are non-zero constants
- (c.) $u = \sum_{n=1}^{\infty} a_n \cos(nx + b_n) e^{-n^2 t}$ where a_n are not all zero and $b_n = 0$ for $n \geq 1$
- (d.) $u = \sum_{n=1}^{\infty} a_n \cos(nx + b_n) e^{-n^2 t}$ where $a_1 \neq 0, a_n = 0$ for $n > 1$, and $b_n = 0$ for $n \geq 1$

44. Let the heat equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \frac{\partial^2 u}{\partial x_3^2}$, $t \geq 0$, $\bar{x} = (x_1, x_2, x_3) \in \mathbb{R}^3$ admit an exponential function $\exp(i(\bar{k} \cdot \bar{x} + wt))$ as its solution, where $\bar{k}$ is a non zero constant real vector, and w is constant. Then the solution

- (a.) Remain constants on certain planes in $\mathbb{R}^3$
- (b.) Repeats itself after a certain length L
- (c.) Has, in general, an amplitude decaying exponentially with time t
- (d.) Is bounded uniformly for $\bar{x} \in \mathbb{R}^3$ for a fixed t

45. Let $u(x, t)$ satisfy for $x \in \mathbb{R}, t > 0$ $\frac{\partial^2 u}{\partial t^2} + \frac{\partial u}{\partial t} + 2 \frac{\partial^2 u}{\partial x^2} = 0$. A solution of the form $u = e^{ix} v(t)$ with $v(0) = 0$ and $v'(0) = 1$

- (a.) Is necessary Bounded
- (b.) Satisfies $|u(x, t)| < e^t$
- (c.) Is necessarily unbounded
- (d.) Is oscillatory in x

46. The Green's function $G(x, \zeta)$, $0 \leq x, \zeta \leq 1$ of the boundary value problem $y'' + \lambda y = 0$, $y(0) = 0 = y(1)$ is

- (a.) Symmetric in x and ζ
- (b.) Continuous at $x = \zeta$
- (c.) $\frac{\partial G(x, \zeta)}{\partial x} \Big|_{x=\zeta^-} - \frac{\partial G(x, \zeta)}{\partial x} \Big|_{x=\zeta^+} = -1$
- (d.) $\frac{\partial G(x, \zeta)}{\partial x} \Big|_{x=\zeta^-} - \frac{\partial G(x, \zeta)}{\partial x} \Big|_{x=\zeta^+} = 1$

47. The PDE

$$\left. \begin{array}{l} u_{xx} + u_{yy} + \lambda u = 0, \quad 0 < x, y < 1 \\ u(x, 0) = u(x, 1) = 0; \quad 0 \leq x \leq 1 \\ u(0, y) = u(1, y) = 0, \quad 0 \leq y \leq 1 \end{array} \right\} \text{ has}$$

- (a.) A unique solution u for any $\lambda \in \mathbb{R}$
- (b.) Infinitely many solutions for some $\lambda \in \mathbb{R}$
- (c.) A solution for countably many values of λ
- (d.) Infinitely many solutions for all $\lambda \in \mathbb{R}$.

48. A bounded solution to the partial differential

equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + e^{-t}$ is

- (a.) $u(x, t) = -e^{-t}$
(b.) $u(x, t) = e^{-x}e^{-t}$
(c.) $u(x, t) = e^{-x} + e^{-t}$
(d.) $u(x, t) = x - e^{-t}$

49. If the initial value problem for partial

differential equation $\frac{\partial u}{\partial t} + \frac{\partial^2 u}{\partial x^2} = 0;$

$u(x, 0) = \sin(\pi x)$, has a solution of the form

$u(x, t) = \phi(t)\sin(\pi x)$, then

- (a.) ϕ is always negative.
(b.) ϕ is always positive.
(c.) ϕ is an increasing function.
(d.) ϕ is a decreasing function.

50. Let u be of the boundary value problem

$$\left. \begin{aligned} u'' + \frac{1}{t}u' &= f(t), t \in (0, 1) \\ u'(0) &= a, u(1) = b \end{aligned} \right\} \quad \text{Define for}$$

$x^2 + y^2 \leq 1, v(x, y) = u(\sqrt{x^2 + y^2})$ and

$g(x, y) = f(\sqrt{x^2 + y^2})$, then v is a solution of

the PDE $\left. \begin{aligned} v_{xx} + v_{yy} &= g \text{ in } \{(x, y) : x^2 + y^2 \leq 1\} \\ v(x, y) &= 0 \text{ on } \{(x, y) : x^2 + y^2 = 1\} \end{aligned} \right\} \text{ if}$

(a.) $a > 0$ and $b > 0$

(b.) $a > 0$ and $b = 0$

(c.) $a = 0$ and $b = 0$

(d.) $a < 0$ and $b = 0$

51. Consider the boundary value problem (BVP)

$u'' = -f, u(0) = u'(1) = 0$ on $[0, 1]$, Where

$u' \equiv \frac{du}{dx}, u'' \equiv \frac{d^2u}{dx^2}$. Assume $f(x)$ is real-

valued continued function on $[0, 1]$. Then, which of the following are correct?

- (a.) The Green's function

$G(x, \zeta), (x, \zeta) \in [0, 1] \times [0, 1]$, for the above BVP is

(b.) $G(x, \zeta) = \begin{cases} x & \text{for } 0 \leq x \leq \zeta \\ \zeta & \text{for } \zeta \leq x \leq 1 \end{cases}$ Both G and

$\frac{\partial G}{\partial x}$ are continuous on $[0, 1] \times [0, 1]$ with

$\frac{\partial^2 G}{\partial x^2}$ having a discontinuity along $x = \zeta$

- (c.) $G(x, \zeta)$ satisfies the homogenous equation $u'' = 0$ for $0 \leq x < \zeta$ and $\zeta < x \leq 1$

- (d.) The solution of the given BVP is

$u(x) = \int_0^x \zeta f(\zeta) d\zeta + \int_x^1 \zeta f(\zeta) d\zeta$.

